SCHMEISSER'S CONJECTURE FOR POLYNOMIALS OF DEGREE AT LEAST 6251

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Abstract. We prove Schmeisser's conjecture for polynomials of degree at least 6251.

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1. Introduction

Let p be a complex polynomial of degree $n \geq 2$, and let $Z(p)$ denote its zeros, counted with algebraic multiplicity. The Gauss–Lucas theorem places the critical points $Z(p')$ in the convex hull $\operatorname{conv} Z(p)$. Schmeisser's conjecture asserts that, when $Z(p) \subseteq \overline{\mathbb{D}} = \{z \in \mathbb{C} : |z| \leq 1\}$, the closed unit disks centered at the critical points cover this entire convex hull. We prove the following degree range.

Theorem 1.1. *Let p be a complex polynomial of degree $n \geq 6251$, all of whose zeros lie in $\overline{\mathbb{D}}$. Then*

$$\operatorname{conv} Z(p) \subseteq \bigcup_{\xi \in Z(p')} \{z \in \mathbb{C} : |z - \xi| \leq 1\}.$$

The radius is sharp in every degree: for $p(z) = z^n - 1$, the only critical point is zero, while every zero has modulus one.

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1.1. **The conjecture and its early development.** Sendov's conjecture concerns the same distance bound at each zero of p. Sendov proposed it in 1958; Hayman's 1967 collection included it as Problem 4.5 under Ilieff's name, through whom the problem had been communicated [6, 13, 15]. Rubinstein [11] proved the assertion at zeros on the unit circle and for polynomials of degrees three and four. Meir and Sharma [9] subsequently treated degree five.

A stronger localization already appeared in Schmeisser's early work. For degrees at most four, he proved that for every zero a there is a critical point ξ satisfying

$$|\xi - a/2| \le 1 - |a|/2 \tag{1.1}$$

[12]. Gacs extended this localization to degree five [5]. It implies the convex-hull covering directly. Indeed, if $w = \sum_i t_i z_i$, with $t_i \ge 0$ and $\sum_i t_i = 1$, then

$$\sum_i t_i \operatorname{Re}(w\overline{z_i}) = |w|^2.$$

Choose a zero a for which $\operatorname{Re}(w\overline{a}) \ge |w|^2$. Then

$$|w - a/2|^2 = |w|^2 - \operatorname{Re}(w\overline{a}) + |a|^2/4 \le |a|^2/4.$$

Together with (1.1), this gives $|w - \xi| \le 1$.

Schmeisser also established geometric and algebraic special cases. His 1969 paper treats hulls whose vertices lie on the unit circle; interior zeros are allowed [12]. His 1971 paper develops coefficient estimates and proves Sendov's conjecture for trinomials and for monic polynomials with real nonpositive nonleading coefficients [13]. In Section 3 of [14], he formulated the convex-hull conjecture and observed that the earlier proofs give this stronger conclusion in degrees at most five, for hulls with all vertices on the unit circle, and for trinomials. The same paper proves the assertion for triangular hulls of arbitrary degree. Problem 4.30 of Hayman and Lingham [7] records the convex-hull question and its special cases $p(0) = 0$ and monic polynomials with real nonpositive nonleading coefficients.

1.2. **Subsequent distance estimates.** For Sendov's conjecture, Brown and Xiang [2] proved the result in degrees at most eight, and in arbitrary degree when there are at most eight distinct zeros. The large-degree approach began with Dégot's theorem [4]: for every fixed $a \in (0, 1)$ there is a degree bound depending on a such that a polynomial with $p(a) = 0$ and all zeros in $\overline{\mathbb{D}}$ has a critical point within distance one of a. Chalebgwa [3] made this dependence explicit, obtaining uniform bounds when a ranges over a compact subinterval of $(0, 1)$. Tao [17] proved a single absolute, unspecified degree threshold valid for all zeros. His argument treats separately sequences of distinguished zeros whose moduli tend to zero, to an interior point of $(0, 1)$, or to one. An explicit absolute bound $n \ge 10^{200000}$ was subsequently given by Zhang [20]. Mazur's all-degree argument and Tao's exposition concern the distance from each actual zero to the critical points [8, 18]; the convex-hull assertion appears as Conjecture 18 in [18].

A related line of work studies the centroid

$$G = \frac{1}{n} \sum_{z \in Z(p)} z = \frac{1}{n-1} \sum_{\xi \in Z(p')} \xi.$$

Pawłowski [10] proved, for zeros in $\overline{\mathbb{D}}$,

$$\min_{\xi \in Z(p')} |G - \xi| \le \frac{2n^{1/(n-1)}}{n^{2/(n-1)} + 1} < 1.$$

Zhang improved this bound to $\sqrt{(n-2)/(n-1)}$ [19]. For $Z(p) \subseteq \overline{\mathbb{D}}$, his later paper [21] gives

$$\min_{\xi \in Z(p')} |a - \xi| \le \sqrt{1 - |a|^2} \qquad (a \in \operatorname{conv} Z(p')).$$

The domain in this estimate is the critical-point hull. For example, its radius follows from a convex representation $a = \sum_j t_j \xi_j$:

$$\sum_j t_j |\xi_j - a|^2 = \sum_j t_j |\xi_j|^2 - |a|^2 \le 1 - |a|^2.$$

Schmeisser's question concerns the larger set $\operatorname{conv} Z(p)$.

Two recent estimates at an actual zero are useful here. For zeros in $\overline{\mathbb{D}}$, Zhang [22] proved the quadratic Tang–Zhang inequality

$$\sum_{\xi \in Z(p')} \frac{1}{|a-\xi|^2} \ge n-1 \qquad (a \in Z(p)),$$

with an infinite summand at a common zero of p and p'. For arbitrary complex polynomials, Bouthat and Fallat [1], and Zhang [23], proved Borcea's quadratic variance bound

$$\min_{\xi \in Z(p')} |a-\xi| \le \sigma_2(p) \qquad (a \in Z(p)), \qquad \sigma_2(p)^2 = \frac{1}{n} \sum_{z \in Z(p)} |z-G|^2.$$

Section 3 proves these estimates for $n \ge 6251$ and $n \ge 2048$, respectively.

1.3. **The explicit degree bound.** An earlier argument established Schmeisser's conjecture for all sufficiently large degrees [16]. Theorem 1.1 gives the explicit threshold 6251. Its proof consists of finite polynomial identities, analytic inequalities uniform for $n \ge 6251$, and a finite collection of rational and interval verifications.

A point missed by all the critical-point unit disks can be moved along a ray to a supporting edge of the zero hull. At a rightmost zero b we study the reciprocal critical points $(b-\xi)^{-1}$. A matrix representation bounds their first two moments. A saddle estimate and local primitive estimates sharpen these bounds and produce nearby zeros of the original polynomial. Near the origin, a finite Cayley–Hermite–Biehler identity supplies a source estimate; the radial identity couples it to the same polynomial's moments. In the other regions, equal-value fibers, product estimates and positive sums of root counts yield a contradiction. Section 11 combines the parameter ranges.

The paper gives the analytic estimates, the complete remainders, the parameter domains and the finite verification criteria. The electronic appendix supplies the rational data and verification programs. All zeros and critical points in sums and products are counted with multiplicity. Terminating decimals denote exact rational numbers. Each certified strict inequality holds on the indicated closed parameter cell, including its boundary.

2. Normalization and finite identities

Write $D(c,r) = \{z \in \mathbb{C} : |z-c| < r\}$ for the open disk, and $t_+ = \max(t,0)$ for real t. For vectors, $\|\cdot\|$ is the Euclidean norm; for matrices it is the associated operator norm. We use A^* for the conjugate transpose, $\|A\|_F^2 = \sum_{i,j} |A_{ij}|^2$ for the squared Frobenius norm, and $A \ge 0$ for a Hermitian positive semidefinite matrix. The symbols I and $\mathbf{1}$ denote the identity matrix and the all-ones vector of the required size; tr and diag denote the trace and diagonal-matrix operations.

Write $N = n-1$, and list the zeros of p as $z_1, \dots, z_n$ and the critical points as $\xi_1, \dots, \xi_N$. A *strict hole* is a point $a \in \operatorname{conv} Z(p)$ for which $|a-\xi_j| > 1$ for every j. The negation of Theorem 1.1 is precisely the existence of such a point. We may multiply p by a nonzero constant whenever convenient.

Lemma 2.1 (Gauss–Lucas)**.** *Every critical point belongs to* $\operatorname{conv} Z(p)$.

Proof. If $w \notin \operatorname{conv} Z(p)$, a separating line and a rotation give $\operatorname{Re}(e^{-i\theta}(w-z_k)) > 0$ for every k. Thus every $e^{i\theta}/(w-z_k)$ has positive real part. Their sum is $e^{i\theta}p'(w)/p(w)$ and cannot vanish. A zero shared by p and p' already lies in the zero hull. □

Lemma 2.2. *Suppose $n \geq 6251$ and all zeros lie in $\overline{\mathbb{D}}$. If a strict hole exists, after a rotation there is a strict hole a with $0 < a < 1$ on an actual supporting edge of a two-dimensional zero hull. For every critical point,*

$$\operatorname{Re} \xi_j < a/2. \tag{2.1}$$

There is an actual simple rightmost zero $b = x + iy$ with $x \geq a$. After conjugation we may take $y \geq 0$.

Proof. Gauss–Lucas implies $|\xi_j| \leq 1$, so zero cannot be a strict hole. Rotate a strict hole onto the positive real axis. Then

$$2a \operatorname{Re} \xi_j < a^2 + |\xi_j|^2 - 1 \leq a^2.$$

Let A be the last point of the zero hull on that ray. For $a \leq A \leq 1$,

$$|A - \xi_j|^2 - |a - \xi_j|^2 = (A - a)(A + a - 2 \operatorname{Re} \xi_j) \geq 0.$$

Thus the hole persists at A. If A is an actual zero, it is either multiple, in which case it is critical, or simple. In the latter case Proposition 3.1 gives $\sum_j |A - \xi_j|^{-2} \geq N$, contrary to all the distances being greater than one.

A hull consisting of one point has no hole. If the hull is a segment, Gauss–Lucas puts all critical points on it. The distance between any two consecutive distinct critical points is at most two, since the whole segment lies in the unit disk. Their closed unit intervals therefore have connected union. Each endpoint of the hull belongs to that union by the preceding actual-zero argument, so the whole segment is covered. In the remaining case A lies on a supporting edge whose endpoints are actual zeros. Strict convexity of the unit disk excludes $A = 1$. Rename A as a and repeat the displayed half-plane inequality at this point. A zero maximizing real part has real part at least a; it cannot be multiple by (2.1). Conjugation makes its imaginary part nonnegative. □

For an actual simple rightmost zero b, put

$$v_j = (b - \xi_j)^{-1}, \quad S = \sum_j v_j = \frac{p''(b)}{p'(b)}, \quad H = |S|, \quad Q = \sum_j |v_j|^2, \quad d = \min_j |b - \xi_j|. \tag{2.2}$$

Rightmostness and Gauss–Lucas give $\operatorname{Re} v_j \geq 0$ and $|v_j| \leq 1/d$. The equality $S = p''(b)/p'(b)$ counts the complete critical divisor. If the other original zeros are $u_1, \ldots, u_N$, factoring $p(z) = (z - b)g(z)$ gives the further identity

$$S = 2 \sum_{k=1}^{N} \frac{1}{b - u_k}. \tag{2.3}$$

Lemma 2.3. *The data in (2.2) satisfy*

$$N = 2 \operatorname{Re}(bS) + (1 - |b|^2)Q - \sum_j \frac{1 - |\xi_j|^2}{|b - \xi_j|^2}. \tag{2.4}$$

If all zeros lie in $\overline{\mathbb{D}}$, $a > 0$ is a strict hole, and $b = x + iy$ is rightmost with $x \geq a$, then $\operatorname{Re} S > 0$ and

$$Q < \frac{2 \operatorname{Re} S}{2x - a}. \tag{2.5}$$

Proof. For each ξ expand

$$2 \operatorname{Re} \frac{b}{b - \xi} = 1 + \frac{|b|^2 - |\xi|^2}{|b - \xi|^2}$$

and sum. For the second assertion, Gauss–Lucas and the strict hole give $2a \operatorname{Re} \xi < a^2 + |\xi|^2 - 1 \leq a^2$. Thus $\operatorname{Re}(b - \xi)^{-1} = (x - \operatorname{Re} \xi)/|b - \xi|^2 > (x - a/2)/|b - \xi|^2$. Summing proves the assertion. □

Lemma 2.4. *In the disk $|z| < d$, with the branch of each logarithm equal to zero at the origin,*

$$\frac{p'(b+z)}{p'(b)} = \prod_{j=1}^{N}(1+zv_j), \qquad p(b+z) = p'(b)\int_0^z \prod_{j=1}^{N}(1+tv_j)\,dt.$$

For every integer $r \geq 2$ and $|z| < d$,

$$\left|\sum_j \operatorname{Log}(1+zv_j) - \sum_{k=1}^{r-1}\frac{(-1)^{k+1}z^k}{k}\sum_j v_j^k\right| \leq \frac{Q|z|^r}{rd^{r-2}(1-|z|/d)}. \tag{2.6}$$

Proof. The product is the factorization of p'; integration uses $p(b) = 0$. The logarithms are analytic because $|zv_j| < 1$. For $k \geq r$, $\sum_j |v_j|^k \leq Qd^{2-k}$ and $1/k \leq 1/r$. Summing the entire remaining geometric series proves the bound. □

When $H > 0$ it is useful to write

$$D = Hd, \qquad \mu = D^{-1}, \qquad q = Q/H^2, \qquad \sigma = S/H, \qquad w_j = v_j/H. \tag{2.7}$$

Then $|\sigma| = 1$, $\sum w_j = \sigma$, $\sum |w_j|^2 = q$, and $\max |w_j| = \mu$. For $r > 0$, the polynomial $p_r(z) = p(rz)$ at the corresponding zero b/r has field data $d/r, rS, rH, r^2Q$. Thus D, q, μ, σ are invariant under positive real rescaling. Rescaled coordinates will be specified when used.

3. Estimates at an actual zero

We prove the quadratic Tang–Zhang inequality [22] and the quadratic variance bound [1, 23] in the degree ranges used below. Both estimates concern an actual zero of the polynomial.

3.1. The reciprocal-square estimate.

Proposition 3.1. *Let p have degree $n \geq 6251$, with all its zeros in $\overline{\mathbb{D}}$. For each simple zero a of p, one has*

$$E_a := \sum_{p'(\xi)=0}\frac{1}{|a-\xi|^2} \geq N.$$

The inequality is strict if $|a| < 1$. At a multiple zero we interpret the corresponding energy as $+\infty$.

Proof. Multiply by a constant to make p monic. Rotation reduces the interior case to $0 < a < 1$. Write

$$p(z) = (z-a)\prod_{j=1}^{N}(z-z_j), \quad q_j = (a-\xi_j)^{-1}, \quad \mu = \frac{1}{N}\sum_j q_j = x+iy,$$

$$s = \frac{1}{N}\sum_j |q_j|^2, \qquad v = s - |\mu|^2.$$

Suppose $s \leq 1$. Thus $|\mu| \leq 1$ and $0 \leq v \leq 1 - |\mu|^2$. Put

$$F(t) = \prod_j (1-atq_j), \qquad J = \prod_j z_j q_j, \qquad \delta = 1 - a^2.$$

Integration of p' from a to 0, followed by its factorization, gives

$$(N+1)\int_0^1 F(t)\,dt = (-1)^N J, \qquad |J| \leq s^{N/2} \leq 1. \tag{3.1}$$

Integration from a to $1/a$ gives a second identity:

$$\prod_j \frac{1-az_j}{a-z_j} = \int_0^1 \prod_j (a+\delta t q_j)\,dt. \tag{3.2}$$

The denominators are nonzero because a is simple. Since

$$|1-az_j|^2-|a-z_j|^2=\delta(1-|z_j|^2)\geq 0,$$

the arithmetic–geometric mean inequality yields

$$1\leq \int_0^1 \left(a^2+2a\delta xt+\delta^2 st^2\right)^{N/2}dt. \tag{3.3}$$

Define

$$\beta_s(t)=1-2axt+a^2st^2,\quad \beta(t)=1-2axt+a^2t^2,\quad \alpha=\frac{N\delta}{2}.$$

For $0<t<1$, the integrand in (3.3) is strictly smaller than $\exp\{\alpha[-1+(2-\beta(1))t]\}$. It follows first that $\beta(1)<1$, and hence $x>a/2$. Writing $g=1-\beta(1)>0$ and evaluating the exponential integral gives

$$e^{\alpha g}-e^{-\alpha}>\alpha(1+g). \tag{3.4}$$

Consequently

$$0<\beta(1)<\frac{\alpha}{1+\alpha},\qquad g>\frac{\log\alpha}{\alpha}\quad(\alpha\geq 1). \tag{3.5}$$

For the first bound, put $z=\alpha/(1+\alpha)$. Since $z<\log(1+\alpha)$,

$$e^z-1-z<\frac{z^2e^z}{2}<\frac{\alpha^2}{2(1+\alpha)}.$$

On the other hand, integrating $1-e^{-t}>t/(1+t)\geq t/(1+\alpha)$ gives $\alpha-1+e^{-\alpha}>\alpha^2/[2(1+\alpha)]$. Thus (3.4) excludes $\alpha g\leq z$. The second bound follows directly from $e^{\alpha g}>\alpha$.

There is also a useful estimate retaining s. Put

$$\rho=\delta s-1+2ax,\qquad \Lambda=\frac{(N+2)\delta}{2}.$$

The quadratic in (3.3) is bounded above by its chord $a^2+\delta(1+\rho)t$. Necessarily $\rho>0$; otherwise its integral is strictly less than one. Integration of this chord to the power $N/2$ and $\log(1+u)\leq u$ give, for $\Lambda>1$,

$$\rho\geq\frac{\log\Lambda}{\Lambda},\quad \beta_s(1)=s-\rho\leq 1-\rho,\quad x^2\geq\rho. \tag{3.6}$$

Indeed $e^{\Lambda\rho}\geq a^{N+2}+\Lambda(1+\rho)>\Lambda$, and $\rho\leq 2ax-a^2=x^2-(x-a)^2$.

Two integration estimates. For nonnegative b_j, the arithmetic–geometric mean inequality implies

$$\prod_{k\neq j}b_k\leq c_N\left(\frac{1}{N}\sum_k b_k^2\right)^{(N-1)/2},\qquad c_N=\left(\frac{N}{N-1}\right)^{(N-1)/2}<\frac{5}{3}.$$

The exact differentiation formula

$$F'=-Na\mu F-a^2t\sum_j q_j^2\prod_{k\neq j}(1-atq_k)$$

therefore yields

$$1\leq|F(1)|+\frac{N}{N+1}a+|R|,\quad |R|\leq\frac{5}{3}Na^2\int_0^1 t\beta_s(t)^{(N-1)/2}\,dt,\quad |F(1)|\leq\beta_s(1)^{N/2}. \tag{3.7}$$

For a second estimate, set $w_j=q_j-\mu$, $D(t)=1-at\mu$, and $G_\theta(t)=\prod_j[D(t)-at\theta w_j]$. Since $\sum_j w_j=0$,

$$D\partial_\theta G_\theta=-a^2t^2\theta\sum_j w_j^2\prod_{k\neq j}(D-at\theta w_k).$$

Moreover $D \neq 0$, and the mean squared modulus of the factors is at most $\beta_s(t)$. Integration in θ gives

$$|F - D^N| \leq \frac{5Na^2t^2v}{6|D|}\beta_s(t)^{(N-1)/2}.$$

Integrate once more, use (3.1), and use $|D| \geq 1 - axt$. With $r = |\mu|$, the result is

$$1 \leq ar + Cr(1-r^2) + E, \quad C = \frac{5}{6}a^3N(N+1)\int_0^1 \frac{t^2\beta(t)^{(N-1)/2}}{1-axt}\,dt, \quad E = \beta(1)^{(N+1)/2}. \tag{3.8}$$

If $2C < a$, then $ar + Cr(1-r^2) \leq a$, since

$$a - ar - Cr(1-r^2) = (1-r)[a - Cr(1+r)] \geq 0.$$

The range $a^2 \geq 9/10$. Let $\nu = (N-3)/2$. Here $\alpha \leq N/20$. For $0 < \alpha \leq 3$, (3.5) gives $\nu g > (N-3)/8 > 4\log N$. For $3 \leq \alpha \leq N/20$, it gives

$$\nu g > 10(1 - 3/N)\log(N/20) > 4\log N.$$

The last inequalities hold already at $N = 700$ and persist because the derivatives of the differences are

$$\frac{1}{8} - \frac{4}{N} > 0, \qquad \frac{6 - 30/N}{N} + \frac{30\log(N/20)}{N^2} > 0.$$

Also $9N/160 > 4\log N$. Thus

$$\beta(1)^\nu < N^{-4}, \qquad (133/160)^\nu < N^{-4}.$$

On $0 \leq t \leq 1/4$, $\beta(t) \leq e^{-3a^2t/4}$ and $1 - axt \geq 3/4$. Extending the positive integral to infinity gives

$$C_{[0,1/4]} \leq \frac{10240}{243}\frac{N(N+1)}{(N-1)^3a^3} < \frac{60}{N}.$$

For $1/4 \leq t \leq 1$, convexity bounds $\beta(t)$ by $\max(133/160, \beta(1))$, and $\beta(t) \leq 2(1-axt)$. Consequently

$$C_{[1/4,1]} < \frac{10}{3}N(N+1)N^{-4} < \frac{7}{N^2}, \qquad C < \frac{61}{N} < \frac{a}{2}.$$

If $\alpha \geq 1$, then $E/\delta < 1/(2N^3) < 1/4$. If $0 < \alpha \leq 1$, the bound $\beta(1) < \min(\alpha, 1/2)$ gives

$$\frac{E}{\delta} < \frac{N}{2}\frac{\min(\alpha, 1/2)^{(N+1)/2}}{\alpha} \leq N2^{-(N+1)/2} < \frac{1}{4}.$$

For the middle inequality, the quotient increases with α on $(0, 1/2]$ and decreases on $[1/2, 1]$. The latter inequality holds for every integer $N \geq 16$ by its endpoint and the ratio test. Equation (3.8) would now give $1 \leq a + E < a + (1-a^2)/4 < 1$.

The range $0 < a \leq \sqrt{9/10}$. Now $\delta \geq 1/10$. Split the integral in (3.7) at the minimum of β_s, clipping this point to $[0, 1]$. On its decreasing part $\beta_s(t) \leq e^{-axt}$, while on its increasing part $\beta_s(t) \leq 1 - \rho$. Using (3.6) gives

$$|R|_{\text{dec}} \leq D_N(a) := \min\left(\frac{28}{Na^2}, \frac{7\Lambda}{N\log\Lambda}\right), \tag{3.9}$$

$$|R|_{\text{inc}} \leq U_N(a) := \frac{5}{3}a^2N\Lambda^{-(N-1)/((N+2)\delta)}, \qquad |F(1)| \leq V_N(a) := \Lambda^{-N/((N+2)\delta)}. \tag{3.10}$$

The first alternative in D_N is omitted at $a = 0$. Both alternatives in D_N decrease with N. The exponent defining V_N has positive derivative

$$\frac{N + 2\log\Lambda}{(N+2)^2\delta}.$$

For $a > 0$, the logarithmic derivative of U_N is

$$\frac{(N+2)^2\delta - N(N-1) - 3N\log\Lambda}{N(N+2)^2\delta} < 0,$$

because $\delta \le 1$, $\log\Lambda > 2$, and its numerator is at most $5N + 4 - 3N\log\Lambda < 4 - N$. It therefore suffices to bound $D_N + U_N + V_N$ at $N_0 = 6250$.

Here are explicit bounds on seven closed intervals, each intersected with $[0, \sqrt{9/10}]$; the right column is a strict upper bound for the sum on that whole intersection.

l	h	upper bound
0	1/4	659/1000
1/4	3/8	396/1000
3/8	1/2	346/1000
1/2	5/8	192/1000
5/8	3/4	54/1000
3/4	7/8	11/1000
7/8	19/20	6/1000

For a row $[l, h]$, put

$$\delta_- = \max(1/10, 1 - h^2), \quad \delta_+ = 1 - l^2, \quad L_\pm = (N_0 + 2)\delta_\pm/2.$$

The three whole-interval upper bounds are

$$\min\left(\frac{28}{N_0 l^2}, \frac{7L_+}{N_0 \log L_+}\right), \quad \frac{5}{3}h^2 N_0 \exp\left(-\frac{N_0 - 1}{N_0 + 2}\frac{\log L_-}{\delta_+}\right), \quad \exp\left(-\frac{N_0}{N_0 + 2}\frac{\log L_-}{\delta_+}\right).$$

At $l = 0$ omit the first alternative. Their sum is below the displayed rational upper bound in each row. These seven inequalities are checked by the rational logarithm and exponential formulas in Section 12; the complete checker is `verify_actual_roots.py`. Every upper bound is smaller than $1 - h$, and $(19/20)^2 > 9/10$. Thus $D_N + U_N + V_N < 1 - a$, contradicting (3.7).

At $a = 0$, factorization gives $\prod_j |q_j| = n/\prod_j |z_j| \ge n$, hence $s > 1$. At $a = 1$, the identity

$$\sum_j \frac{1}{1 - \xi_j} = 2\sum_j \frac{1}{1 - z_j}$$

has real part at least N, since each summand on the right has real part at least $1/2$. Cauchy–Schwarz gives $E_a \ge N$. Rotation treats all boundary zeros and proves the proposition. □

3.2. **A variance bound.**

Proposition 3.2. *Let p have degree $n \ge 2048$ and zeros $z_1, \dots, z_n$, with*

$$\bar{z} = \frac{1}{n}\sum_i z_i, \qquad \sigma_2(p)^2 = \frac{1}{n}\sum_i |z_i - \bar{z}|^2.$$

Define $d(a) = \min_{p'(\xi)=0} |a - \xi|$. For every zero a, one has $d(a) \le \sigma_2(p)$. In particular, if all zeros lie in $\overline{\mathbb{D}}$, then

$$d(a)^2 \le 1 - \frac{1 - |a|^2}{n}. \tag{3.11}$$

Proof. The cases $\sigma_2(p) = 0$ and a multiple zero are immediate. Translation, scaling, rotation, and multiplication by a nonzero constant reduce a contrary example to a monic polynomial with

$$\sum_i z_i = 0, \quad \sum_i |z_i|^2 = n, \quad z_1 = a \ge 0, \quad |a - \xi_j| > 1 \quad (1 \le j \le N).$$

Let $D = \operatorname{diag}(z_i)$, $e = n^{-1/2}(1,\dots,1)^T$, and let U be an isometry onto $e^\perp$. The characteristic polynomial of U^*DU is p'/n: its determinant is the e-cofactor of $zI-D$, namely $p(z)e^*(zI-D)^{-1}e = p'(z)/n$. The identity extends across the zeros by polynomial continuation. Schur triangularization and the Frobenius norm therefore give

$$\sum_j |\xi_j|^2 \le \|U^*DU\|_F^2 = \sum_i |z_i|^2 - 2\|De\|^2 + |e^*De|^2 = n-2.$$

The critical centroid is zero, since $N^{-1}\sum_j \xi_j = n^{-1}\sum_i z_i = 0$. Thus, writing $\kappa = N^{-1}\sum_j |\xi_j|^2$, we have $\kappa \le (N-1)/N < 1$, and $a > 0$. The original centroid also implies $a^2 \le N$.

Set

$$q_j = (a-\xi_j)^{-1}, \quad \mu = \mathbb{E}q, \quad s = \mathbb{E}|q|^2 < 1, \quad v = s - |\mu|^2, \quad \Pi = \prod_j |q_j|, \quad h = 1-\sqrt{1-s}.$$

Here $\mathbb{E}$ denotes the average over all N critical labels. Since $\mathbb{E}(q^{-1}) = a$, $\mathbb{E}|q|^{-2} = a^2+\kappa$, and $\mathbb{E}\xi = 0$,

$$|1-a\mu|^2 \le \kappa v. \tag{3.12}$$

Indeed $a\mu - 1 = \mathbb{E}[\xi(q-\mu)]$. Applying weighted Cauchy–Schwarz to $q^{-1} - \bar{q} = (1-|q|^2)/q$ also gives

$$|a-\mu|^2 \le (1-s)(a^2+\kappa-1) \le (1-s)(a^2-1/N). \tag{3.13}$$

Consequently $\operatorname{Re}\mu \ge ah$ and $|\mu| \le a(1+\sqrt{1-s})$.

The product factorizations and the arithmetic–geometric mean inequality give

$$\Pi \le s^{N/2}, \quad |p(0)| \le a[(n-a^2)/N]^{N/2} \le \min(1, \sqrt{e}\,a),$$

$$|p'(a)| = n/\Pi, \qquad \sum_{i\ne 1} |a-z_i|^2 = n(1+a^2).$$

In particular,

$$\frac{n}{s^{N/2}} \le \frac{n}{\Pi} = |p'(a)| \le \left(\frac{n(1+a^2)}{N}\right)^{N/2}.$$

It follows that

$$1 \le \frac{N}{n} n^{2/N} \le (1+a^2)s \le 2(1+\sqrt{1-s}), \qquad a^2 > \frac{2\log n - 1}{n}. \tag{3.14}$$

For the upper bound use $ah \le |\mu| \le \sqrt{s}$ and $s/h = 1+\sqrt{1-s}$. Since $s < 1$ and $e^u \ge 1+u$,

$$1 + a^2 > \frac{N}{n} n^{2/N} \ge \frac{N + 2\log n}{n} = 1 + \frac{2\log n - 1}{n}.$$

In particular $M = Na^2 > 14$ for $n \ge 2048$.

Put

$$F(t) = \prod_j (1 - atq_j), \quad \beta(t) = 1 - 2a\operatorname{Re}\mu\, t + a^2 s t^2, \quad \gamma(t) = s\big(a^2(1-t)^2 + \kappa\big).$$

On the entire unit interval,

$$\beta(t) \le 1 - a^2h^2t, \qquad \beta(t) \le 1 - a^2 st(1-t). \tag{3.15}$$

Integration from a to 0 and the differentiation identity used above give

$$1 \le |F(1)| + \frac{N}{n}|\mu|\Pi|p(0)| + |R|, \tag{3.16}$$

where $c_N = (N/(N-1))^{(N-1)/2} < 5/3$ and

$$|F(1)| \le \min(\beta(1), \kappa s)^{N/2},$$
$$|R| \le c_N N a^2 s \int_0^1 t \min(\beta(t), \gamma(t))^{(N-1)/2}\, dt. \tag{3.17}$$

The γ bound follows by rewriting the jth differentiated term as

$$q_j^2 \prod_{k\neq j}(1 - atq_k) = q_j\Big(\prod_k q_k\Big)\prod_{k\neq j}(a(1-t) - \xi_k)$$

and using the critical centroid and second moment before applying the arithmetic–geometric mean inequality.

We also need the centered identity. Write $D(t) = 1 - at\mu$ and $\omega_j = a(\mu - q_j)$. Then $\sum \omega_j = 0$ and $\sum |\omega_j|^2 = Na^2 v$. Comparing F to D^N gives

$$1 \le |\mu|\Pi|p(0)| + |D(1)|^n + E_c, \qquad E_c = na|\mu| \int_0^1 |F(t) - D(t)^N|\, dt. \tag{3.18}$$

The interpolation $\prod_j (D + t\theta\omega_j)$ gives both bounds

$$|F - D^N| \le \frac{c_N N a^2 t^2 v}{2|D|}\beta^{(N-1)/2} \qquad (D \neq 0), \tag{3.19}$$

$$|F - D^N| \le \frac{c_{N,2} N(N-1) a^2 t^2 v}{2}\beta^{(N-2)/2}, \quad c_{N,2} = \left(\frac{N}{N-2}\right)^{(N-2)/2} < 3. \tag{3.20}$$

For the second estimate differentiate twice in θ, use $\sum_{j\neq k} |\omega_j \omega_k| \le N(N-1)a^2 v$, and integrate the Taylor remainder with weight $1 - \theta$. Thus it remains valid when $D = 0$.

We divide the parameter range into three cases. The numerical endpoint bounds and their degree monotonicities are checked at the end of the proof.

Case 1: $s \le 3/4$. Equation (3.14) gives $a^2 s \ge 1/4$. On $[0, 3/4]$, (3.15) gives $\beta(t) \le e^{-a^2 st/4}$, so

$$|R|_{[0,3/4]} \le \frac{64 c_N N}{(N-1)^2 a^2 s} < \frac{432}{N} < \frac{1}{4}.$$

On $[3/4, 1]$, $\gamma(t) \le 15s/16 + 2(1 + \sqrt{1-s})/16 \le 57/64$. Since $a^2 s \le 4$,

$$|R|_{[3/4,1]} \le \frac{5}{3} N (57/64)^{(N-1)/2} < \frac{1}{16}.$$

Each of the other two terms in (3.16) is less than $(3/4)^{N/2} < 1/16$. Their sum is less than $7/16$, a contradiction.

Case 2: $a \le 1/4$. Now $s \ge 16/17$, $h > 3/4$, and $|\mu| \le 5a/4$. The product term in (3.16) is at most $25a^2/12 \le 25/192$. On $[0, 1/2]$ use $\beta(t) \le 1 - a^2 t$; on $[1/2, 1]$ use convexity and $\beta \le 1 - a^2/2$. These give

$$|R|_{[0,1/2]} < \frac{27}{4M} < \frac{27}{56}, \qquad |R|_{[1/2,1]} \le \frac{5}{8} M e^{-(N-1)M/(4N)} < \frac{35}{128}.$$

Also $|F(1)| \le e^{-9M/32} < 1/48$. Thus the right side of (3.16) is at most

$$\frac{25}{192} + \frac{27}{56} + \frac{35}{128} + \frac{1}{48} = \frac{2437}{2688} < 1.$$

Case 3: $s > 3/4$ *and* $a > 1/4$. We have $h > 1/2$, $a < 2$, and $|\mu| < 3a/2$. Suppose first that $v > 0$. From (3.12), $\beta(1) = |D(1)|^2 + a^2 v$, and (3.15),

$$|D(1)|^n \le v\beta(1)^{(n-2)/2} < v/32.$$

For $1/4 < a \le 1/2$, we have $h \ge 1-a$ and $a^2h^3 \ge 27/1024$. On $[0,1/4]$ use $|D| \ge 7/8$ and $\beta \le e^{-3a^2ht/2}$ in (3.19), obtaining

$$E_{c,[0,1/4]}/v \le \frac{512c_N nN(|\mu|/a)}{189(N-1)^3a^2h^3} < \frac{300}{N} < \frac{1}{4}.$$

At the two endpoints of $[1/4,1]$,

$$\beta(1/4) \le 1-a^2(7/16-a/2) \le 1-5/256, \qquad \beta(1) \le 1-a^2(1-a)^2 \le 1-9/256.$$

The two losses increase on $[1/4,1/2]$. Convexity gives $\beta \le 1-5/256$ on $[1/4,1]$, where $|D| \ge 1/2$. In (3.19) the coefficient after multiplication by $na|\mu|$ is at most $c_N nNa^3|\mu| \le 5nN/32 < 3n^2/16$. Hence

$$E_{c,[1/4,1]}/v \le \frac{3}{16}n^2e^{-5(n-2)/512} < \frac{1}{32}.$$

For $a > 1/2$, instead use $a^2h^3 > 1/32$, $|D| > 1/2$ on $[0,1/4]$, and $|\mu|/a < 3/2$. This gives

$$E_{c,[0,1/4]}/v \le \frac{128c_N nN(|\mu|/a)}{27(N-1)^3a^2h^3} < \frac{400}{N} < \frac{1}{4}.$$

For $a > 1/2$, the endpoint bounds are

$$\beta(1/4) \le 1-3a^2h/8 < 1-3/64, \qquad \beta(1) \le 1-a^2h^2 < 1-1/16.$$

Convexity gives $\beta < 1-3/64$ on $[1/4,1]$. Since $a < 2$, $|\mu| < 1$, and $c_{N,2} < 3$, the coefficient in (3.20) after multiplication by $na|\mu|$ is less than $12nN(N-1) < 12n^3$. Thus

$$E_{c,[1/4,1]}/v \le 12n^3e^{-3(n-3)/128} < \frac{1}{32}.$$

In either subcase $|D(1)|^n + E_c < 5v/16$, whereas

$$|\mu|\Pi|p(0)| \le \sqrt{s-v} \le \sqrt{1-v} \le 1-v/2.$$

This contradicts (3.18). If $v = 0$, all $q_j = \mu$, and $a\mu = 1$, $D(1) = 0$, $F = D^N$. The same identity becomes $1 \le |\mu|^{N+1} < 1$.

Here are elementary checks for the constants used in the three cases. The inequality $\log 2 > 56/81$ follows from the first two terms of the positive atanh series. The expression $(1-1/n)(2\log n - 1)$ increases and exceeds 14 at $n = 2048$. The bounds $e < 25/9$ and $e > 5/2$ follow from the exponential series. For $N \ge 2047$,

$$\frac{N^2}{(N-1)^2} < \frac{101}{100}, \qquad \frac{(N+1)N^2}{(N-1)^3} < \frac{101}{100},$$

and both ratios decrease. The two centered coefficients are bounded by

$$\frac{512}{189}\frac{5}{3}\frac{3}{2}\frac{101}{100}\frac{1024}{27} = \frac{6619136}{25515} < 300, \qquad \frac{128}{27}\frac{5}{3}\frac{3}{2}\frac{101}{100}32 = \frac{51712}{135} < 400.$$

Positive exponential sums through degrees 7 and 12 give $e^{7161/2047} > 32$ and $e^{63/16} > 48$. The remaining endpoint estimates follow from $2^{17} > 32 \cdot 2048$, $(5/2)^{19} > 6 \cdot 2048^2$, and $2^{47} > 384 \cdot 2048^3$. Each function $n^k e^{-c(n-j)}$ used above decreases on $n \ge 2048$, as its logarithmic derivative is $k/n - c < 0$ there. Finally, in the unit-disk case,

$$\sigma_2(p)^2 = \frac{1}{n}\sum_i |z_i|^2 - |\bar{z}|^2 \le \frac{n-1+|a|^2}{n},$$

which proves (3.11). □

4. Reciprocal matrices and local fields

Throughout this section b is an actual simple rightmost zero. The notation d, S, H, Q is that of (2.2). The estimates are independent of the degree.

4.1. **A matrix envelope.**

Proposition 4.1. *Put $\eta = d\,\mathrm{Re}\,S$, $\tau = d\,\mathrm{Im}\,S$, and*

$$\Delta = 3\pi^2 - 8, \quad K_0 = \frac{9\pi^2}{\Delta}, \quad A_0 = 2 + \frac{24\pi}{\Delta}, \quad \gamma = \frac{40 - 3\pi^2}{2\Delta}.$$

Then

$$Qd^2 \le K_0 + A_0\eta + \gamma\eta^2 - \tau^2/2. \tag{4.1}$$

In particular, in the normalization (2.7), writing $\sigma = c + is$,

$$q < \frac{21}{5}\mu^2 + \frac{11}{2}\mu c + \frac{c^2}{4} - \frac{s^2}{2}. \tag{4.2}$$

Proof. Translate and scale to $b = 0, d = 1$. Write $v_j = -1/\xi_j$ and let $w_k = -1/u_k$, where the u_k are all the other original zeros. Set $M = I + \mathbf{1}\mathbf{1}^*$ and $A = M^{1/2}\,\mathrm{diag}(w_k)M^{1/2} = K + iJ$, with K, J Hermitian. Since $\mathrm{Re}\,w_k \ge 0$, we have $K \ge 0$. The rank-one determinant identity and differentiation of $p(z) = zp'(0)\prod_k(1 + zw_k)$ give

$$\det(I + z\,\mathrm{diag}(w_k)M) = \frac{p'(z)}{p'(0)} = \prod_j (1 + zv_j).$$

Thus the eigenvalues of A, including multiplicities, are the v_j.

For a Hermitian matrix B define $\mathcal{T}(B)_{ij} = i\,\mathrm{sgn}(i - j)B_{ij}$, with zero diagonal. If x is a unit vector and $b_k = |x_k|^2$, diagonal unitary conjugation removes the coordinate phases and gives

$$\det(tI - \mathcal{T}(xx^*)) = \frac{1}{2}\left\{\prod_k (t + ib_k) + \prod_k (t - ib_k)\right\}. \tag{4.3}$$

Indeed $tI - \mathcal{T}(xx^*)$ is the midpoint of an upper and a lower triangular matrix with diagonals $t + ib_k$ and $t - ib_k$; their difference has rank one. The determinant is affine along a rank-one line. Zero coordinates follow by continuity. After omitting those coordinates, the positive roots in (4.3) are the solutions of $\sum_k \arctan(b_k/t) = (2j - 1)\pi/2$. The left side strictly decreases from $p\pi/2$ to zero, where p is the number of nonzero coordinates. Since $\arctan u \le u$ and $\sum b_k = 1$, the jth positive eigenvalue is at most $L_j = 2/((2j - 1)\pi)$. The spectrum is symmetric about zero.

Let $\eta = \mathrm{tr}\,K$ and $k = \mathrm{tr}\,K^2$. The trace pairing $\mathrm{tr}(K\mathcal{T}(xx^*)) = -x^*\mathcal{T}(K)x$, followed by diagonalization of either sign of $\mathcal{T}(xx^*)$, shows that for every $\lambda > 0$,

$$\|\mathcal{T}(K)\| \le \lambda\eta + \sqrt{k}\left(\sum_{j\ge 1}(L_j - \lambda)_+^2\right)^{1/2}. \tag{4.4}$$

Here the diagonal entries of the conjugated K are nonnegative, sum to η, and have sum of squares at most k. This proves the inequality by Cauchy–Schwarz and includes the whole spectrum.

In an upper triangular Schur basis for A, $J = \mathrm{diag}(\mathrm{Im}\,v_j) + \mathcal{T}(K)$, whence $\|J\| \le 1 + \|\mathcal{T}(K)\|$. Write $w_j = a_j + ib_j$ and $\beta = \sum b_j = (\mathrm{Im}\,S)/2$. Direct multiplication gives

$$k = 3\sum a_j^2 + \left(\sum a_j\right)^2, \quad \mathrm{tr}\,J^2 = 3\sum b_j^2 + \beta^2, \quad \|Je\|^2 = \frac{N+1}{N}\left(\sum b_j^2 + \beta^2\right), \qquad e = N^{-1/2}\mathbf{1}.$$

In particular,

$$\mathrm{tr}\,J^2 = \frac{3N}{N+1}\|Je\|^2 - 2\beta^2 \le 3\|J\|^2 - 2\beta^2.$$

In a Schur basis the off-diagonal squared contributions to $\operatorname{tr} K^2$ and $\operatorname{tr} J^2$ agree, so

$$\operatorname{tr} K^2 - \operatorname{tr} J^2 = \sum_j ((\operatorname{Re} v_j)^2 - (\operatorname{Im} v_j)^2).$$

Also $0 \le \operatorname{Re} v_j \le 1$, whence $\sum_j (\operatorname{Re} v_j)^2 \le \eta$. Therefore

$$Q = 2\sum_j (\operatorname{Re} v_j)^2 - k + \operatorname{tr} J^2 \le 2\eta + 3(1 + \|\mathcal{T}(K)\|)^2 - k - 2\beta^2.$$

Take

$$\lambda = \frac{24\pi + (40 - 3\pi^2)\eta}{6\pi(3\pi + 4\eta)}, \qquad a_\lambda^2 = (2/\pi - \lambda)^2 + (2/(3\pi) - \lambda)^2.$$

For $\eta \ge 0$, one has $L_3 < \lambda < L_1$ and $3a_\lambda^2 < 1$. These follow by multiplying the positive denominators; the endpoint values of the fractional linear function $\lambda(\eta)$ suffice. Thus the sum in (4.4) is at most a_λ^2, including when $\lambda > L_2$. Maximizing the resulting quadratic in $\sqrt{k}$ gives

$$3(1 + \|\mathcal{T}(K)\|)^2 - k \le \frac{3(1 + \lambda\eta)^2}{1 - 3a_\lambda^2}.$$

Put

$$R_\eta = 18\pi^2 + 48\pi\eta + (40 - 3\pi^2)\eta^2.$$

Direct substitution gives

$$1 + \lambda\eta = \frac{R_\eta}{6\pi(3\pi + 4\eta)}, \qquad 1 - 3a_\lambda^2 = \frac{\Delta R_\eta}{6\pi^2(3\pi + 4\eta)^2}.$$

Consequently

$$2\eta + \frac{3(1 + \lambda\eta)^2}{1 - 3a_\lambda^2} = 2\eta + \frac{R_\eta}{2\Delta} = K_0 + A_0\eta + \gamma\eta^2.$$

This proves (4.1). Finally $\pi > 157/50$ implies $K_0 < 21/5$, $A_0 < 11/2$, and $\gamma < 1/4$. Divide by $(Hd)^2$ to obtain (4.2). □

4.2. **Logarithmic bounds and saddle moments.** The next inequality controls the logarithm on a disk.

Lemma 4.2. *For $0 < r < 1$ put*

$$h(r) = \frac{r - \log(1 + r)}{r^2}, \qquad j(r) = \frac{-r - \log(1 - r)}{r^2}.$$

For every $z = x + iy$ with $|z| \le r$,

$$\log|1 + z| \le x - h(r)x^2 + j(r)y^2.$$

Proof. The analytic identity $(z - \operatorname{Log}(1 + z))/z^2 = \int_0^1 t/(1 + tz)\,dt$ holds in $|z| < 1$. For fixed t, the image of $|z| \le r$ under $t/(1 + tz)$ lies in the disk with center $t/(1 - t^2r^2)$ and radius $t^2r/(1 - t^2r^2)$. The difference of squared radius and squared distance from the center is

$$\frac{t^4(r^2 - x^2 - y^2)}{(1 - t^2r^2)((1 + tx)^2 + t^2y^2)} \ge 0.$$

Integration therefore puts the preceding integral in the disk with center $c_r = -\log(1 - r^2)/(2r^2)$ and radius $e_r = (\operatorname{atanh} r - r)/r^2$. Multiplication by z^2 and taking real parts gives $\operatorname{Re}\operatorname{Log}(1 + z) \le x - (c_r - e_r)x^2 + (c_r + e_r)y^2$. The two coefficients are $h(r)$ and $j(r)$. □

Lemma 4.3. *Let* $\operatorname{Re} v_j \ge 0$, $|v_j| \le 1$, $\eta = \sum \operatorname{Re} v_j$, *and* $Q = \sum |v_j|^2 > 0$. *Suppose* $z_0 = -X + iY$, $|z_0| = s < 1$, $X \ge 0$, *and* $\sum_j v_j/(1 + z_0 v_j) = 0$. *Then*

$$\frac{\eta}{Q} \ge \frac{X(1-s)^2}{(1+|Y|)^2 - X^2}.$$

Proof. Conjugate to make $Y \ge 0$, and put $B = X/(1 + 2Y + s^2)$, $A = (1 - 2BX)/(1 - s)^2$. For $v = a + ib$, $r = |v| \le 1$, and $d_v = |1 + z_0 v|^2$, expansion gives

$$(Aa - Br^2)d_v - (a - Xr^2) = a(Ad_v - 1 + 2BXr^2) + Br^2[2Y(1 + b) + s^2(1 - r^2)].$$

The second summand is nonnegative. Also $d_v \ge (1 - sr)^2$ and $2BX \le 2s^2/(1 + s^2) \le s < 1$. The function $A(1 - sr)^2 + 2BXr^2$ decreases on $[0, 1]$ to its value one at $r = 1$, since $As(1 - s) - 2BX = (s - 2BX)/(1 - s) \ge 0$. Thus $\operatorname{Re}(v/(1 + z_0 v)) \le Aa - B|v|^2$. Sum and use the saddle equation to obtain $\eta/Q \ge B/A$, the stated bound. □

Lemma 4.4. *Let* $\operatorname{Re} v_j \ge 0$, $|v_j| \le 1$, $Q = \sum |v_j|^2 > 0$, *and* $\sum_j v_j/(1 + z_0 v_j) = 0$, *with* $\operatorname{Re} z_0 \le 0$. *Write* $u_j = v_j/(1 + z_0 v_j)$ *and* $Q_u = \sum |u_j|^2$. *Suppose* $|z_0| < s < 1$, $-\operatorname{Re} z_0 < X$, $|\operatorname{Im} z_0| < Y$, *and* $X < 1 - Y$. *Here* s *is an upper bound for the modulus. Put* $U = (1 - s)^{-1}$. *Then*

$$\frac{1}{1 + 2Y + s^2} \le \frac{Q_u}{Q} \le q_m := \min\left(U^2, \frac{1}{1 - 2X^2 - 2Y}\right)$$

when the last denominator is positive, and

$$\frac{\sum (\operatorname{Re} u_j)^2}{Q_u} \le \beta := \min\left\{X\left(\frac{1+X}{1-s^2} + XU^2\right), \frac{X(1 - Y + X)}{(1 - Y)^2 + X^2}\right\}.$$

Proof. Write $z_0 = x_0 + iy_0$, $v_j = a_j + ib_j$, and $d_j = |1 + z_0 v_j|^2$. Then

$$\operatorname{Re} u_j = \frac{a_j + x_0|v_j|^2}{d_j}, \quad \operatorname{Im} u_j = \frac{b_j - y_0|v_j|^2}{d_j}, \quad |u_j|^2 = \frac{|v_j|^2}{d_j}.$$

Since $a_j \ge 0$, $x_0 \le 0$, and $|v_j| \le 1$, $d_j \le 1 + 2Y + s^2$. The zero real sum gives $\sum a_j/d_j = -x_0 Q_u$. In the exact identity

$$Q = Q_u + 2x_0 \sum a_j|u_j|^2 - 2y_0 \sum b_j|u_j|^2 + |z_0|^2 \sum |v_j|^2|u_j|^2$$

we have $0 \le \sum a_j|u_j|^2 \le -x_0 Q_u$ and $|\sum b_j|u_j|^2| \le Q_u$. This proves the mass bounds.

The inverse disk inequality $(1 - |z_0|^2)|u_j|^2 + 2\operatorname{Re}(z_0 u_j) \le 1$ gives $\operatorname{Re} u_j \le (1 + X)/(1 - s^2)$, while $\operatorname{Re} u_j \ge -X|u_j|^2 \ge -XU^2$. Split the zero real sum into its positive and negative terms. The total absolute negative part is at most XQ_u. Multiplying each part by its respective amplitude bound proves the first variance bound.

For the second, write $z_0 = -X_0 + iY_0$. Give $r_j = 1/\overline{v_j}$ weights $|u_j|^2/Q_u$, omitting a zero v_j if necessary. Since $1/\overline{u_j} = r_j + \overline{z_0}$ and $\sum u_j = 0$, their weighted mean is $X_0 + iY_0$. Each r_j has nonnegative real part and modulus at least one. Put $L = 1 - |Y_0|$ and $c = X_0/L < 1$. The numbers $r'_j = (r_j - iY_0)/L = a'_j + ib'_j$ have $a'_j \ge 0$, $a'^2_j + b'^2_j \ge 1$, and weighted mean real part c. On this unbounded domain,

$$\frac{(a - c)^2}{(a - c)^2 + b^2} \le \frac{a + c^2}{1 + c^2}.$$

For $a \ge 1$ the right side is at least one. For $0 \le a \le 1$ replace b^2 by $1 - a^2$ and use

$$(a + c^2)(1 - 2ac + c^2) - (1 + c^2)(a - c)^2 = a(1 - a)(1 + c)^2 \ge 0.$$

The denominator cannot vanish. Taking weighted means proves $\sum (\operatorname{Re} u_j)^2/Q_u \le c(1 + c)/(1 + c^2)$. This function increases on $[0, 1]$, with derivative $(1 + 2c - c^2)/(1 + c^2)^2 > 0$. Substituting $c \le X/(1 - Y)$ proves the second bound. □

4.3. **An affine reciprocal-field estimate.**

Proposition 4.5. *For every actual simple rightmost zero,*

$$Qd^2 \le 20 + 11Hd. \tag{4.5}$$

Consequently $Qd^2 \le 13(1 + Hd)$ *whenever* $Hd \ge 7/2$.

Proof. Normalize to $b = 0, d = 1$, and suppose $Q > 20 + 11H$. Set $B = -\sum v_j^2$, $\eta = \operatorname{Re} S$. The matrix envelope implies $\eta > 0$. Conjugate so that $\operatorname{Im} S \ge 0$, and write

$$t = \frac{\operatorname{Im} S}{\eta}, \quad x = \frac{\operatorname{Re} B}{Q}, \quad y = \frac{|\operatorname{Im} B|}{Q}.$$

We have $B - Q = -2\sum(\operatorname{Re} v_j)v_j$, hence $|B - Q| \le 2\eta$ and $|B| \le Q$. Also $H/Q < 1/11$. The envelope gives $t^2 < 2\gamma < (7/10)^2$: if its quadratic coefficient were nonpositive, its constant and linear coefficients would already be smaller than $20 + 11H$. Thus

$$(t, y) \in [0, 7/10] \times [0, 1/4], \quad (x - 1)^2 + y^2 < \frac{4}{121(1 + t^2)}, \quad x^2 + y^2 \le 1. \tag{4.6}$$

Write $B/Q = x + i\epsilon y$, with $\epsilon \in \{-1, 1\}$. Both signs are used when $y = 0$.

Here and below directed rational rounding is denoted by $\lfloor u \rfloor_m = \lfloor mu \rfloor/m$ and $\lceil u \rceil_m = \lceil mu \rceil/m$. Write $\operatorname{sf}_m(u) = \lfloor \sqrt{u} \rfloor_m$ and $\operatorname{sc}_m(u) = \lceil \sqrt{u} \rceil_m$; these are computed by integer square roots. Unless indicated, square-root meshes have $m = 10^6$. Use $p_0 = 6283/2000 < \pi < 22/7$ and put $k_U = 9p_0^2/(3p_0^2 - 8)$, $a_U = 2 + 24p_0/(3p_0^2 - 8)$, $g_U = (40 - 3p_0^2)/(2(3p_0^2 - 8))$. These are upper bounds for K_0, A_0, γ, since each decreases on the indicated interval for π.

Each row of the certificate consists of six rationals $(t_l, t_h, y_l, y_h, \rho, k)$, with $\rho > 0$, $k > 0$, and the phase sign ϵ. They specify a closed rectangle and a contour. The following construction gives every bound used to check that row. Put

$$d_l = \operatorname{sf}(1 + t_l^2), \quad d_h = \operatorname{sf}(1 + t_h^2), \quad e_0 = \frac{2}{11d_l}, \quad g_0 = g_U - t_l^2/2.$$

If $y_l > e_0$ the cell is impossible. Otherwise define $r_0 = 1 - \operatorname{sc}(e_0^2 - y_l^2)$. If $r_0^2 + y_l^2 \ge 1$ or $g_0 \le 0$ the cell is also impossible under (4.6). In a retained cell cap y_h by $\widehat{y}_h = \min(y_h, e_0, \operatorname{sc}(1 - r_0^2))$. The retained rows have $r_0 > \widehat{y}_h \ge y_l$ and $g_0 > 0$. Let h_0 be the largest $1/100$ mesh point strictly below the positive root of

$$P(h) = k_U - 20 + (a_U - 11d_l)h + g_0h^2.$$

The checks $h_0 > 0$, $P(h_0) < 0$, and $P(h_0 + 1/100) \ge 0$ imply $\eta > h_0$. Set

$$Q_0 = 20 + 11d_lh_0, \quad r_Q = \operatorname{sf}_{10^4}(Q_0), \quad b_m = \operatorname{sf}(r_0^2 + y_l^2).$$

Choose the first 10^{-6} mesh point $a \in [1, 2]$ with

$$(ab_m - 1)(11 - a) > a^2. \tag{4.7}$$

Put $s = a/11$, $U = (1 - s)^{-1}$, $e = a^2/[b_m(11 - a)]$. For $L = p''/p'$ the geometric series gives

$$L(z) = S + Bz + E(z), \qquad |E(z)| \le \frac{Q|z|^2}{1 - |z|}.$$

On $|z| = aH/Q < s$, (4.7) gives $|Bz| > |S + E(z)|$. Rouché's theorem gives an actual zero z_0 of p'' with $|z_0| < aH/Q$ and

$$|z_0 + S/B| < (H/Q)e, \qquad \operatorname{Re} z_0 \le 0.$$

The latter assertion is Gauss–Lucas twice; p' is nonzero in the whole disk $|z| < 1$.

For phase bounds set

$$q_U = 1/d_l, \ c_U = t_h/d_h, \quad q_L = 1/\operatorname{sc}(1 + t_h^2), \ c_L = t_l/\operatorname{sc}(1 + t_l^2),$$

$$R_f = \frac{r_0}{r_0^2 + y_l^2}, \quad Y_f = \frac{\widehat{y}_h}{r_0^2 + \widehat{y}_h^2}, \quad R_L = \mathrm{sf}(1 - \widehat{y}_h^2).$$

On the unit-disk fiber, differentiation in $x > y$ gives $R_L \le x/(x^2+y^2) \le R_f$ and $y_l \le y/(x^2+y^2) \le Y_f$. For $\epsilon = 1$ take

$$D_c = q_U R_f + c_U Y_f + e, \qquad I_c = \max(0, c_U R_f - q_L y_l, q_U Y_f - c_L R_L) + e;$$

for $\epsilon = -1$ take

$$D_c = \max(0, q_U R_f - c_L y_l) + e, \qquad I_c = c_U R_f + q_U Y_f + e.$$

These follow by multiplying S/B explicitly, and give $-\operatorname{Re} z_0 < (H/Q)D_c$, $|\operatorname{Im} z_0| < (H/Q)I_c$. Initially put $X = \min(s, D_c/11)$, $Y = \min(s, I_c/11)$. Lemma 4.3 permits replacing X by its minimum with

$$\frac{\mathrm{sc}((1-s)^4 + 4h_*^2(1+Y)^2) - (1-s)^2}{2h_*}, \qquad h_* = \frac{1}{11 d_l}.$$

Apply Lemma 4.4 and denote its upper mass and variance bounds by q_m, β. Each row checks $1 - 2X^2 - 2Y > 0$, $X < 1 - Y$, and $\beta < 1/2$.

For $u_j = v_j/(1 + z_0 v_j)$ let $B_0 = -\sum u_j^2$. The geometric series gives $|B_0 - B|/Q < U^2 - 1 =: p_*$. The mass and variance bounds give

$$\operatorname{Re}(B_0/Q) \ge \frac{1 - 2\beta}{1 + 2Y + s^2}, \quad |\operatorname{Im}(B_0/Q)| \le 2q_m\sqrt{\beta(1-\beta)}, \quad |B_0/Q| \le q_m.$$

Use the rational enclosures

$$R = \left\lfloor \max\left(r_0 - p_*, \frac{1 - 2\beta}{1 + 2Y + s^2}\right)\right\rfloor_{10^4},$$

$$I = \lceil \min(\widehat{y}_h + p_*, 2q_m \,\mathrm{sc}(\beta(1-\beta)))\rceil_{10^4}, \quad M = \lceil q_m \rceil_{10^4}, \quad G = \mathrm{sf}_{10^4}(2p_0/M).$$

Thus $R > 0$, and

$$\left| \int_{-\infty}^{\infty} e^{-B_0 t^2/(2Q)}\, dt \right| = \sqrt{\frac{2\pi}{|B_0/Q|}} \ge G.$$

The contour. Define

$$\lambda = \left\lfloor \frac{\rho}{k/r_Q + D_c/11} \right\rfloor_{10^4}, \quad T_0 = \lfloor \rho r_Q \rfloor_{10^4}, \quad \delta = \mathrm{sc}_{10^5}(\rho^2(1 + \lambda^{-2})).$$

Check $\lambda > 1$, $T_0 > 0$, $U\delta < 1$. Put

$$T = \max\{T_0, \lambda[k + (H/\sqrt{Q})D_c]\}, \qquad D_c' = k - \sqrt{Q}\operatorname{Re} z_0.$$

Then $0 < D_c' < T/\lambda$ and $T/\sqrt{Q} < \rho$. Indeed $Q > r_Q^2$ and $H/Q < 1/11$, so

$$\begin{aligned} \frac{T}{\sqrt{Q}} &= \max\left\{\frac{T_0}{\sqrt{Q}}, \lambda\left(\frac{k}{\sqrt{Q}} + \frac{H}{Q}D_c\right)\right\} \\ &< \max\left\{\frac{T_0}{r_Q}, \lambda\left(\frac{k}{r_Q} + \frac{D_c}{11}\right)\right\} \le \rho. \end{aligned}$$

Take the vertical segment $z_0 + it/\sqrt{Q}$, $-T \le t \le T$, and both horizontal segments joining its ends to $\operatorname{Re} z = k/\sqrt{Q}$. Relative to z_0 every point has modulus less than δ.

Use the directed logarithm coefficients

$$\underline{h}(r) = \left\lfloor \sum_{\ell=0}^{15} \frac{(-r)^\ell}{\ell + 2} \right\rfloor_{10^6}, \qquad \overline{j}(r) = \left\lceil \sum_{\ell=0}^{15} \frac{r^\ell}{\ell + 2} + \frac{r^{16}}{18(1-r)} \right\rceil_{10^6}. \tag{4.8}$$

The alternating and geometric remainders prove $\underline{h} \le h$ and $\overline{j} \ge j$. Replace the two coefficients in Lemma 4.2 before expanding its quadratic form. At $r = U\rho$ and $r = U\delta$ write the respective pairs as (h_v, j_v) and (h_c, j_c). Set

$$q_l = \lfloor (1 + 2Y + s^2)^{-1} \rfloor_{10^6}, \quad A_v = \lfloor h_v - (h_v + j_v)\beta \rfloor_{10^6},$$

$$A_c = \lfloor h_c - (h_c + j_c)\beta \rfloor_{10^6}, \quad B_c = \lceil j_c - (h_c + j_c)\beta \rceil_{10^6}.$$

Every row has $q_l, A_v > 0$ and $B_c \ge 0$.

The paired vertical integral. For $|t| \le T$ put $f(t) = \sum_j \operatorname{Log}(1 + iu_j t/\sqrt{Q})$ and $g(t) = -B_0 t^2/(2Q)$. For a symmetric shell $a_j \le |t| \le b_j$, put $\rho_j = \min(\rho, b_j/r_Q)$, taking $\rho_j = \rho$ if $b_j = \infty$, and $r_j = U\rho_j$. With $h_j = \underline{h}(r_j)$ and $j_j = \overline{j}(r_j)$, set

$$\alpha_j = \lfloor \min\{R/2, q_l[h_j - (h_j + j_j)\beta]\} \rfloor_{10^5} > 0.$$

Both $\operatorname{Re} f(\pm t)$ and $\operatorname{Re} g(t)$ are at most $-\alpha_j t^2$. Writing the even and odd parts of $f - g$ as $E(t), O(t)$, the logarithmic series gives

$$|O(t)| \le \frac{q_m U |t|^3}{\sqrt{Q}} \left(\frac{1}{3} + \frac{r_j^2}{5(1 - r_j^2)} \right), \quad |E(t)| \le \frac{q_m U^2 t^4}{Q} \left(\frac{1}{4} + \frac{r_j^2}{6(1 - r_j^2)} \right).$$

These bounds retain all odd orders at least five and all even orders at least six. For arbitrary complex A, O,

$$e^{A+O} + e^{A-O} - 2e^A = O^2 \int_{-1}^{1} (1 - |s|) e^{A+sO}\, ds.$$

Apply this with $A = g + E$, and apply the first-difference integral to $e^{g+E} - e^g$. The real-part bounds hold along both line segments, so

$$|e^{f(t)} + e^{f(-t)} - 2e^{g(t)}| \le (|O(t)|^2 + 2|E(t)|) e^{-\alpha_j t^2}.$$

Use the shell endpoints $0, 1, 2, 3, 4, 5, 6, 8, 10, 12, \infty$. If $\Delta M_{k,j}$ is an upper bound for $\int_{a_j}^{b_j} t^k e^{-\alpha_j t^2}\, dt$, the paired error is at most

$$\mathcal{E} = \left\lceil \sum_j \frac{q_m^2 U^2 c_{o,j}^2 \Delta M_{6,j} + 2 q_m U^2 c_{e,j} \Delta M_{4,j}}{r_Q^2} \right\rceil_{10^4},$$

where $c_{o,j}$ and $c_{e,j}$ are the parenthesized coefficients. Extending a shell past the actual endpoint T only enlarges its positive error integral.

Here the moments have a purely rational verification. For $M_k(a) = \int_a^\infty t^k e^{-\alpha t^2}\, dt$,

$$M_k(a) = \frac{a^{k-1} e^{-\alpha a^2}}{2\alpha} + \frac{k - 1}{2\alpha} M_{k-2}(a).$$

For $a > 0$, $a e^{-\alpha a^2}/(1 + 2\alpha a^2) \le M_0(a) \le e^{-\alpha a^2}/(2\alpha a)$. Differentiating these proposed bounds and integrating to infinity proves the two signs. At zero, $M_4(0) = 3\sqrt{\pi}/(8\alpha^{5/2})$ and $M_6(0) = 15\sqrt{\pi}/(16\alpha^{7/2})$. Use directed square roots and $\pi < 22/7$. For each finite endpoint $x = \alpha a^2 < 82$,

$$E_{40}(x) \le e^x \le E_{80}(x) + \frac{x^{81}}{81!(1 - x/82)}, \qquad E_m(x) = \sum_{\ell=0}^{m} \frac{x^\ell}{\ell!}.$$

Taking reciprocals in the correct directions gives rational moment bounds; subtract a lower right tail from an upper left tail. The final infinite shell is included. The certificate checks every domain and positivity condition in this construction.

The two horizontal segments. There are two valid bounds, either of which may be used on a row. For the translated bound, expansion of Lemma 4.2 at $w = (d + i\epsilon_c T)/\sqrt{Q}$, with $\epsilon_c \in \{-1, 1\}$ and $0 \le d \le T/\lambda$, gives

$$\operatorname{Re}\sum \operatorname{Log}(1 + u_j w) \le -q_l(A_c - B_c/\lambda^2)T^2 + (h_c + j_c)IT^2/\lambda.$$

Indeed, if $\nu = \sum(\operatorname{Re} u_j)^2/Q_u \le \beta$, the quadratic part before substitution is

$$\frac{Q_u}{Q}\{-[h_c - (h_c + j_c)\nu]T^2 + [j_c - (h_c + j_c)\nu]d^2\},$$

which increases with ν since $T^2 \ge d^2$. The mixed term has modulus at most $(h_c + j_c)IdT$. When $A_c - B_c/\lambda^2 > 0$, the downward-rounded damping is

$$\kappa_{\mathrm{tr}} = \left\lfloor q_l(A_c - B_c/\lambda^2) - (h_c + j_c)I/\lambda \right\rfloor_{10^5}.$$

For the original-coordinate bound put $l = 1/\lambda$, $J = 1 + I_c/(\lambda D_c)$, and $r_o = \mathrm{sc}_{10^5}(\rho^2(l^2 + J^2))$. If $J < 2$ and $r_o < 1$, take $h_o = \underline{h}(r_o)$, $j_o = \overline{j}(r_o)$, and $A_o = h_o - (h_o + j_o)/(11d_l)$. Writing $z = x_c + iy_c$, $V = \sum(\operatorname{Re} v_j)^2 \le \eta$ and $W = \sum \operatorname{Re} v_j \operatorname{Im} v_j$, the exact expanded bound is

$$\log|p'(z)/p'(0)| \le \eta x_c - (\operatorname{Im} S)y_c + [j_o Q - (h_o + j_o)V]x_c^2 + [(h_o + j_o)V - h_o Q]y_c^2 + 2(h_o + j_o)W x_c y_c.$$

On both segments, $|x_c|\sqrt{Q}/T \le l$, $|y_c|\sqrt{Q}/T \in [2 - J, J]$, $x_c \le k/\sqrt{Q}$, and $H/(\sqrt{Q}T) \le 1/(\lambda D_c)$. As $2|W| \le \widehat{y}_h Q$, the normalized upper bound is

$$j_o l^2 + \frac{q_U k}{\lambda D_c T_0} + \max_{2-J \le v \le J}(-A_o v^2 + Lv), \qquad L = (h_o + j_o)\widehat{y}_h l + \frac{c_U}{\lambda D_c}.$$

When $A_o > 0$, the maximum is attained at an endpoint or at $L/(2A_o)$ if that point belongs to the interval. For normalization at the saddle, its exact equation gives

$$\ell_0 = \sum_j \left[\operatorname{Log}(1 + z_0 v_j) - \frac{z_0 v_j}{1 + z_0 v_j}\right].$$

The series has coefficient $(-1)^m(m-1)/m$ in degree $m \ge 2$. Hence $|\ell_0| \le Q|z_0|^2 f(s)$, where $f(s) = 1/(1-s) - \sum_{j\ge 0} s^j/(j+2)$. Use $f_s = \lceil 1/(1-s) - \sum_{j=0}^{15} s^j/(j+2)\rceil_{10^6}$. Since $Q|z_0|^2/T^2 \le a^2/(\lambda^2 D_c^2)$, a second valid damping is

$$\kappa_{\mathrm{or}} = \left\lfloor -\max_{[2-J,J]}(-A_o v^2 + Lv) - j_o l^2 - \frac{q_U k}{\lambda D_c T_0} - \frac{f_s a^2}{\lambda^2 D_c^2} \right\rfloor_{10^5}.$$

Choose the larger valid positive damping κ. Every retained row has at least one.

The final finite predicates. Put $F_0 = \lfloor (G - \mathcal{E} - 1/1000 - 1/100)/2 \rfloor_{10^4}$ and $J_0 = \lceil 1 + I_c/(\lambda D_c)\rceil_{10^4}$. In addition to the conditions above, each row verifies

$$F_0 > 0, \qquad E_2(RT_0^2/2) > \frac{2000}{RT_0}, \qquad \kappa T_0^2 > \frac{1}{2}, \tag{4.9}$$

$$E_{24}(\kappa T_0^2) > \frac{200 T_0}{\lambda}, \qquad \kappa(k^2 + J_0^2 T_0^2) > J_0^2, \tag{4.10}$$

$$E_{24}(\kappa T_0^2) > \frac{11}{10}\frac{k^2 + J_0^2 T_0^2}{k F_0}. \tag{4.11}$$

The two Gaussian tails are bounded by

$$2\int_T^\infty e^{-Rt^2/2}\,dt \le \frac{2e^{-RT^2/2}}{RT} < \frac{1}{1000}.$$

The sum of the moduli of the two horizontal integrals, divided by $|p'(z_0)|/\sqrt{Q}$, is bounded by

$$\frac{2T}{\lambda}e^{-\kappa T^2} < \frac{1}{100}.$$

The first bound decreases for $T > 0$; the second decreases for $\kappa T^2 > 1/2$. Thus the finite comparisons at T_0 prove both bounds on the whole ray $T \geq T_0$. The endpoint primitive values therefore differ by more than $2F_0|p'(z_0)|/\sqrt{Q}$. At one endpoint $z_{\epsilon_c} = k/\sqrt{Q} + i(\operatorname{Im} z_0 + \epsilon_c T/\sqrt{Q})$,

$$|p(z_{\epsilon_c})| > F_0|p'(z_0)|/\sqrt{Q}, \quad |p'(z_{\epsilon_c})| \leq e^{-\kappa T^2}|p'(z_0)|, \quad \sqrt{Q}|\operatorname{Im} z_{\epsilon_c}| \leq J_0 T.$$

But every original zero lies in $\operatorname{Re} z \leq 0$, and zero itself is a simple zero. Therefore $\operatorname{Re}(p'/p)(z) \geq \operatorname{Re}(1/z)$ in $\operatorname{Re} z > 0$, which would imply

$$\frac{k}{k^2 + J_0^2 T^2} < \frac{e^{-\kappa T^2}}{F_0}.$$

The last two predicates make $(k^2 + J_0^2 T^2)e^{-\kappa T^2}$ decreasing for $T \geq T_0$ and already smaller than $(10/11)kF_0$ at T_0. This is a contradiction.

The finite input consists of the four tables `C11_A20_PHASE_1_DIRECT_QUADRATIC_METRIC_CELLS.csv`, `C11_A20_PHASE_-1_DIRECT_QUADRATIC_METRIC_CELLS.csv`, and their corresponding `SKIPS.csv` tables. They contain respectively 77,198 retained rectangles and 14,14 excluded rectangles. On each sign they cover exactly $[0, 7/10] \times [0, 1/4]$. This is checked by sorting every rational vertical edge, merging the horizontal intervals on each intervening open slab, and repeating the check on each closed edge. All excluded rectangles satisfy the explicit exclusion above. The program `affine/replay_affine_c11_a20.py` reconstructs every quantity from the six rational entries and verifies (4.7)–(4.11) with rational arithmetic, including all 2750 shells. This proves the required finite assertion and the proposition. The final consequence follows from $20 + 11D \leq 13(1 + D)$ for $D \geq 7/2$. □

4.4. **Closing the unbounded field range.** The two signed-curvature bounds from Proposition 5.2 give the following estimate.

Proposition 4.6. *If $D = Hd \geq 270$, then $Qd/H < 9/4$.*

Proof. Put $y = Qd/H = qD$. Proposition 4.5 implies

$$y \leq 13(1 + 1/D) \leq Y_0 := 3523/270, \qquad q \leq 13(D + 1)/D^2 \leq 3523/72900 < 97/2000.$$

We also have $\mu = 1/D \leq 1/270$. For $0 \leq r \leq 97/2000$ define

$$v(r) = r + 17r^2 + 137r^3 + 313r^4, \quad m(r) = 1 - v(r),$$

$$A(r) = \frac{1 + 17r + 137r^2 + 313r^3}{m(r)}, \qquad B(r) = \frac{18 + 154r + 450r^2 + 313r^3}{m(r)},$$

$$C(r) = 1 + \left(2 + \frac{100}{199}\right) rA(r)^2 + 3rB(r) + \frac{600}{199}r^2A(r)^2 + 4rA(r) + 29r + 436r^2.$$

Let $K = p''/p'$. The coarse quartic row gives

$$\frac{\operatorname{Re} K'(b)}{H^2} < C(q)q^2 + \frac{50000}{3}q^5 + \frac{650}{9}q\mu.$$

On the other hand, rightmostness gives $\operatorname{Re} K'(b) = Q - 2\sum(\operatorname{Re} v_j)^2 \geq Q - 2H/d$. Consequently

$$0 < P_D(y) := y^2 C(y/D) + \frac{50000}{3}\frac{y^5}{D^3} + \frac{650}{9}y - D(y - 2). \tag{4.12}$$

On the whole interval $0 \le r \le 97/2000$, $v(r) < 1$. The convergent series $1/m = \sum_{k\ge 0} v^k$ and all numerators above have nonnegative coefficients. Thus $C, C', C'' \ge 0$ there. For $3 \le y \le Y_0$ and $D \ge 270$,

$$\partial_y^2 P_D = 2C + \frac{4y}{D}C' + \frac{y^2}{D^2}C'' + \frac{1000000}{3}\frac{y^3}{D^3} > 0,$$

$$\partial_D P_D = -\frac{y^3}{D^2}C' - 50000\frac{y^5}{D^4} - (y-2) < 0.$$

All occurrences of C and its derivatives have argument y/D. Direct rational substitution gives

$$P_{270}(3) = -\frac{34307701791336525292 0829}{10130931628072339534650} < 0, \qquad P_{270}(Y_0) < -197.$$

Convexity bounds P_{270} above by its endpoint chord. Monotonicity in D then contradicts (4.12) if $y \ge 3$.

Now $y < 3$, hence $q < 1/90 < 7/625$. The fine quartic row gives $\operatorname{Re} K'(b)/H^2 < (9/4)q^2 + 13q\mu$. It follows that $0 < (9/4)y^2 + 13y - D(y-2)$. At $D = 270$ this convex quadratic has endpoint values $-1719/64$ and $-843/4$ at $y = 9/4$ and $y = 3$. It decreases in D for every $y > 2$. Thus $y \ge 9/4$ is impossible. All the rational endpoint inequalities, including the two companion rows, are checked in `quartic/verify_d270.py`. □

4.5. **A direct finite-product obstruction.** The following criterion applies to a closed four-dimensional parameter cell. It uses the original-root factorization in the right half-plane.

Lemma 4.7. *Suppose* $\operatorname{Re} w_j \ge 0$, $|w_j| \le \mu$, $\sum w_j = \sigma = e^{i\phi}$, *and* $\sum |w_j|^2 = q$, *with* $0 \le \phi \le \pi/2$. *Put* $T = -\sum w_j^2 = q(1 - \alpha + i\beta)$. *Then*

$$\alpha^2 + \beta^2 \le (2\mu\cos\phi/q)^2, \qquad (1-\alpha)^2 + \beta^2 \le 1.$$

If $c = \cos\phi > 0$, *put* $A = q\alpha/2$, $C = -q\beta/2$. *For any upper bound* $\mu_h \ge \mu$, *the third moment* $P_3 = \sum w_j^3$ *satisfies*

$$\frac{4A^2}{c} - 3\mu_h^2 c \le \operatorname{Re} P_3 \le \min\left(\mu_h A, \mu_h^2 c - \frac{4C^2}{c}\right), \tag{4.13}$$

$$\left|\operatorname{Im} P_3 - \frac{3AC}{c}\right| \le \frac{3}{2}\left(\mu_h^2 c - \frac{A^2 + C^2}{c}\right) + \mu_h(q - A). \tag{4.14}$$

Proof. Write $w_j = x_j + iy_j$. The identity $T - q = -2\sum x_j w_j$, weighted Cauchy–Schwarz, and $\sum x_j = c$ give $|T - q| \le 2\mu c$; also $|T| \le q$. Here $A = \sum x_j^2$, $C = \sum x_j y_j$. Give the points (x_j, y_j) probability weights x_j/c. Their mean is $(A, C)/c$, and the trace of their covariance matrix is at most $\mu_h^2 - (A^2 + C^2)/c^2$. Now $\operatorname{Re} P_3 = 4\sum x_j^3 - 3\sum x_j|w_j|^2$ gives the lower bound, while $\operatorname{Re} P_3 \le \sum x_j^3 \le \mu_h A$ and $\operatorname{Re} P_3 = \sum x_j|w_j|^2 - 4\sum x_j y_j^2$ give the two upper bounds. Finally $\operatorname{Im} P_3 = 3\sum x_j^2 y_j - \sum y_j^3$. The covariance bound controls the first sum about AC/c, and $|\sum y_j^3| \le \mu_h \sum y_j^2 = \mu_h(q - A)$ controls the second. Thus points with $x_j = 0$ are included. □

Lemma 4.8. *Let* b *be an actual simple rightmost zero, assume* $H = |S| > 0$, *and use* (2.7). *Conjugate so that* $\sigma = e^{i\phi}$ *with* $0 \le \phi \le \pi/2$. *Define the actual normalized primitive*

$$J(w) = \frac{S\,p(b + w/S)}{p'(b)}, \quad \theta = \frac{-\sum_j w_j^2}{\sigma^2}, \quad \kappa = \frac{\sum_j w_j^3}{\sigma^3}.$$

Suppose a closed parameter cell has bounds $q \le q_h$, $\mu \le \mu_h$, $\phi_l \le \phi \le \phi_h$. *At a nonzero test point* w, *put* $W = |w|$ *and assume* $\mu_h W < 1$. *For recorded exact coefficients* θ_0, κ_0, *let* $|\theta - \theta_0| \le h_2$,

$|\kappa-\kappa_0|\le h_3$ *hold on the whole cell. Put*

$$B(t)=\sum_{k=4}^{8}\frac{q_h\mu_h^{k-2}W^k}{k}t^k+\frac{q_h\mu_h^7W^9}{9(1-\mu_hW)}t^9,$$

$$J_0=w\int_0^1 e^{tw+\theta_0w^2t^2/2+\kappa_0w^3t^3/3}\,dt,\qquad g_0=e^{w+\theta_0w^2/2+\kappa_0w^3/3},$$

$$E_J=W\int_0^1 e^{A_0t+B_0t^2+C_0t^3}\left(e^{h_2W^2t^2/2+h_3W^3t^3/3+B(t)}-1\right)dt,$$

$$E_g=|g_0|\left(e^{h_2W^2/2+h_3W^3/3+B(1)}-1\right),\quad (A_0,B_0,C_0)=(\operatorname{Re}w,\operatorname{Re}(\theta_0w^2/2),\operatorname{Re}(\kappa_0w^3/3)).$$

Write $\phi_m=(\phi_l+\phi_h)/2$, $\eta_\phi=2\sin((\phi_h-\phi_l)/4)$, $r=|J_0|$, $g=|g_0|$, *and*

$$P_-=\operatorname{Re}(we^{-i\phi_m})-\eta_\phi W,\quad C_+=\operatorname{Re}(e^{i\phi_m}g_0\overline{J_0})+\eta_\phi gr.$$

If

$$P_->0,\qquad r>E_J,\qquad \frac{C_++gE_J}{r^2-E_J^2}+\frac{E_g}{r-E_J}-\frac{P_-}{W^2}<0, \tag{4.15}$$

no actual configuration belongs to that parameter cell. For a quadratic model, set $\kappa_0=h_3=0$ *and begin the remainder sum at* $k=3$*; alternatively use* $B(t)=q_h\mu_hW^3t^3/[3(1-\mu_hW)]$.

Proof. For $\operatorname{Re}(w/\sigma)>0$, the original-root factorization and rightmostness imply

$$\operatorname{Re}\{\sigma(J'(w)/J(w)-1/w)\}\ge 0. \tag{4.16}$$

Indeed, after multiplication by H, the expression is $p'(z)/p(z)-1/(z-b)=\sum_{u\ne b}(z-u)^{-1}$ at $z=b+w/S$. Each summand has positive real part. The critical-point product gives $J'(w)=\exp(w+\theta w^2/2+\kappa w^3/3+R_4(w))$. Lemma 2.4, with each order through eight kept separately, gives $|R_4(tw)|\le B(t)$, including every order at least nine. Integrating the exponential perturbation proves $|J-J_0|\le E_J$, $|J'-g_0|\le E_g$.

If $|Z-v|\le E<|v|$, inversion maps that disk exactly to

$$\left|\frac{1}{Z}-\frac{\overline{v}}{|v|^2-E^2}\right|\le\frac{E}{|v|^2-E^2}.$$

This follows by multiplying the inequality by $|Z|(|v|^2-E^2)$ and expanding squares. Apply it to J, use the derivative error and $|e^{i\phi}-e^{i\phi_m}|\le\eta_\phi$, and then subtract $\operatorname{Re}(\sigma/w)\ge P_-/W^2$. The resulting upper bound for (4.16) is exactly the last expression in (4.15). The denominators are positive because $r>E_J$. □

We specify the coefficient enclosures used in this criterion. For a cell in (q,ϕ,α,β) take exact midpoints $q_m,\phi_m,\alpha_m,\beta_m$, set $C_m=1-\alpha_m+i\beta_m$, $d_q=(q_h-q_l)/2$, and $d_C=\sqrt{((\alpha_h-\alpha_l)/2)^2+((\beta_h-\beta_l)/2)^2}$. Then a valid quadratic radius is

$$h_2=d_q(|C_m|+d_C)+q_md_C+2q_m|C_m|\sin((\phi_h-\phi_l)/2)+|q_mC_me^{-2i\phi_m}-\theta_0|.$$

Indeed,

$$\begin{aligned}&q(1-\alpha+i\beta)e^{-2i\phi}-q_mC_me^{-2i\phi_m}\\&\quad=(q-q_m)(1-\alpha+i\beta)e^{-2i\phi}+q_m(\alpha_m-\alpha+i(\beta-\beta_m))e^{-2i\phi}\\&\qquad+q_mC_m(e^{-2i\phi}-e^{-2i\phi_m}).\end{aligned}$$

Apply the triangle inequality and $|e^{-2i\phi}-e^{-2i\phi_m}|\le 2\sin((\phi_h-\phi_l)/2)$. For the cubic radius put $c_l=\cos\phi_h$, $c_h=\cos\phi_l$, $A_l=q_l\alpha_l/2$, $A_h=q_h\alpha_h/2$, and let C_l,C_h be the extrema of $-q\beta/2$ over its four endpoint pairs. Division is used only when $c_l>0$. Put $C_*^2=\operatorname{dist}(0,[C_l,C_h])^2$ and

$$R_l=4A_l^2/c_h-3\mu_h^2c_h,\qquad R_h=\min(\mu_hA_h,\mu_h^2c_h-4C_*^2/c_h).$$

Let I_l^0, I_h^0 be the minimum and maximum of $3AC/c$ over the eight endpoint triples. Set

$$V_* = \max(0, \mu_h^2 c_h - (A_l^2 + C_*^2)/c_h), \qquad R_I = \tfrac{3}{2} V_* + \mu_h(q_h - A_l).$$

Equations (4.13)–(4.14) enclose P_3 in $[R_l, R_h] + i[I_l^0 - R_I, I_h^0 + R_I]$. Subdivide its entire imaginary interval into m equal closed strips. For a strip with midpoint P_m and midpoint-to-corner radius R_m, use

$$h_3 = R_m + 2|P_m| \sin(3(\phi_h - \phi_l)/4) + |P_m e^{-3i\phi_m} - \kappa_0|.$$

Every quantity is enclosed in the indicated outward direction. Every strip is retained, with its endpoints. The values $m = 4, 8$ suffice for the two cubic certificates below.

There are three further exclusions for a closed parameter cell. Let y_* be the threshold 3 or 9/4 in the contrary assumption $q/\mu \geq y_*$, and let μ_0 be the entrance cap, so that $\mu \leq \min(\mu_0, q/y_*)$. Assume $q_l > 0$, $0 \leq \phi_l \leq \phi_h \leq \pi/2$ and $0 \leq \alpha_l \leq \alpha_h < 1$, and put $b_*^2 = \operatorname{dist}(0, [\beta_l, \beta_h])^2$. On the branch $q \leq y_* \mu_0$, use $R_{\mathrm{mom}} = 2\cos\phi_l / y_*$ and $\mu_h = q_h / y_*$; on the branch $q \geq y_* \mu_0$, use $R_{\mathrm{mom}} = 2\mu_0 \cos\phi_l / q_l$ and $\mu_h = \mu_0$. The cell is excluded if any of

$$\alpha_l^2 + b_*^2 > R_{\mathrm{mom}}^2, \qquad (1 - \alpha_h)^2 + b_*^2 > 1, \qquad q_l > \mathcal{E}(\mu_h, \phi_l) \tag{4.17}$$

holds, where

$$\mathcal{E}(\mu, \phi) = K_0 \mu^2 + A_0 \mu \cos\phi + \gamma \cos^2\phi - \tfrac{1}{2} \sin^2\phi.$$

Here K_0, A_0, γ are the constants in Proposition 4.1. The first two tests use lower bounds for the squared distances in the two moment disks. For the third, (4.1) gives $q \leq \mathcal{E}(\mu, \phi)$, while

$$\partial_\mu \mathcal{E} = 2K_0 \mu + A_0 \cos\phi \geq 0, \qquad \partial_\phi \mathcal{E} = -\sin\phi \{A_0 \mu + (2\gamma + 1)\cos\phi\} \leq 0.$$

All three tests therefore exclude the whole cell; their strict inequalities are checked with outward enclosures.

Proposition 4.9. *If $D = Hd \geq 30$, then $Qd/H < 3$.*

Proof. Proposition 4.6 treats $D \geq 270$. Suppose $30 \leq D \leq 270$ and $q/\mu \geq 3$. Proposition 4.5, in its consequence for $D \geq 7/2$, and Lemma 4.7 put the contrary domain in

$$q \in [1/90, 403/900], \quad \mu \leq \min(1/30, q/3), \quad 0 \leq \phi \leq \pi/2, \quad 0 \leq \alpha \leq 2/3, \quad |\beta| \leq 2/3,$$

with the two moment-disk inequalities and the exact matrix envelope. Divide the q interval at $1/10$. The fixed dyadic trees in `fields/tang_cubic_family_D30_y3.json` have 9613 and 16784 closed leaves. A leaf is excluded by (4.17) or Lemma 4.8. The certificate has $3237,239,390$ leaves of the first three types, 3423 quadratic-model leaves, 8 single cubic-model leaves, and 19100 complete cubic-strip families. The latter have 79392 members. Thus there are 82823 point obstructions in all.

The verifier reconstructs each leaf from the exact dyadic path; both halves of every subdivision are closed. Bottom-up sibling coalescence reduces each tree to its root, rejecting duplicate leaves, missing siblings, ancestor conflicts, and inconsistent splitting coordinates. All inequalities and all the point predicates are checked on the entire leaves by `fields/tang_verify_cubic_family_cover_D30_frozen.py`. The source file records the exact test points and coefficient centers. Stored finite decimals defining those centers are interpreted as their exact binary rational values; their mismatch from the mathematical centers is included in the radii above. The integrands are entire, with integration remainders bounded as in Section 12. The certificate excludes the stated closed domain. Together with $D \geq 270$ this proves the proposition. □

Proposition 4.10. *If $D = Hd \geq 25$, then $Qd/H < 3$.*

Proof. Only $25 \leq D \leq 30$ remains after Proposition 4.9. Under the contrary ratio $q/\mu \geq 3$, the affine estimate gives

$$1/10 \leq q \leq 59/125, \quad \mu \leq \min(1/25, q/3), \quad 0 \leq \alpha \leq 2/3, \quad |\beta| \leq 2/3.$$

In this range $\phi < \pi/4$. Indeed if $\phi \ge \pi/4$, (4.2), $\cos\phi < 71/100$, and $q \ge 3\mu$ give

$$0 < \frac{21}{5}\mu^2 + \frac{181}{200}\mu - \frac{1}{8} \le -\frac{513}{6250} < 0.$$

Use the closed ϕ interval $[0, \pi/4]$ and divide the q interval at $3/25$. The two dyadic trees in `fields/tang_D25_cubic_family_cover.json` have 15303 and 35311 leaves. The exclusions consist of 4752 moment-disk, 393 quadratic-disk, and 219 matrix leaves, using (4.17). There are 2246 quadratic-model leaves, 15 single cubic-model leaves, and 42989 complete strip families; the families contain 184216 members. Every one of the 186477 point obstructions satisfies (4.15). The checker `fields/tang_verify_D25_cubic_family_frozen.py` uses the preceding finite-product and covariance formulas and reconstructs the two closed trees. Thus it excludes this domain. Its common boundary with $D \ge 30$ is included. □

Proposition 4.11. *If $D = Hd \ge 100$, then $Qd/H < 9/4$.*

Proof. By Proposition 4.6, it remains to consider $100 \le D \le 270$ and $q/\mu \ge 9/4$. The contrary domain is contained in

$$q \in [1/120, 1313/10000], \quad \mu \le \min(1/100, 4q/9), \quad 0 \le \phi \le \pi/2, \quad 0 \le \alpha \le 8/9, \quad |\beta| \le 8/9,$$

with the same two moment disks and matrix envelope. Split the q interval at $9/400$. The two trees in `fields/point_D100_y225.json` have 5200 and 4545 leaves. There are 1507 moment-disk exclusions, 57 quadratic-disk exclusions, and 21 matrix exclusions from (4.17). On each of the remaining 8160 leaves the quadratic version of Lemma 4.8 applies, with $B(t) = q_h\mu_h W^3 t^3/[3(1-\mu_h W)]$. The coefficient radius is the full-cell h_2 displayed above. The checker `fields/tang_independent_D100.py` reconstructs both trees, all coefficient radii, and every strict point obstruction. It evaluates the equivalent entire integral $J_0/g_0 = w\int_0^1 e^{(t-1)w+(t^2-1)\theta_0 w^2/2}\,dt$; this follows by dividing the defining integral by g_0. Thus every point of the contrary domain is excluded, which proves the proposition. □

5. Companion zeros and local fields

Throughout this section b is a simple actual zero, $S = p''(b)/p'(b) \ne 0$, and $H, Q, d, D, q, \mu, \sigma$ have the meanings in (2.2) and (2.7). Put

$$\alpha_j = \frac{1}{S(b-\xi_j)}, \qquad J(w) = \frac{Sp(b+w/S)}{p'(b)}, \qquad K = p''/p'.$$

Thus $\sum \alpha_j = 1$, $\sum |\alpha_j|^2 = q$ and $\max|\alpha_j| = \mu$. In particular $q > 0$. The exact identities are

$$J'(w) = \prod_j (1+\alpha_j w), \qquad J(w) = \int_0^w \prod_j (1+\alpha_j v)\,dv. \tag{5.1}$$

When a phase or curvature conclusion is asserted, b is assumed rightmost among all original zeros. The constructed companions are zeros of p.

5.1. **Logarithmic bounds and circles.** For $m \ge 2$ and $\mu|w| < 1$, the logarithm vanishing at zero satisfies

$$\log J'(w) = \sum_{k=1}^{m-1} \frac{(-1)^{k+1}w^k}{k} \sum_j \alpha_j^k + E_m(w), \qquad |E_m(w)| \le \frac{q\mu^{m-2}|w|^m}{m(1-\mu|w|)}. \tag{5.2}$$

Indeed $\sum_j |\alpha_j|^k \le q\mu^{k-2}$ for $k \ge 2$, so the entire remaining logarithmic series is bounded by a geometric series. Also, throughout $|w| \le Z < 1/\mu$,

$$\left|\frac{J''}{J'}(w)\right| \le 1 + qZ + \frac{q\mu Z^2}{1-\mu Z}. \tag{5.3}$$

This follows from $J''/J' = 1 - w\sum \alpha_j^2 + w^2 \sum \alpha_j^3/(1+\alpha_j w)$.

Lemma 5.1. *Suppose J' is nonzero on $|w| \le Z_{\text{out}}$, $|J''/J'| \le L$ there, and $|\zeta| + R < Z_{\text{out}}$. If*

$$\frac{|J(\zeta)|}{|J'(\zeta)|} + \frac{e^{LR} - 1 - LR}{L} < R,$$

then $|w - \zeta| < R$ contains exactly one zero of J, which is simple.

Proof. Since $J''(0)/J'(0) = \sum_j \alpha_j = 1$, the hypothesis gives $L \ge 1$. On every segment $\zeta + tv$, $|v| \le R$, integration of the logarithmic derivative gives $|J'(\zeta+tv)/J'(\zeta) - 1| \le e^{Lt|v|} - 1$. A further integration gives

$$|J(\zeta + v) - J(\zeta) - vJ'(\zeta)| \le |J'(\zeta)| \frac{e^{L|v|} - 1 - L|v|}{L}.$$

On $|v| = R$, Rouché's theorem compares $J(\zeta + v)$ with $vJ'(\zeta)$. The critical-free containing disk makes its unique zero simple. □

5.2. **Gaussian quartic centers.** Put $\theta = -\sum \alpha_j^2 = K'(b)/S^2$, so $|\theta| \le q$. For $s = \pm 2\pi i$ define

$$\begin{aligned} \zeta_s = s &+ (s - s^2/2)\theta + (3s - 2s^2 + s^3/2)\theta^2 \\ &+ (15s - 21s^2/2 + 23s^3/6 - 5s^4/8)\theta^3 \\ &+ (105s - 72s^2 + 88s^3/3 - 22s^4/3 + 7s^5/8)\theta^4. \end{aligned} \tag{5.4}$$

If $\zeta_s = \sum_{j=0}^4 c_j(s)\theta^j$, write

$$t\zeta_s + \theta t^2 \zeta_s^2/2 = st + \sum_{j=1}^{9} A_j(s,t)\theta^j.$$

Thus $A_j = tc_j + (t^2/2)\sum_{k+l=j-1} c_k c_l$, where a coefficient outside $0 \le j \le 4$ is zero. Define

$$E_0 = 1, \qquad E_j = \frac{1}{j} \sum_{k=1}^{\min(j,9)} kA_k E_{j-k}, \qquad I_0 = 0, \quad I_m = \frac{1 - mI_{m-1}}{s} \quad (m \ge 1).$$

For a polynomial $f(t) = \sum f_m t^m$ set $\mathcal{I}_s f = \sum f_m I_m$. Since $e^s = 1$, $\mathcal{I}_s f = \int_0^1 e^{st} f(t)\,dt$ at the two indicated values of s. The coefficients of the Gaussian primitive at the center are

$$a_j(s) = \mathcal{I}_s \left(\sum_{k=0}^{\min(4,j)} c_k E_{j-k} \right).$$

The recurrence gives $a_0 = \cdots = a_4 = 0$ exactly. It can alternatively be used to determine $c_1, \ldots, c_4$ successively: the new center coefficient enters the corresponding residual with coefficient one.

The following strict bounds hold for $6 \le p \le 44/7$, $0 \le t \le 1$, with $s = \pm ip$:

polynomials	bounds in increasing order
$c_1, \ldots, c_4$	$21, 133, 1030, 9000$
$A_1, \ldots, A_9$	$63/10, 45, 400, 3400, 86000, 320000,$ $1700000, 9100000, 39000000$
$a_5, \ldots, a_{11}$	$82000, 350000, 1900000, 13000000,$ $120000000, 3100000000, 23000000000$

Here and below a bound for a complex polynomial means a bound for its modulus. To verify these inequalities, substitute $s = ip$ in the recurrences and subtract the squared modulus from the square of the proposed bound. Convert the resulting rational polynomial to tensor Bernstein form on $[6, 44/7] \times [0, 1]$, omitting the second variable when absent. If every control coefficient is positive, accept the box; otherwise bisect it and use the exact de Casteljau averages. At depth d

the path-variable axis is chosen unless $d \equiv 2 \pmod 3$, when the p-axis is chosen. For one-variable polynomials bisect the p-axis. All twenty inequalities terminate with positive coefficients on a total of twenty-four closed boxes, at depth at most twenty. The recurrences and bounds specify the certificate; `check_gaussian_domains_new.py` implements it with rational arithmetic. The coefficient identities $a_0 = \cdots = a_4 = 0$ are checked by exact polynomial arithmetic. They apply to the primitive at $s = \pm 2\pi i$, where $e^s = 1$. The resulting residual polynomials are bounded on $6 \le p \le 44/7$.

Set

$$\begin{aligned} Z(q) &= 63/10 + 21q + 133q^2 + 1030q^3 + 9000q^4, \\ M(q) &= (63/10)q + 45q^2 + 400q^3 + 3400q^4 + 86000q^5 \\ &\quad + 320000q^6 + 1700000q^7 + 9100000q^8 + 39000000q^9. \end{aligned}$$

Let P_{11} be the degree-eleven Taylor polynomial of $Z(q)e^{M(q)}$, and let $B_5, \dots, B_{11}$ denote the seven residual bounds in the table. For $J_\theta(w) = \int_0^w e^{z+\theta z^2/2}\,dz$ we have

$$|J_\theta(\zeta_s)| < \sum_{j=5}^{11} B_j q^j + Z(q)e^{M(q)} - P_{11}(q). \tag{5.5}$$

Every coefficient in the last difference is nonnegative. Expand the exponential along the path, bound the retained coefficients by the table and every higher coefficient by Ze^M. For exact numerical bounds we use

$$\mathcal{E}_{24}(x) = \sum_{k=0}^{24} \frac{x^k}{k!} + \frac{x^{25}}{25!(1 - x/26)} > e^x \qquad (0 < x < 26). \tag{5.6}$$

The quotient of the right side of (5.5) by q^5 increases with q. Substituting (5.6) proves the respective coefficients $132200, 126000, 86200, 89000$ at $q = 1/20, 97/2000, 7/625, 9/556$.

Proposition 5.2. *Each row of the following table gives two distinct simple actual zeros, other than b, in the disks $|S(z-b) - \zeta_\pm| < R$. The disks lie in the indicated critical-free containing disk.*

row	q_0	μ_0	$R(q,\mu)$	Z_{out}
coarse	$97/2000$	$1/270$	$300000q^5 + 1300q\mu$	$41/5$
fine	$7/625$	$1/200$	$120000q^5 + 220q\mu$	$659/100$
wide	$9/400$	$1/100$	$250000q^5 + 400q\mu$	$174/25$
origin	$33/1000$	$8/625$	$3/10$	$15/2$
medium	$9/556$	$1/139$	$120000q^5 + 225q\mu$	$168/25$

Every row holds on $0 < q \le q_0$, $0 < \mu \le \mu_0$. At a rightmost zero the fine and medium rows give $|\operatorname{Im} S| < 4Q/H$, and the wide row gives $|\operatorname{Im} S| < (23/5)Q/H$.

Proof. For the first four rows use $J' = e^{w+\theta w^2/2+E_3(w)}$ and (5.2). The following constants are exact:

row	Z_0	M_0	E_0	B	P_0	L	β
coarse	$39/5$	$101/200$	$43/25$	126000	550	$143/100$	$27/100$
fine	$657/100$	$2/25$	$109/100$	86200	175	$11/10$	$1/100$
wide	$343/50$	$43/250$	$5/4$	132200	250	$59/50$	$3/50$
origin	$719/100$	$7/25$	$71/50$	132200	350	$32/25$	$11/50$

At each upper corner, set $r_3 = q_0\mu_0 Z_0^3/[3(1-\mu_0 Z_0)]$. The finite predicates are

$$Z(q_0) < Z_0, \quad M(q_0) < M_0, \quad \mathcal{E}_{24}(M_0 + r_3) < E_0, \quad \frac{Z_0^4 E_0}{12(1-\mu_0 Z_0)} < P_0, \tag{5.7}$$

$$Z_0 + R_0 < Z_{\mathrm{out}} < 1/\mu_0, \quad 1 + q_0 Z_{\mathrm{out}} + \frac{q_0\mu_0 Z_{\mathrm{out}}^2}{1-\mu_0 Z_{\mathrm{out}}} < L,$$

$$\frac{\mathcal{E}_{24}(LR_0) - 1 - LR_0}{LR_0} < \beta, \tag{5.8}$$

where $R_0 = R(q_0, \mu_0)$. The center residual obeys

$$|J(\zeta_s)/J'(\zeta_s)| < E_0 B q^5 + E_0 P_0 q\mu.$$

Indeed the primitive discrepancy is bounded by integrating $q\mu Z_0^3 t^3/[3(1-\mu_0 Z_0)]$ for $0 \le t \le 1$; its integral contains the factor $1/12$, and $|J'(\zeta_s)| > 1/E_0$. For the first three rows the two coefficient inequalities $E_0 B < (1-\beta)[q^5]R$ and $E_0 P_0 < (1-\beta)[q\mu]R$ prove the circle criterion; the brackets denote coefficients of the indicated monomials in R. In the origin row their sum at the corner is less than $0.217280 < (1-\beta)R_0$. Every majorant is increasing, so these predicates hold on the closed rectangle. Source exclusion and separation follow respectively from

$$6 - 21q_0 - 133q_0^2 - 1030q_0^3 - 9000q_0^4 > R_0,$$

$$12(1 - q_0 - 17q_0^2 - 137q_0^3 - 313q_0^4) > 2R_0.$$

All displayed predicates are rational after (5.6) is substituted, and direct evaluation gives positive differences.

For the medium row estimate the discrepancy relative to the endpoint derivative. Write $g(t) = t\zeta_s + \theta t^2 \zeta_s^2/2$. The coefficient of θ in $g(t) - g(1) - (t-1)s$ is $(t-1)s + (t^2-t)s^2/2$ and has modulus less than $63/10$. To see this, put $u = 1-t$ and use $p^2 = (2\pi)^2 < 40$ and

$$1 - u^2[1 + 10(1-u)^2] = (1-u)(10u^3 - 10u^2 + u + 1).$$

The cubic on the right has Bernstein coefficients $(1, 7/6, 1/2, 1/4)$ on $[0, 1/2]$, $(1/4, 1/8, 5/48, 11/32)$ on $[1/2, 3/4]$, and $(11/32, 7/12, 13/12, 2)$ on $[3/4, 1]$. For the remaining coefficients use $(t-1)c_k + (t^2-1)\sum_{i+j=k-1} c_i c_j/2$. This gives the relative-path majorant

$$M_r(q) = (63/10)q + (2653/10)q^2 + (10442/5)q^3 + 18282q^4 + (174349/2)q^5 + 325990q^6 + 1727450q^7 + 9270000q^8 + 40500000q^9.$$

At $q_0 = 9/556$, $\mu_0 = 1/139$, take

$$Z_0 = 6681/1000, \quad M_0 = 29/250, \quad M_{r,0} = 183/1000,$$

$$E_0 = 57/50, \quad L = 28/25, \quad \beta = 3/200.$$

The bounds $Z(q_0) < Z_0$, $M(q_0) < M_0$, $M_r(q_0) < M_{r,0}$ and $e^{M_0 + r_3} < E_0$ follow from the same rational predicates. Dividing the integral discrepancy by the actual $J'(\zeta_s)$ gives

$$\left| \frac{J(\zeta_s) - J_\theta(\zeta_s)}{J'(\zeta_s)} \right| < \frac{q\mu Z_0^4 e^{M_{r,0} + 2r_3}}{12(1-\mu_0 Z_0)} < 215 q\mu. \tag{5.9}$$

The path remainder and endpoint denominator each contribute a factor e^{r_3}. The model contribution is less than $101460q^5$. Now $101460 < (1-\beta)120000$ and $215 < (1-\beta)225$. The other circle, separation and containment predicates in (5.8) hold with $Z_{\mathrm{out}} = 168/25$. These estimates prove the medium row.

At a rightmost b, both companions satisfy $\mathrm{Re}(w/\sigma) \le 0$. Choose the sign of s for which $\mathrm{Re}(s/\sigma) = 2\pi|\operatorname{Im}\sigma|$. The triangle inequality gives

$$2\pi|\operatorname{Im}\sigma| < q\left(21 + 133q + 1030q^2 + 9000q^3 + R/q\right).$$

At the fine corner the bracket is less than 24; at the wide corner it is less than $28.681 < (157/25)(23/5)$; at the medium corner it is less than $4(157/25)$. Since $2\pi > 157/25 > 6$, the phase bounds follow. □

5.3. **Signed curvature from the two companions.** Write $p_0 = 2\pi$ for this subsection only, and define

$$B(\theta) = 1 + \theta + (3 - p_0^2/2)\theta^2 + (15 - 23p_0^2/6)\theta^3 + (105 - 88p_0^2/3 + 7p_0^4/8)\theta^4,$$
$$T(\theta) = \theta + 4\theta^2 + (21 - 5p_0^2/4)\theta^3 + (144 - 44p_0^2/3)\theta^4.$$

Then $\zeta_\pm = (p_0^2/2)T \pm ip_0 B$. If the actual roots lie in disks of radius R about these centers, rightmostness gives

$$\frac{p_0^2}{2}\operatorname{Re}(T/\sigma) + p_0|\operatorname{Im}(B/\sigma)| < R. \tag{5.10}$$

For other centers, add to R an upper bound for their displacements from (5.4).

Lemma 5.3. *For $r = |\theta| \le 1/20$, put*

$$v = r + 17r^2 + 137r^3 + 313r^4, \quad m = 1 - v.$$

$$A_r = \frac{1 + 17r + 137r^2 + 313r^3}{m}, \quad B_r = \frac{18 + 154r + 450r^2 + 313r^3}{m}.$$

$$\mathcal{C}(r) = 1 + (2 + 100/199)rA_r^2 + 3rB_r + (600/199)r^2A_r^2 + 4rA_r + 29r + 436r^2.$$

The two companion disks imply

$$\frac{\operatorname{Re} K'(b)}{H^2} < \mathcal{C}(q)q^2 + \frac{2R}{p_0^2} \qquad (q \le 1/20). \tag{5.11}$$

It also implies the sharper bound with $\mathcal{C}(q)$ replaced by $6/5$. For $q \le 3/50$ the replacement $13/10$ is valid provided an independent circle estimate gives $|\operatorname{Im}\sigma| < 8q$.

Proof. Set $\tau = B/|B| = c + is$ and $\delta = \operatorname{Im}(B/\sigma)$. The coefficient bounds give $|B| \ge m$, $c > 99/100$, $1 - c \le (100/199)s^2$, $|s| \le rA_r$, and $|s - \operatorname{Im}\theta| \le r^2B_r$. For $f(\sigma) = \operatorname{Re}(\theta\sigma^2 - T/\sigma)$,

$$|f(\sigma) - f(\tau)| \le r(3 + 4r + 29r^2 + 436r^3)|\sigma - \tau|.$$

The exact chord identity is

$$\frac{|\sigma - \tau|^2}{\delta^2} = \frac{2}{|B|^2(1 + \operatorname{Re}(\sigma\overline{\tau}))}.$$

On the right semicircle, $\operatorname{Re}(\sigma\overline{\tau}) \ge -|s|$. The endpoint comparison at $r = 1/20$ gives $|\sigma - \tau| \le (9/5)|\delta|$ and

$$(9/5)r(3 + 4r + 29r^2 + 436r^3) < 7/22 < 2/p_0.$$

The case $\delta = 0$ follows by continuity: the antipodal point $-\tau$ has negative real part.

Writing $\theta = x + iy$, expansion of $f(\tau)$ gives

$$y^2 - 4x^2 + x(1 - c) - 2xs^2 - y[(s - y)(2c + 1) + 2y(c - 1)]$$
$$+ 4(x^2 - y^2)(1 - c) - 8xys - (21 - 5p_0^2/4)\operatorname{Re}(\theta^3\overline{\tau}) - (144 - 44p_0^2/3)\operatorname{Re}(\theta^4\overline{\tau}).$$

Use the bounds above, $|xy| \le r^2/2$, and the coefficient bounds $|21 - 5p_0^2/4| < 29$, $|144 - 44p_0^2/3| < 436$. The result is $f(\tau) \le \mathcal{C}(r)r^2$. Every term in $\mathcal{C}$ increases, and $\mathcal{C}(97/2000) < 9$, $\mathcal{C}(7/625) < 11/5$. Combining this estimate with (5.10) cancels δ and proves (5.11).

For the two sharper constants there is a finite polynomial proof. Write $\theta = re^{i\phi}$, $u = \cos\phi$, and let $T_j(u)$ be the Chebyshev polynomials. If $B = \sum b_j\theta^j$ and $T = \sum d_j\theta^j$, define

$$\begin{aligned} M &= \sum_{i,j} b_i b_j r^{i+j} T_{|i-j|}(u), \\ U &= \sum_{i,j} b_i b_j r^{i+j+1} T_{i+j+1}(u), \qquad V = \sum_{i,j} d_i b_j r^{i+j} T_{|i-j|}(u), \\ P &= (U-V)/r^2, \qquad W = V(M-1)/r^2. \end{aligned}$$

The last two expressions are polynomials, including at $r = 0$. For $(r_0, k) = (1/20, 6/5)$ or $(3/50, 13/10)$, let $m_0 = 1 - r_0 - 17r_0^2 - 137r_0^3 - 313r_0^4$. The polynomials

$$A = kM - P, \qquad (1+m_0)A + W \tag{5.12}$$

are positive on $0 \le r \le r_0$, $-1 \le u \le 1$ and

$$3947841760435743/10^{14} \le p_0^2 \le 3947841760435744/10^{14}.$$

Indeed, tensor Bernstein conversion on the three closed u intervals $[-1,0]$, $[0,1/2]$, $[1/2,1]$ gives 4950 positive rational coefficients for each r_0. The conversion is exactly $x^k = \sum_{j=k}^{n} \binom{j}{k}\binom{n}{k}^{-1}\binom{n}{j} x^j (1-x)^{n-j}$ after affine substitution in each variable. These polynomials and the three boxes specify the certificate. Machin's formula and alternating arctangent sums of lengths twenty and twenty-one give the stated enclosure of p_0^2. Now

$$f(\tau)/r^2 = \frac{1}{M}\left(P - \frac{W}{1+\sqrt{M}}\right).$$

If $W \ge 0$ use $A > 0$; if $W < 0$ use $1 + \sqrt{M} \ge 1 + m_0$ and the second inequality in (5.12). Thus $f(\tau) \le kr^2$ on the whole disk, with $r = 0$ supplied by continuity.

At the larger cap $r \le q \le 3/50$, the phase bound gives $\operatorname{Re}\sigma > 7/8$. The coefficient bounds give $|B| > 21/25$, $|\operatorname{Im}\tau| < 1/5$, $\operatorname{Re}\tau > 97/100$, hence $\operatorname{Re}(\sigma\overline{\tau}) > 3011/4000$. The rational comparison $(21/25)^2(1 + 3011/4000)(4/3)^2 > 2$ gives $|\sigma - \tau| < (4/3)|\delta|$. At $r = 3/50$,

$$(4/3)r(3 + 4r + 29r^2 + 436r^3) = 214911/781250 < 7/22 < 2/p_0.$$

The same cancellation proves the second assertion on this larger domain. □

Two consequences needed by the field argument in Section 4 are

$$\operatorname{Re} K'(b)/H^2 < \mathcal{C}(q)q^2 + (50000/3)q^5 + (650/9)q\mu \qquad \text{on the coarse row,} \tag{5.13}$$

$$\operatorname{Re} K'(b)/H^2 < (9/4)q^2 + 13q\mu \qquad \text{on the fine row.} \tag{5.14}$$

These follow from the companion disks, since $p_0 > 6$ and $11/5 + (120000/18)(7/625)^3 < 9/4$, $220/18 < 13$. On the medium row the sharper signed constant instead gives

$$\operatorname{Re} K'(b)/H^2 < (5/4)q^2 + (23/2)q\mu, \tag{5.15}$$

because $2/p_0^2 < 1250/24649$, $6/5 + (1250/24649)120000(9/556)^3 < 5/4$, and $(1250/24649)225 < 23/2$.

5.4. **Centers retaining four moments.** In addition to θ put $\kappa = \sum \alpha_j^3$ and $\lambda = -\sum \alpha_j^4$. Their bounds are $|\kappa| \le q\mu$, $|\lambda| \le q\mu^2$. Assign weights $1, 2, 3$ to θ, κ, λ. For $s = \pm 2\pi i$, the weight-four center is

$$Z_4 = \zeta_s + u\kappa + v\theta\kappa + w\theta^2\kappa + r\kappa^2 + l\lambda + m\theta\lambda, \tag{5.16}$$

where the four pure θ terms are those in (5.4), and

$$u = -s^3/3 + s^2 - 2s, \qquad v = 5s^4/6 - 13s^3/3 + 12s^2 - 20s,$$
$$w = -7s^5/4 + 157s^4/12 - 51s^3 + 131s^2 - 210s, \qquad r = s^5/3 - 13s^4/6 + 26s^3/3 - 22s^2 + 40s,$$
$$l = -s^4/4 + s^3 - 3s^2 + 6s, \qquad m = 3s^5/4 - 19s^4/4 + 18s^3 - 51s^2 + 90s.$$

For higher weights define

$$J_4(z) = \int_0^z e^{w+\theta w^2/2+\kappa w^3/3+\lambda w^4/4}\, dw.$$

Let $T_j(z) = \sum_{k=0}^{j}(-1)^{j-k} j! z^k/k!$. Its formal expansion is

$$J_4(z) = \sum_{a,b,c\ge 0} \frac{\theta^a \kappa^b \lambda^c}{2^a 3^b 4^c a! b! c!} \{e^z T_{2a+3b+4c}(z) - T_{2a+3b+4c}(0)\}.$$

Starting from $Z_0 = s$, define recursively

$$Z_j = Z_{j-1} - [\text{weight } j] J_4(Z_{j-1}), \qquad 1 \le j \le 8, \qquad e^s = 1. \tag{5.17}$$

Here [weight j] selects the homogeneous component of weight j. The new correction has derivative coefficient one, so this cancels all coefficients up to the indicated weight. The first four levels give exactly (5.16). The numbers of new terms in levels $1, \dots, 8$ are $1, 2, 3, 4, 5, 7, 8, 10$. Thus (5.17) uniquely specifies all forty nonconstant terms of Z_8 as rational polynomials in s.

For $Z = Z_m(s)$, where $m = 4$ or 8 and $s = \pm 2\pi i$, put

$$A(t) = t(Z - s) + \theta t^2 Z^2/2 + \kappa t^3 Z^3/3 + \lambda t^4 Z^4/4,$$
$$A_r(t) = (t-1)(Z - s) + \theta(t^2 - 1)Z^2/2 + \kappa(t^3 - 1)Z^3/3 + \lambda(t^4 - 1)Z^4/4.$$

These exponents give

$$J_4(Z) = Z \int_0^1 e^{st + A(t)}\, dt, \qquad \frac{J_4(Z)}{J_4'(Z)} = Z \int_0^1 e^{s(t-1) + A_r(t)}\, dt.$$

For caps $q_0, \mu_0 > 0$ with $q \le q_0$, $\mu \le \mu_0$, use the scaled variables

$$(u_1, u_2, u_3) = \left(\frac{\theta}{q_0}, \frac{\kappa}{q_0 \mu_0}, \frac{\lambda}{q_0 \mu_0^2}\right), \qquad |u_1|, |u_2|, |u_3| \le 1.$$

For $\nu \in \mathbb{Z}^3_{\ge 0}$, write $\mathbf{u}^\nu = u_1^{\nu_1} u_2^{\nu_2} u_3^{\nu_3}$ and $\mathrm{wt}(\nu) = \nu_1 + 2\nu_2 + 3\nu_3$; $\gamma \le \nu$ means coordinatewise comparison. For either exponent A or A_r, let $A_\gamma(t)$ be its coefficient of $\mathbf{u}^\gamma$ after this substitution, and let $E_\nu(t)$ be the coefficient of $\mathbf{u}^\nu$ in its exponential. Each A_γ has degree at most four in t. The exponential coefficients satisfy

$$E_0 = 1, \qquad E_\nu = \frac{1}{\mathrm{wt}(\nu)} \sum_{0 < \gamma \le \nu} \mathrm{wt}(\gamma) A_\gamma E_{\nu - \gamma}, \qquad \nu \ne 0. \tag{5.18}$$

The sum is over nonzero multi-indices $\gamma \le \nu$. This identity is obtained by applying the weighted Euler derivative to the chosen exponential. Since $e^s = 1$, both integrals use the recurrence for I_m above. The integrated coefficients through weight m vanish exactly. For the absolute residual $J_4(Z)$ or the relative residual $J_4(Z)/J_4'(Z)$, let F_* be the sum of the moduli of its coefficients in $\mathbf{u}$ with $m < \mathrm{wt}(\nu) \le M$, where $M = 14$ for $m = 4$ and $M = 28$ for $m = 8$.

On every interval $[j/128, (j+1)/128]$ express each quartic coefficient in Bernstein form. Its values lie in the convex hull of its five complex control coefficients, so the largest control modulus bounds the entire interval. The same applies to its difference from the endpoint value. Alternatively the midpoint Taylor bound $\sum_{k=0}^{4} |f^{(k)}(m_t)| h^k/k!$, where $m_t = (2j+1)/256$ and $h = 1/256$ are the midpoint and half-length, bounds that interval. The four-moment rows use the former method and the weight-eight rows use the latter. There are 253 path polynomials for Z_4 and 1409 for Z_8.

For each exponent coefficient take the largest bound over the 128 intervals. Let a_j be the sum of these bounds in weight j, and z_j the sum of the moduli of the coefficients of Z in $\mathbf{u}$ in that weight, including $z_0 = |s|$. Put $A_*(v) = \sum a_j v^j$ and $Z_*(v) = \sum z_j v^j$. The infinite tail after weight M is bounded by

$$T_* = Z_*(1)e^{A_*(1)} - \sum_{j=0}^{M} [v^j]\{Z_*(v)e^{A_*(v)}\}. \tag{5.19}$$

The positive exponential coefficients are generated by $e_0 = 1$, $e_j = j^{-1}\sum_{k=1}^{j} k a_k e_{j-k}$. All coefficients in the discarded tail are nonnegative. Thus (5.19) bounds every omitted weight, including every mixed parameter term. Its exponential is bounded upward by (5.6), or by an outward complex ball evaluation with an explicit remainder bound.

To pass from the model to the actual finite product, put

$$r_5 = \frac{q_0\mu_0^3 Z_*(1)^5}{5(1-\mu_0 Z_*(1))}, \qquad |Z| \le Z_*(1) < 1/\mu_0.$$

An absolute model residual $F_* + T_*$ has actual derivative-relative bound $(F_* + T_*)e^{M_c + r_5}$, where M_c bounds the center exponent after s. A relative model residual instead has bound $(F_* + T_*)e^{r_5}$. In both cases the actual product discrepancy is bounded by

$$\frac{q\mu^3 Z_*(1)^6 e^{M_r + 2r_5}}{30(1-\mu_0 Z_*(1))}. \tag{5.20}$$

Here M_r bounds the relative exponent along the path, and $1/30$ includes the integral of t^5. The path remainder and endpoint denominator give the two r_5 terms.

There are two scaling rules. If $q = tq_0$, $0 < t \le 1$, and $\mu \le \mu_0$, a monomial $\theta^a\kappa^b\lambda^c$ is bounded by t^{a+b+c} times its corner majorant. The uncancelled Z_4 terms have total degree at least two; the tail above weight fourteen has total degree at least five. For Z_8 the corresponding degrees are at least three and ten. These statements follow from $\mathrm{wt}(a,b,c) \le 3(a+b+c)$ and the exact low-weight cancellation. They hold term by term in the multivariate majorant and hence for its infinite tail. When instead $q \le k/D$, $\mu = 1/D$ and D increases, the three parameters scale with weights $1, 2, 3$.

Proposition 5.4. *For each row and $0 < q \le q_0$, $0 < \mu \le \mu_0$, the disks $|S(z-b) - Z_m(s)| < R$, $s = \pm 2\pi i$, are disjoint from each other and from b, and contain exactly one simple original zero each, denoted $z_\pm$ respectively. Put $w_\pm = S(z_\pm - b)$. Both disks lie in the critical-free disk $|S(z-b)| < Z_{\mathrm{out}}$. The final column gives $|\operatorname{Im} S| < kQ/H$ at a rightmost zero.*

m	q_0	μ_0	$R(q,\mu)$	Z_{out}	k
4	1/20	1/50	$9q$	17/2	7
4	1/25	1/25	$(45/4)q$	17/2	15/2
4	3/70	1/70	$34q^2 + 13400q\mu^3$	17/2	21/4
8	3/50	1/50	$64q^2 + 95000q\mu^3$	19/2	8
8	273/5000	91/5000	$22q^2 + 45000q\mu^3$	9	33/5

The first row also gives $|w_\pm| < 17/2$ and $|(w_+ + w_-)/2| < 19/10$ for the same pair. On the subrectangle $q \le 57/1000$, $\mu \le 1/51$, the fourth row gives

$$|w_\pm| < 8917/1000, \qquad |(w_+ + w_-)/2| < 2109/1000, \qquad |\operatorname{Im} S| < (37/5)Q/H. \tag{5.21}$$

Proof. Use (5.17)–(5.20). For the first two rows the exact caps of the center modulus are respectively 8.030851273687 and 7.706829806638. Upper bounds at the corners are as follows:

(q_0, μ_0)	F_*	T_*	model/actual derivative	product error	circle error
$(.05, .02)$	.091297	.017192	.216053	.008556	.200791
$(.04, .04)$	.079034	.012556	.169285	.049027	.200791

The sums in the last three columns are strictly less than $9/20$. The containing-disk bound is $L = 31/20$ in both cases. The finite sum retains all weights five through fourteen; the infinite sum is exactly bounded by (5.19). The path, derivative and circle inequalities follow from the deterministic recurrences and all 128 closed path intervals specified above. At smaller q, both actual residuals are at most $t = q/q_0$ times their corner values. Since $(e^{LR} - 1 - LR)/L$ is convex and vanishes at zero, the circle error is at most t times its corner value. Thus the radii are $t(9/20)$ throughout the parameter rectangles.

For the third row the same construction at its own corner yields

$$Z_*(1) < 7.629836, \quad |Z_4 - s| < 1.346650,$$
$$M_c < .478584, \quad M_r < .481296, \quad F_* < .033415, \quad T_* < .000911.$$

Using the unrounded rational witnesses produced by the path conversion, the model contribution is less than $30.181q^2$ and the actual product contribution is less than $11961q\mu^3$. Both are below nine tenths of the corresponding radius coefficient. At the corner $R_0 < .064124$, $Z_*(1) + R_0 < 17/2$, (5.3) is less than $31/20$, and the nonlinear fraction is less than $.052 < 1/10$. The model terms and positive tail are bounded by $(q/q_0)^2$ times their corner values; (5.20) retains its explicit $q\mu^3$. Thus the third row holds throughout its parameter rectangle.

For the last two rows use the relative exponent A_r. The upper bounds, including all weights at least twenty-nine, are

row	F_*	T_*	model error	product error	L	nonlinear fraction
$q_0 = .06$	.136221	.012814	.150033	.028511	7/4	1/3
$q_0 = .0546$	.045190	.000674	.046017	.010525	17/10	1/5

In each row the model error divided by q_0^2 is less than $(1 - \beta)$ times the q^2 radius coefficient, and the product error divided by $q_0\mu_0^3$ is less than $(1 - \beta)$ times the other coefficient. The nonlinear fraction is less than β. Containment, source exclusion and separation hold at both corners and improve on the smaller rectangles. Lemma 5.1 therefore applies.

The coefficient polynomials are generated by (5.17); their exponential coefficients are given by (5.18); every path coefficient is bounded on the stated 128 intervals; and the infinite positive tail is (5.19). The electronic appendix contains the exact rational coefficient lists and programs performing these operations at 352 or 448 bits with outward bounds. A proposed rational modulus bound is accepted only if it exceeds its whole interval bound. The vanishing identities are checked separately over $\mathbb{Q}[s]$.

Every center correction has positive total parameter degree, and $R(q, \mu) \le (q/q_0)R_0$. If Γ bounds $|Z_m - s|$ at the corner, rightmostness gives $2\pi|\operatorname{Im}\sigma| < (q/q_0)(\Gamma + R_0)$. The rational or outward strict gates $\Gamma + R_0 < 2\pi q_0 k$ give exactly the last column of the table. Averaging the two centers cancels all odd powers of s. The sum of the absolute even coefficients at $(.05, .02)$, plus $9/20$, is less than $1.890068 < 19/10$. At $(.057, 1/51)$ the corresponding evaluations give respectively 8.916593, 2.108213 and 7.352979, proving (5.21) for the pair (z_+, z_-). □

5.5. **Curvature and the hole field.**

Lemma 5.5. *Assume the unit-disk strict-hole geometry of Lemma 2.2. Suppose $C_0 \ge 0$ and*

$$Qd/H < k, \qquad |\operatorname{Im} S| < K_0 Q/H, \qquad \operatorname{Re} K'(b) < C_0/d^2.$$

Then

$$H > \frac{N - (2C_0 + 2K_0 k)/d}{2(1 + a)}. \tag{5.22}$$

Proof. Put $p_b = 1 - |b|^2 \geq 0$, $\lambda_b = 1 - |b-a|^2 > 0$. For $v_j = (b - \xi_j)^{-1}$, the strict hole gives

$$\lambda_b |v_j|^2 + 2\,\mathrm{Re}((b-a)v_j) < 1.$$

This is an open disk whose greatest real coordinate is $m_R = (1 - x + a)/\lambda_b$. Rightmostness gives $\mathrm{Re}\, v_j \geq 0$, so

$$Q - 2\sum(\mathrm{Re}\, v_j)^2 = \mathrm{Re}\, K'(b) < C_0/d^2, \qquad Q < 2m_R H + C_0/d^2.$$

Since $\lambda_b \geq p_b$, one has $p_b m_R \leq 1 - x + a$. Also $p_b < 2d$, because $d > 1 - |b|$ in the strict crescent. Apply (2.4) and $0 \leq y \leq 1$ to obtain

$$N < 2(x + p_b m_R)H + 2K_0 Q/H + p_b C_0/d^2 \leq 2(1+a)H + (2K_0 k + 2C_0)/d.$$

This also covers $p_b = 0$ without division by it. □

Proposition 5.6. *If $D \geq 70$, then*

$$|\,\mathrm{Im}\, S| < (21/4)Q/H, \qquad Qd/H < Y_{70}(D) < 8/3,$$

$$Y_{70}(D) = \frac{4D}{D - 91/10 + \sqrt{(D - 91/10)^2 - 24D}}.$$

In the strict-hole unit-disk geometry,

$$H > \frac{N - (596/5)/d}{2(1+a)}. \tag{5.23}$$

If $D \geq 100$, the stronger consequence is

$$H > \frac{N - (1899/20)/d}{2(1+a)}. \tag{5.24}$$

Proof. First apply Proposition 4.10, giving $Qd/H < 3$. This enters the third row of Proposition 5.4. The six additional moment terms give $|Z_4 - \zeta_s| < 175q\mu$ on its parameter rectangle. This follows by taking the modulus of each displayed coefficient in (5.16) at $s = 2\pi i$; their total at the corner is less than $.107054 < 175(3/70)(1/70)$. The effective Gaussian radius is therefore

$$R_{\mathrm{eff}} = 34q^2 + (175 + 13400/4900)q\mu.$$

Lemma 5.3 with $6/5$ and the exact comparisons

$$6/5 + (1250/24649)34 < 3, \qquad (1250/24649)(175 + 13400/4900) < 91/10$$

give $\mathrm{Re}\, K'(b)/H^2 < 3q^2 + (91/10)q\mu$. On the other hand $\sum(\mathrm{Re}\, v_j)^2 \leq H/d$, so $\mathrm{Re}\, K'(b)/H^2 \geq q - 2\mu$. With $y_0 = Qd/H$ this yields

$$3y_0^2 - (D - 91/10)y_0 + 2D > 0.$$

At $D = 70$ the polynomial is positive at 2, negative at 3, decreasing on $[2,3]$, and equals $-16/15$ at $8/3$. These signs persist for all $D \geq 70$; the bound $y_0 < 3$ removes the upper quadratic branch. The smaller root is $Y_{70}(D)$. For $Y_{70} = 2 + v$ its inverse relation is $D = 151/(5v) + 211/10 + 3v$, decreasing on $0 < v \leq 2/3$. This proves the assertion for all $D \geq 70$. For the hole field use $k = 8/3$, $K_0 = 21/4$ and $C_0 = 3k^2 + (91/10)k = 228/5$ in Lemma 5.5. The resulting constant is $596/5$. For $D \geq 100$, Proposition 4.11 gives $k = 9/4$ instead. Then $C_0 = 2853/80$ and the constant is $1899/20$. □

Proposition 5.7. *If $D \geq 139$, then*

$$|\,\mathrm{Im}\, S| < 4Q/H, \qquad Qd/H < Y(D) := \frac{4D}{D - 23/2 + \sqrt{(D - 23/2)^2 - 10D}} < 9/4.$$

In the unit-disk strict-hole geometry,

$$H > \frac{N - (2637/32)/d}{2(1+a)}. \tag{5.25}$$

Proof. Proposition 4.11 first gives $Qd/H < 9/4$, which enters the medium row of Proposition 5.2. Equation (5.15) gives

$$(5/4)y_0^2 - (D - 23/2)y_0 + 2D > 0, \qquad y_0 = Qd/H.$$

At $D = 139$ this is positive at 2, negative at $9/4$, and decreases on $[2, 9/4]$; the signs persist on the degree-independent ray. The bound $y_0 < 9/4$ excludes the upper branch. The inverse relation at $Y = 2 + v$ is $D = 28/v + 33/2 + (5/4)v$, decreasing for $0 < v \le 1/4$. Finally apply Lemma 5.5 with $k = 9/4$, $K_0 = 4$ and $C_0 = (5/4)k^2 + (23/2)k$. Its constant is $2637/32$. □

Proposition 5.8. *Suppose $Qd/H < 3$ and $D \ge 5000/91$. Then the second weight-eight row gives $|\operatorname{Im} S| < (33/5)Q/H$ and $\operatorname{Re} K'(b) < 85/d^2$. In the strict-hole unit-disk geometry it follows that*

$$H > \frac{N - 210/d}{2(1+a)}. \tag{5.26}$$

Proof. The assumed ratio and field enter $q \le 273/5000$, $\mu \le 91/5000$. The difference $Z_8 - \zeta_s$ has coefficient-norm sum less than .397022 at this corner. Adding the radius gives a total effective Gaussian radius less than .477420. Every term of this center difference has weight at least two. As D increases, with $t = (5000/91)/D$, its parameters lie in the corner polydisk scaled by t, t^2, t^3. Consequently the effective radius is at most t^2 times its corner bound. Since $33/5 < 8$, the phase hypothesis for the 13/10 bound in Lemma 5.3 holds. The coefficient sum in (5.17) gives

$$D^2 \operatorname{Re} K'(b)/H^2 < 84.718 < 85.$$

This comparison uses all forty terms and the whole radius; the coefficient bound is a finite directed evaluation of those explicitly defined polynomials. Lemma 5.5 with $k = 3$, $K_0 = 33/5$ and the sharper displayed coefficient gives a field constant less than $2(84.718) + 6(33/5) < 210$. □

6. Inverse logarithmic counting and transport

Use (2.2) and (2.7), with b an actual simple zero and $S \ne 0$.

6.1. **Inverse rectangles.**

Lemma 6.1. *Suppose $Q/H < V$, $d \ge d_0 > R > 0$ and $H \ge H_0 > 0$. Put*

$$\Lambda = \frac{VR}{1 - R/d_0}, \qquad e_R = V\left(\frac{R^2}{2} + \frac{R^3}{3(d_0 - R)}\right), \qquad \widehat{R} = R - e_R,$$

$$B_R = \frac{V}{(1 - R/d_0)^2(1-\Lambda)^3}, \qquad p_* = (1+\Lambda)^{-1}, \qquad A_* = (1-\Lambda)^{-1}.$$

Assume $0 < \Lambda < 1$ and $\widehat{R} > 0$. Let $0 < \nu < 1$, $W > 0$, and choose positive e, ϵ, g such that

$$e > e^{-W}, \qquad B_R/H_0 < \epsilon, \qquad \nu B_R \widehat{R} < g, \qquad W^2 < (1-\nu^2)(H_0\widehat{R})^2, \tag{6.1}$$

$$\epsilon(1+e) + ge < 1 - A_* e, \quad \epsilon(1+e) + ge^2 < p_* - e, \quad \epsilon + ge < p_*. \tag{6.2}$$

For any positive odd integer ℓ with $\ell\pi \le \nu H\widehat{R}$, the disk $|z - b| < R$ contains at least ℓ distinct simple original zeros. If $\nu H\widehat{R} \ge \pi$, its number of original zeros is strictly greater than

$$\nu H\widehat{R}/\pi - 2 > \nu(7/22)H\widehat{R} - 2. \tag{6.3}$$

The source b is included once in this count.

Proof. Let $L(z) = \log(p'(z)/p'(b))$ on $|z-b| < d$, with $L(b) = 0$. The finite logarithms give on $|z-b| \le R$

$$|L'(z)/S - 1| < \Lambda, \qquad |L(z) - S(z-b)| < He_R.$$

For the second bound retain the quadratic term and bound the entire tail from order three by the displayed geometric series. The first bound integrated along each joining segment proves injectivity: the quotient $[L(z) - L(w)]/[S(z-w)]$ lies in $|u-1| < \Lambda$. Rouché's theorem on $|z-b| = R$ shows that the image contains $|v| < H\widehat{R}$. Its analytic inverse ϕ satisfies $\phi(0) = b$. Put $A(v) = S\phi'(v)$. Differentiation gives

$$A(0) = 1, \qquad \operatorname{Re} A > p_*, \qquad |A| < A_*, \qquad |A'| < B_R/H.$$

The last assertion uses $|L''| \le Q/(1 - R/d_0)^2$ and $A' = -SL''(\phi)/(L'(\phi))^3$.

The whole rectangle $|\operatorname{Re} v| \le W$, $|\operatorname{Im} v| \le Y = \ell\pi$ lies strictly inside the inverse disk by (6.1). The actual primitive is

$$\frac{Sp(\phi(v))}{p'(b)} = \int_0^v e^u A(u)\,du = e^v A(v) - 1 - E(v), \qquad E(v) = \int_0^v e^u A'(u)\,du.$$

Use the broken path $0, -W, -W + i\operatorname{Im} v, v$. For $v = x + iy$,

$$|E(v)| < \epsilon(1 + e^x) + ge^{-W}.$$

For $G = e^v A - 1$, the left side has $|G| > 1 - A_* e^{-W}$, the right side has $e^{-W}|G| > p_* - e^{-W}$, and each horizontal side has $|G| > 1 + p_* e^x$, since ℓ is odd. Equations (6.2) give $|E| < |G|$ on all four sides, including every corner. Hence Rouché compares the actual primitive with G.

For the count, use the homotopy $A_t = 1 - t + tA$, $0 \le t \le 1$. Then $\operatorname{Re} A_t \ge p_*$ and $|A_t| \le A_*$. The same left and right estimates, and the strict negative real part on the odd-π sides, keep $e^v A_t - 1$ nonzero throughout the boundary. It has the same number of zeros as $e^v - 1$, namely ℓ. Injectivity of ϕ and the nonvanishing of p' in the physical disk give distinct simple original zeros. Finally, the largest permitted odd integer is $1 + 2\lfloor(\nu H\widehat{R}/\pi - 1)/2\rfloor$, which is strictly larger than $\nu H\widehat{R}/\pi - 2$. □

Lemma 6.2. *Use the notation and hypotheses of Lemma 6.1, replacing the third inequality in (6.2) by*

$$\epsilon + ge < \frac{p_*}{1 + p_* + A_*}.$$

For an integer $k \ge 0$ with $Y = (2k + 1/2)\pi \le \nu H\widehat{R}$, put $\mathcal{R}_Y = \{v : |\operatorname{Re} v| < W,\ |\operatorname{Im} v| < Y\}$. Then $\phi(\mathcal{R}_Y) \subset \{z : |z-b| < R\}$ contains $2k+1$ simple original zeros. In particular, when $\nu H\widehat{R}/\pi \ge 1/2$, the number of original zeros in $|z-b| < R$ is strictly greater than $\nu H\widehat{R}/\pi - 3/2$.

Proof. On these horizontal sides $e^{iy} = \pm i$. For $u = e^x > 0$,

$$|e^v A(v) - 1| \ge \max(1 - A_* u, p_* u), \qquad \frac{\max(1 - A_* u, p_* u)}{1 + u} \ge \frac{p_*}{1 + p_* + A_*}.$$

The two quotients meet at $u = 1/(p_* + A_*)$. This gives the horizontal Rouché comparison. On $y = \pm Y$, the amplitude homotopy satisfies

$$\operatorname{Im}(e^v A_t - 1) = e^x \sin y\, \operatorname{Re} A_t,$$

which is positive on the upper side and negative on the lower side. Here $\operatorname{Re} A_t > 0$. The model $e^v - 1$ has exactly $2k + 1$ zeros. Choose the largest permitted k to obtain the last bound. □

The same inverse lemmas allow a sharper logarithmic radius. For any integer $M \ge 2$ one may replace e_R by

$$V\left(\sum_{k=2}^{M} \frac{R^k}{k d_0^{k-2}} + \frac{R^{M+1}}{(M+1)d_0^{M-1}(1 - R/d_0)}\right). \tag{6.4}$$

This retains every logarithmic order beyond M; all the other constants and the full-boundary proof are unchanged. To use exact integer counts, it is sufficient to verify $\ell\pi_{\mathrm{up}} \le \nu H_0 \widehat{R}$ for a rational upper bound $\pi_{\mathrm{up}} > \pi$. If the source root is inserted separately in a later weighted sum, use $\ell - 1$ other roots, since b was already counted. The one-sided and narrow-strip refinements needed near the origin are proved in Section 8.

6.2. **Schur coupling and positive cumulative sums.**

Lemma 6.3. *There is a nonnegative matrix (c_{ij}), indexed by all original zeros and all critical points, with*

$$\sum_j c_{ij} = 1/n, \qquad \sum_i c_{ij} = 1/N, \qquad N\sum_i c_{ij} z_i = \xi_j.$$

If $|z_i| \le 1$, then $N\sum_i c_{ij}|z_i - \xi_j|^2 \le 1 - |\xi_j|^2$.

Proof. Let $T = \mathrm{diag}(z_i)$, $e = n^{-1/2}(1,\dots,1)^T$ and compress T to $e^\perp$. Its characteristic polynomial is p'/n after making p monic. Indeed its cofactor in the e direction is $p(z)e^*(zI - T)^{-1}e = p'(z)/n$, first off the zeros and then identically. Lift a unitary Schur basis of the compression to an n by N matrix U. Its columns are orthonormal, $UU^* = I - ee^*$, and the diagonal entries of U^*TU are ξ_j. Set $c_{ij} = |U_{ij}|^2/N$. All three identities follow. Expanding around the barycenter gives

$$N\sum_i c_{ij}|z_i - \xi_j|^2 = \sum_i |U_{ij}|^2|z_i|^2 - |\xi_j|^2 \le 1 - |\xi_j|^2.$$

□

Lemma 6.4. *Assume the unit-disk strict-hole geometry of Lemma 2.2, with the chosen simple rightmost zero $b = x + iy$, $x \ge a$. For $0 < a \le A < 1$, define*

$$\Gamma = \frac{2}{2 - A^2}, \quad u = \Gamma A, \quad q_A = \frac{u}{1-u}, \quad E_A = \frac{2\Gamma}{(1-u)^2},$$

and assume $u < 1/2$. For $\xi_j = r_j\omega_j$, $|\omega_j| = 1$, put

$$g = \frac{1}{N}\sum_j \frac{1}{b - \omega_j}, \quad k = \mathrm{Re}\, g, \quad h = H/N, \quad \eta = \sum_{i,j} c_{ij}\frac{|z_i - \xi_j|^2}{|b - \xi_j|^2}.$$

Then $k > 0$, $h > (1 - q_A)k$, and $\eta < E_A a k$.

Proof. The hole gives $1 - a < r_j \le 1$, and with $t_j = a/2 - \mathrm{Re}\,\omega_j$ it gives

$$(1 - r_j)(1 + r_j - a^2) < 2ar_jt_j.$$

In particular $t_j > 0$. Since $2r/(1 + r - a^2)$ increases on $0 < r \le 1$, $1 - r_j < \Gamma a t_j$. Also $t_j < x - \mathrm{Re}\,\omega_j \le |b - \omega_j|$. Hence

$$|b - r_j\omega_j| > (1 - u)|b - \omega_j|, \qquad \left|\frac{1}{b - r_j\omega_j} - \frac{1}{b - \omega_j}\right| < q_A\,\mathrm{Re}\,\frac{1}{b - \omega_j}.$$

Averaging gives $|S/N - g| < q_A k$ and the field bound. The variance part of Lemma 6.3, together with $1 - r_j^2 < 2\Gamma a t_j$, gives

$$\eta \le \frac{1}{N}\sum_j \frac{1 - r_j^2}{|b - r_j\omega_j|^2} < \frac{2\Gamma a}{(1-u)^2 N}\sum_j \frac{t_j}{|b - \omega_j|^2} \le E_A a k.$$

□

Lemma 6.5. *Suppose the number L_j of original roots in nested disks $|z-b|<t_jd$, where $0<t_1<\cdots<t_m<1$, satisfies $L_j>s_jD-2$ with $0=s_0<s_1<\cdots<s_m$. If $D\ge D_0>0$, $N\ge N_0$, and*

$$c=\frac{N_0}{N_0+1}\left(\sum_{j=1}^{m}(1-t_j)^2(s_j-s_{j-1})-\frac{2}{D_0}\right)>0,$$

then $\eta>chd$. Under Lemma 6.4, $d<E_Aa/[(1-q_A)c]$.

Proof. Every root in the jth disk has $|z_i-\xi_k|>(1-t_j)|b-\xi_k|$ for every k. Its entire row in the coupling contributes more than w_j/n, where $w_j=(1-t_j)^2$. Positive summation of cumulative counts gives

$$n\eta>\sum_{j<m}(w_j-w_{j+1})L_j+w_mL_m>D\sum_j w_j(s_j-s_{j-1})-2w_1.$$

Use $w_1<1$, $N/(N+1)\ge N_0/(N_0+1)$, and $D\ge D_0$. Combining the result with $\eta<E_Aak<E_Aah/(1-q_A)$ proves the last assertion. The cumulative counts enter the Abel sum with nonnegative coefficients. □

For root-product weights, let $B_p=\prod_{i=1}^n f_i$, where $0\le f_i\le 1$ for every original zero, and set $-\log 0=+\infty$. If every root in the jth nested region has $f_i\le e^{-w_j}$, where $w_1\ge\cdots\ge w_m\ge 0$, and its cumulative count is at least K_j, then

$$-\log B_p\ge w_mK_m+\sum_{j<m}(w_j-w_{j+1})K_j. \tag{6.5}$$

This is the pointwise step-function lower estimate followed by summation over every root. If $K_j=s_jH_*-2$, the budget is $H_*\sum_j w_j(s_j-s_{j-1})-2w_1$.

6.3. **Crescent geometry and half-disk factors.**

Lemma 6.6. *Under the strict-hole geometry with $0<a\le x$, $b=x+iy$, $y\ge 0$, $|b|\le 1$, put $t=\sqrt{1-a^2/4}$. Every critical point satisfies*

$$|b-\xi|^2>(x-a/2)^2+(t-y)^2. \tag{6.6}$$

Consequently $d>x-a/2$, $d>1-y$, $d>1-|b|$, and

$$d^2>(1-|b|)^2+a^2/4. \tag{6.7}$$

Proof. Let $e=a/2+it$, write $\xi=e-u+iv$ and put $\lambda=y/t$. Then $u>0$ by (2.1), and $0\le\lambda<1$ because $y\le\sqrt{1-x^2}\le\sqrt{1-a^2}<t$. The exact identity

$$|b-\xi|^2-|b-e|^2=(1-\lambda)(|\xi-a|^2-1)+[2(x-a)+a\lambda]u+\lambda(u^2+v^2)>0$$

proves the crescent comparison. Its difference from $(1-y)^2$ is $x(x-a)+2y(1-t)\ge 0$. For fixed $r=|b|$ and a, the expression $1+r^2-ax-2t\sqrt{r^2-x^2}$ increases for $a\le x<r$: its derivative is $-a+2tx/\sqrt{r^2-x^2}>0$. Continuity treats $x=r$. Thus its minimum occurs at $x=a$. Since $2r-5a^2/4>0$ and

$$(2r-5a^2/4)^2-4(1-a^2/4)(r^2-a^2)=a^2(r-1)(r-4)+9a^4/16\ge 0,$$

squaring in the positive direction gives (6.7). This includes $r=1$. □

Lemma 6.7. *Under the preceding hypotheses put $c=x/2$. Then*

$$D>\frac{Nc}{2\sqrt{2-2c+c^2}-2\sqrt{2}c(1-c)}.$$

In particular, if $N\ge 6250$ and $x\ge 30501/500000$, then $D>70$.

Proof. For fixed b, the squared right side of (6.6) decreases as a increases to x, since its derivative is $-x+ay/(2\sqrt{1-a^2/4})<0$ for $0<a\le x\le 1$. Put $t=\sqrt{1-c^2}$, $v=t-y\ge 0$ and $G=\sqrt{c^2+v^2}$. Thus $d>G$ and $p_b=1-|b|^2=-3c^2+2tv-v^2\le 2v-3c^2-v^2$. Since $\operatorname{Re} S>cQ$, the radial identity implies

$$D>\frac{NcG}{2|b|c+p_b}.$$

The identities $c+(1-c)v\le\sqrt{1+(1-c)^2}G$ and $(3c^2+v^2)^2-8c^2G^2=(v^2-c^2)^2\ge 0$, together with $|b|\le 1-p_b/2$, give

$$\begin{aligned}2|b|c+p_b&\le 2[c+(1-c)v]-(1-c)(3c^2+v^2)\\&\le[2\sqrt{2-2c+c^2}-2\sqrt{2}c(1-c)]G.\end{aligned}$$

The bracket is positive for $0<c\le 1/2$. For $c\ge c_0=30501/1000000$, one has $2-2c+c^2<49/25<2$. Therefore the denominator is less than $(14/5)(1-c+c^2)$ and

$$D>\frac{5N}{14}\frac{c}{1-c+c^2}\ge\frac{476578125000000}{6793005177007}>70.$$

The rational function increases with c because its derivative is $(1-c^2)/(1-c+c^2)^2>0$. □

Lemma 6.8. *Assume all original zeros lie in $\overline{\mathbb{D}}$ and $a>0$ satisfies $|a-\xi_j|>1$ for every critical point. Let $b=x+iy$ be a simple rightmost zero with $x\ge a$, $y\ge 0$, and $|b|\le 1$, as in Lemma 6.6. Put $c=x-a/2$ and $p_b=1-|b|^2$. If $|\operatorname{Im} S|<K_0Q/H$ and $Nc>2K_0$, then*

$$D>\frac{Nc^2-2K_0c}{2xc+p_b}.$$

If $N\ge N_0$, $a\le a_U$, $x\ge x_0>a_U/2$, $p_b\le p_U$, and $N_0(x_0-a_U/2)>2K_0$, then the uniform bound is obtained by putting $N=N_0$, $c=x_0-a_U/2$, $x=x_0$, $p_b=p_U$.

Proof. The half-plane inequality gives $d>c$, $Q/H<1/c$. Hence $N<(2x+p_b/c)H+2K_0/c$ and the first assertion follows. With x,p_b fixed, the derivative numerator in c is $2[Nxc^2+p_b(Nc-K_0)]>0$. After writing $c=x-a_U/2$, the derivative numerator in x is $2[(Na_U/2+2K_0)c^2+p_b(Nc-K_0)]>0$. The quotient decreases with p_b and increases with N. These signs also hold at $p_b=0$. □

Lemma 6.9. *Let $b=x+iy$ be an actual simple rightmost zero, with $x>0$, $y\ge 0$, $0<B_L\le|b|\le B_U<1$, $x\le X$, and $\gamma=b-a/2$, $0<a\le a_U$. For a disk radius R put*

$$P=1-B_U^2,\qquad T=\frac{a_U}{2P},\qquad U=\frac{R}{P},\qquad L=1+2TX.$$

Suppose $R^2<P$ and $0<f<1$ satisfy

$$f^2-(1-f^2B_L^2)(T^2+U^2)>2Uf^2LB_U. \tag{6.8}$$

Then every original zero z with $|z-b|\le R$ obeys $|(\gamma-z)/(1-\overline{z}\gamma)|<f$.

Proof. Write $\tau=a/2$, $p_b=1-|b|^2$, $z=b+v$, $L_\gamma=1-|\gamma|^2=p_b+2\tau x-\tau^2$ and $A_0=p_b+\tau\overline{b}$. Rightmostness gives $\operatorname{Re} v\le 0$. The gate implies

$$\begin{aligned}T^2&<\frac{f^2}{1-f^2B_L^2}<\frac{1}{1-B_L^2},\\ \tau^2\le P^2T^2&<\frac{P^2}{1-B_L^2}\le P\le p_b.\end{aligned}$$

Thus $L_\gamma>0$, $|\gamma|<1$, and $1-f^2|\gamma|^2>0$. Direct expansion gives

$$f^2|1-\overline{z}\gamma|^2-|\gamma-z|^2=f^2|A_0|^2-\tau^2-(1-f^2|\gamma|^2)|v|^2-2\operatorname{Re}(Vv),$$

where $V = f^2 L_\gamma \bar{b} + \tau(1-f^2)$. Thus $\operatorname{Re} V > 0$, $\operatorname{Im} V = -f^2 L_\gamma y \le 0$. On the half disk $|v| \le R$, $\operatorname{Re} v \le 0$, $\operatorname{Re}(Vv) \le f^2 L_\gamma y R$. The resulting lower bound is

$$f^2 p_b^2 - (1-f^2|b|^2)(\tau^2+R^2) + 2f^2\tau x(p_b - R^2) + f^2\tau^2 R^2 - 2Rf^2 L_\gamma y.$$

The two terms containing $2f^2\tau x$ and $f^2\tau^2$ are nonnegative. After division by p_b^2, use $p_b \ge P > R^2$ and

$$\frac{L_\gamma}{p_b} = 1 + \frac{2\tau x - \tau^2}{p_b} \le 1 + 2TX = L.$$

The resulting bound is at least $f^2 - (1-f^2 B_L^2)(T^2+U^2) - 2Uf^2 L B_U > 0$. The positive difference also excludes a zero denominator. □

For $N \ge N_0$, if a field estimate has the form $H > (N-c)/F$, with $c \ge 0$, $F > 0$ and $(N_0 - c)/F > H_* > 0$, then

$$\begin{aligned} \frac{H}{N} > \frac{1-c/N}{F} \ge \frac{1-c/N_0}{F} > \frac{H_*}{N_0}, \\ Q \ge \frac{H^2}{N} > \frac{N}{N_0^2} H_*^2 \ge \frac{H_*^2}{N_0}. \end{aligned} \tag{6.9}$$

The lower bound $(1-c/N)/F$ is nondecreasing in N, so these estimates apply on the whole ray $N \ge N_0$.

7. The boundary and the compact range

We prove in this section the assertion of the main theorem whenever the modulus of the proposed exceptional point is at least $1/10$. Some of the moment estimates obtained along the way will also be useful in the remaining range. Throughout the section, P is monic of degree $n \ge 6251$, $N = n-1$, and all its zeros $z_1, \dots, z_n$ belong to $\overline{\mathbb{D}}$. Its critical points, with multiplicity, are denoted by $\xi_1, \dots, \xi_N$. We put

$$\mu = \frac{1}{n}\sum_i z_i = \frac{1}{N}\sum_j \xi_j, \qquad Q = \frac{1}{N}\sum_j |\xi_j|^2, \qquad L(z) = \frac{1}{N}\sum_j \log|z-\xi_j|. \tag{7.1}$$

The letter Q in this section denotes a second moment; reciprocal energy will always be written E_b.

Suppose, towards a contradiction, that a real point $0 < a \le 1$ in the convex hull of the zeros satisfies

$$|a - \xi_j| > 1 \quad (1 \le j \le N). \tag{7.2}$$

Choose an actual zero $b = x + iy$ of largest real part, so $x \ge a$; reflection permits us to assume $y \ge 0$. Write

$$r = |b-a|, \quad d = \min_j |b-\xi_j|, \quad S = \sum_j \frac{1}{b-\xi_j}, \quad H = |S|, \quad E_b = \sum_j \frac{1}{|b-\xi_j|^2}. \tag{7.3}$$

Gauss–Lucas and (7.2) imply

$$|\xi_j| \le 1, \quad \operatorname{Re}\xi_j < a/2, \quad r^2 \le 1 - a^2 < 1, \quad d > 1 - r, \quad \operatorname{Re} S > \frac{a}{2} E_b. \tag{7.4}$$

In particular b is simple. We use the actual-zero energy inequality $E_b \ge N$ of Proposition 3.1. It excludes $r = 0$ and also excludes a strict witness which is itself an original zero. Consequently all applications below have $0 < a < 1$ and $r > 0$.

7.1. **A product inequality retaining the original zeros.** The fiber arguments use the following fact for arbitrary equal-value fibers.

Lemma 7.1 (Equal-value perpendicular bisector)**.** *If $P(v) = P(w)$ and $v \neq w$, the critical points of P cannot all belong to either one of the two open half-planes bounded by the perpendicular bisector of $[v, w]$.*

Proof. We prove the needed half-plane polarization fact first. Let f be a nonzero polynomial of degree at most m with no zero in $\{\operatorname{Re} z > 0\}$. For $\operatorname{Re} w > 0$ put

$$T_w f(z) = f(z) + \frac{w - z}{m} f'(z).$$

This polynomial has degree at most $m - 1$ and is nonzero throughout $\{\operatorname{Re} z > 0\}$. To see the latter assertion, write the zeros of f as $\zeta_1, \dots, \zeta_d$, where $d \leq m$, and fix z with $x = \operatorname{Re} z > 0$. Each $v_j = (z - \zeta_j)^{-1}$ satisfies $\operatorname{Re} v_j \geq x|v_j|^2$. The disk $\{v : \operatorname{Re} v \geq x|v|^2\}$ is convex and contains zero; hence $v = m^{-1} \sum_{j=1}^{d} v_j$ is in that disk as well. If $v = 0$, then $T_w f(z) = f(z) \neq 0$. Otherwise $T_w f(z) = 0$ would imply $w = z - 1/v$, whose real part is at most zero, a contradiction.

Let $F(\zeta_1, \dots, \zeta_m)$ be symmetric and affine in each variable separately. If $f(z) = F(z, \dots, z)$ has no zero in the right half-plane, neither does F when all its variables are in that half-plane. Indeed, replacing one of the remaining equal variables z by w applies exactly the operation T_w, with m replaced by the number of remaining variables. Iteration proves the assertion. Reflection proves its left-half-plane version.

An affine change of variable reduces the asserted bisector to the imaginary axis and the two equal-value points to $-1, 1$. Consider

$$F(\zeta_1, \dots, \zeta_N) = \int_{-1}^{1} \prod_{j=1}^{N} (t - \zeta_j)\, dt.$$

Its diagonal is

$$F(z, \dots, z) = \frac{(1 - z)^{N+1} - (-1 - z)^{N+1}}{N + 1}.$$

Every zero of this polynomial has $|1 - z| = |-1 - z|$, and hence is on the imaginary axis. Half-plane polarization therefore makes F nonzero if all the critical points are strictly on the same side of that axis. On the other hand, $F(\xi_1, \dots, \xi_N)$ is a nonzero constant multiple of $P(1) - P(-1) = 0$. This contradiction proves the lemma. □

For $|\zeta| < 1$ define

$$\mathcal{B}_P(\zeta) = \prod_{i=1}^{n} \left| \frac{\zeta - z_i}{1 - \overline{z_i}\zeta} \right|.$$

Every factor is at most one. The following lower bound for the product retains the reciprocal energy E_b at the original zero b.

Lemma 7.2 (Adaptive product inequality)**.** *Under (7.2)–(7.3), put*

$$u = \min(r^2, 1/2), \qquad t = 1 - u, \qquad \gamma = b - ta, \qquad C = ta\big(2x - (1 + t)a\big). \tag{7.5}$$

Then $|\gamma| < 1$, $C \geq a^2 u(1 - u) > 0$, and

$$C \exp(CE_b/2) < 2(1 + |b|)(1 + |\gamma|)\mathcal{B}_P(\gamma) \leq 8\mathcal{B}_P(\gamma). \tag{7.6}$$

In particular, if $n \geq 6251$, then

$$C < C_* := \frac{1}{388}. \tag{7.7}$$

Proof. Set $k = 1 - |\gamma|^2$. Since $|\gamma|^2 = r^2 + 2ua(x-a) + u^2a^2 > u$ and $|b|^2 - |\gamma|^2 = ta(2x - ta) > 0$, we have $0 < k < t < 1$. Direct expansion gives, for every critical point,

$$|b-\xi|^2 - |1-\overline{\gamma}\xi|^2 = t(|\xi - a|^2 - 1) + C + (t-k)(1-|\xi|^2) > C. \tag{7.8}$$

Also $|1-\overline{\gamma}\xi|^2 - |\gamma - \xi|^2 = k(1-|\xi|^2) \geq 0$.

Let $M = 1 + |b|$ and $\eta = C/(2M)$. For every $z \in D(b,\eta)$,

$$|z-\xi|^2 > (|b-\xi| - \eta)^2 \geq |b-\xi|^2 - 2M\eta + \eta^2 > |\gamma - \xi|^2.$$

Here $\eta < |b-\xi|/2$, by (7.8), and $\eta < ta = |b-\gamma|$. Lemma 7.1 therefore forbids two equal-value points, one in $D(b,\eta)$ and the other equal to γ. In particular $P(\gamma) \neq 0$. If $|P(\gamma)| < \eta|P'(b)|/n$, the logarithmic derivative of $P(z) - P(\gamma)$ at b would give

$$\left| \sum_{P(\zeta)=P(\gamma)} \frac{1}{b-\zeta} \right| = \frac{|P'(b)|}{|P(\gamma)|} > \frac{n}{\eta}.$$

At least one of these n zeros, counted with multiplicity, would be in $D(b,\eta)$, again a contradiction. Hence

$$|P(\gamma)| \geq \frac{\eta}{n}|P'(b)|. \tag{7.9}$$

Let $q = P'/n$, and introduce the reciprocal polynomials

$$P^*(z) = \prod_i (1 - \overline{z_i}z), \qquad q^*(z) = \prod_j (1 - \overline{\xi_j}z).$$

From (7.8) and $-\log(1-v) > v$ for $0 < v < 1$,

$$\frac{|P'(b)|}{n|q^*(\gamma)|} > \exp\left(\frac{C}{2} \sum_j |b-\xi_j|^{-2} \right). \tag{7.10}$$

The exact reversed derivative identity is

$$\frac{q^*(\gamma)}{P^*(\gamma)} = \frac{1}{n} \sum_i \frac{1}{1 - \overline{z_i}\gamma}.$$

For $|w| \leq g < 1$ one has $\mathrm{Re}(1-w)^{-1} \geq (1+g)^{-1}$, as follows by multiplying by $|1-w|^2$. Thus the modulus of the last display is at least $(1+|\gamma|)^{-1}$. Combining it with (7.9) and (7.10) proves (7.6). Finally

$$\frac{6250}{776} > \log(8 \cdot 388)$$

implies (7.7), since $C\exp(NC/2)$ increases strictly in $C > 0$ and in N. This numerical comparison is certified by the positive exponential series, as described below. □

The normalization at the origin uses the following movement estimate.

Lemma 7.3 (Movement of an equal-value fiber)**.** *Let $\omega = b/|b|$, let $a_0 \leq a$, and suppose $2\,\mathrm{Re}(\overline{\omega}\xi_j) < d_0$ for all j. If positive numbers s, δ satisfy*

$$a_0 - s > \delta, \qquad (a_0 - s)(a_0 + s - d_0) > \delta(4+\delta), \tag{7.11}$$

then $|P(s\omega)| \geq \delta|P'(b)|/n$.

Proof. Writing $v = |b| \geq a_0$, the squared-distance difference at b is

$$|b-\xi|^2 - |s\omega - \xi|^2 = (v-s)(v+s-2\,\mathrm{Re}(\overline{\omega}\xi)) > (a_0 - s)(a_0 + s - d_0).$$

Moving b by at most δ decreases this difference by at most $\delta(4+\delta)$. Thus every point of $D(b,\delta)$ is farther from every critical point than $s\omega$ is. The first condition in (7.11) puts $s\omega$ outside that disk. The same fiber and logarithmic-derivative argument as in (7.9) proves the assertion. □

7.2. **Normalization, annuli, and moment estimates.**

Proposition 7.4 (Three moment estimates)**.** *Assume $n \geq 6251$. Under (7.2), each row of Table 1 implies $P(0) \neq 0$. With*

$$B = -\frac{1}{n}\log|P(0)|, \qquad \mathcal{E}(r;a,N) = \frac{8r}{aN} + \frac{r^2}{2(1-r)(1-2r)}, \tag{7.12}$$

its conclusions are

$$B < \frac{\ell}{n} + \mathcal{E}(r;a,N), \tag{7.13}$$

$$|\mu| \leq \kappa B + n^{-2}, \tag{7.14}$$

$$Q < 1 - a^2 + w\left(\frac{\ell}{N} + \mathcal{E}(r;a,N)\right) + 2n^{-2}. \tag{7.15}$$

The same formulas hold with the parameters in the four boundary rows of the tables below,

TABLE 1. The moment estimates. The hypotheses use the actual a, r.

Row	$a \geq$	$r \leq$	ℓ	κ	w
M0	.9	.057	4.42	2.5482	5.098
M1	.93	.055	4.22	2.3817	4.765
M2	.95	.054	4.11	2.3108	4.623

whenever their respective hypotheses hold.

We give the normalization argument with parameters; this also supplies the estimates needed to prove the boundary assertions. A row specifies lower a_0, upper displacement $e < 1/2$, a squared critical-distance bound $R_c^2 < (1-e)^2$, and the constants listed in Tables 2–4. The names B, A, G1, G2 stand, respectively, for the boundary range $a \geq .975$, and the small displacement ranges $(a,r) \in [.85,1) \times [0,.02]$, $[.96,1) \times [0,.04]$, and $[.965,1) \times [0,.045]$. For row B the displacement bound is a conclusion of Lemma 7.2: indeed

$$(.975)^2(.053)^2(1-(.053)^2) > 1/388,$$

and $u(1-u)$ increases on $[0,1/2]$; hence $r < .053$.

Proof of Proposition 7.4. First,

$$|\omega - 1|^2 = \frac{2y^2}{|b|(|b|+x)} \leq r^2/a^2, \qquad 2\,\mathrm{Re}(\overline{\omega}\xi_j) < a + 2r/a.$$

All rows satisfy $2e < a_0^2$. The function $a + 2e/a$ therefore increases on $[a_0, 1]$, and is at most $1 + 2e < d_0$ there. The two pairs $(s_1,\delta_1), (s_2,\delta_2)$ in the tables satisfy (7.11). Lemma 7.3 applies to both.

Put $k = N/n$, $A_n = \delta_1^{2/n}$, and $A_- = 1 - \alpha_0/n$. The inequality $\alpha_0/2 > \log(1/\delta_1)$ gives $A_n > A_- > 0$. Original-zero AM–GM at $s_1\omega$ gives

$$2s_1\,\mathrm{Re}(\overline{\omega}\mu) \leq 1 + s_1^2 - A_n e^{2kL(b)}. \tag{7.16}$$

On $R_c \leq s \leq 2$ the logarithmic chord

$$s^2 \leq R_c^2 + \frac{22}{5}\log(s/R_c) \tag{7.17}$$

follows by convexity of e^{2v} in $v = \log s$ and its endpoint inequality. The latter follows already from $\log(1/R_c) \ge (1 - R_c^2)/2$ and the displayed certificate below. Average (7.17) at $s = |b - \xi_j|$ and use (7.16). For $v = |b|$ this yields

$$Q \le F(L(b)) := \frac{v}{s_1}(1 + s_1^2 - A_n e^{2kL(b)}) + R_c^2 - v^2 + \frac{22}{5}(L(b) - \log R_c). \tag{7.18}$$

Let

$$T = \frac{1 + s_1^2 - A_- R_c^2}{s_1}, \qquad F_0 = a_0 T + R_c^2 - a_0^2.$$

The row comparisons $T < 2a_0$ and $22/5 - 2a_0 k A_- R_c^2/s_1 < 0$ show that F decreases on the entire half-line $L(b) \ge \log R_c$. Indeed $F'' < 0$, while $R_c^{2k} \ge R_c^2$; the quadratic $vT - v^2$ decreases for $v \ge a_0$. The resulting chain is

$$Q \le F(L(b)) \le F(\log R_c) \le F_0 < q_c, \qquad |\mu| < m_c, \tag{7.19}$$

where the last inequality uses $q_c < m_c^2$.

The same calculation at every $z\omega$, $0 \le z \le s_2$, bounds the average squared critical distance by

$$z^2 + \frac{v - z}{s_1}(1 + s_1^2 - A_n e^{2kL(b)}) + R_c^2 - v^2 + \frac{22}{5}(L(b) - \log R_c).$$

Subtract $.9e^{2L(b)}$. Its derivative in $L(b)$ is negative throughout the same half-line, because

$$2R_c^2\left(kA_-\frac{a_0 - s_2}{s_1} + .9\right) > 22/5.$$

At $L(b) = \log R_c$ it is at most $z^2 - zT + F_0 - .9R_c^2$. This is a convex quadratic in z; the two endpoint values, at $0, s_2$, are negative. Critical-zero AM–GM consequently gives

$$|P'(z\omega)| < (.95)^N |P'(b)| \qquad (0 \le z \le s_2). \tag{7.20}$$

Since $e^{N/20} > 200n$ and $s_2 < 100\delta_2$, integration and the second movement inequality give

$$|P(0)| \ge \frac{\delta_2}{2n}|P'(b)| > 0, \qquad L(b) \le \frac{\ell}{N} - \frac{n}{N}B, \tag{7.21}$$

where $e^\ell > 2/\delta_2$. The inequalities $R_c > 1.001p$ and $(1.001)^N > 2p/\delta_2$ also give $|P(0)| > p^n$. In particular $0 \le B < -\log R_c + \ell/n < B_c$.

For the annulus iteration, suppose $|\mu| \le m$, $Q < q_c$, and

$$\rho^2 + 2\rho m + q_c < s_*^2, \qquad \rho < s_* < p, \qquad n(s_*/p)^n < 1. \tag{7.22}$$

For every $|z| \le \rho$, AM–GM gives $|P'(z)| < n s_*^N$. Integrating on the segment from 0 to z,

$$\frac{|P(z) - P(0)|}{|P(0)|} < \frac{\rho}{s_*} n (s_*/p)^n < 1.$$

Thus every original zero has modulus greater than ρ. For $0 < \rho \le s \le 1$, the function $2\sinh t/t$, $t = -\log s$, increases with $t \ge 0$. It follows that

$$s^{-1} - s \le \frac{\rho^{-1} - \rho}{-\log \rho}(-\log s).$$

Since $-P'(0)/P(0) = \sum_i z_i^{-1}$, any $\kappa > (\rho^{-1} - \rho)/(-\log \rho)$ therefore gives

$$|\mu| \le \kappa B + \frac{|P'(0)|}{n|P(0)|}. \tag{7.23}$$

The residual in this formula is bounded by

$$\frac{m_c^N}{p^n} = \frac{(m_c/p)^n}{m_c} < 2d_s^n < n^{-2},$$

using $m_c/p < d_s$ and $1/m_c < 2$. Beginning with $m = m_c$, each successive triple (ρ, κ, m') in Table 4 satisfies (7.22) and $\kappa B_c + n^{-2} < m'$. The last triple supplies (7.14).

The geometric bounds (7.4) and $E_b \geq N$ give

$$H > Na/2, \qquad E_b/H < 2/a, \qquad E_b/H^2 < 4/(Na_0^2), \qquad (Hd)^{-1} < 2/[Na_0(1-e)].$$

The last two bounds are respectively below $7/625$ and $1/270$. The Fine companion estimate of Proposition 5.2 therefore gives

$$|\operatorname{Im} S| < 4E_b/H < 8/a. \tag{7.24}$$

For $w_0 = b-a$, the complete logarithmic series at b converges absolutely, because $r/d < r/(1-r) < 1$. Its linear term satisfies

$$-\operatorname{Re}(w_0 S) = -(x-a)\operatorname{Re} S + y \operatorname{Im} S \leq r|\operatorname{Im} S|.$$

The terms of order at least two have total absolute value, after averaging, at most $E_b r^2/[2N(1-r/d)]$. Using $E_b/N < (1-r)^{-2}$ proves

$$L(a) \leq L(b) + \mathcal{E}(r; a, N). \tag{7.25}$$

Since $L(a) > 0$, (7.21) proves (7.13).

Finally $s^2 \leq 1 + w \log s$ on $1 \leq s \leq 2$, because $w \log 2 > 3$ and the exponential is convex. Apply this chord at $s = |a - \xi_j|$ and use (7.21), (7.23), and (7.25). We obtain

$$Q < 1 - a^2 + w\left(\frac{\ell}{N} + \mathcal{E}(r; a, N)\right) + 2n^{-2} + (2a\kappa - wn/N)B.$$

The coefficient of B is negative, since $w > 2\kappa$ and $a \leq 1$. This proves (7.15). □

TABLE 2. Preliminary parameters. In every row $s_1 = .25$ and $s_2 = .45$.

Row	a_0	e	R_c^2	d_0	δ_1	δ_2	α_0	ℓ
B	0.975	0.053	0.896708	1.10611	0.01976	0.03838	7.86	3.97
A	0.85	0.02	0.960299	1.04011	0.00823	0.02383	9.61	4.44
G1	0.96	0.04	0.921499	1.08011	0.02114	0.03857	7.73	3.96
G2	0.965	0.045	0.911924	1.09011	0.02047	0.03836	7.79	3.97
M0	0.9	0.057	0.889148	1.11411	0.00534	0.02433	10.48	4.42
M1	0.93	0.055	0.892924	1.11011	0.01089	0.0297	9.05	4.22
M2	0.95	0.054	0.894815	1.10811	0.01474	0.03346	8.45	4.11

TABLE 3. Coarse normalization and annulus parameters.

Row	q_c	m_c	p	s_*	B_c	d_s	w
B	0.59717	0.77287	0.94595	0.94414	0.055158	0.81705	4.494
A	0.59041	0.76849	0.97891	0.9771	0.020976	0.78506	4.33
G1	0.54582	0.7389	0.95893	0.95712	0.041521	0.77056	4.33
G2	0.56641	0.75271	0.95394	0.95213	0.046745	0.78907	4.354
M0	0.70869	0.84194	0.94195	0.94014	0.059463	0.89384	5.098
M1	0.66376	0.81482	0.94395	0.94214	0.057312	0.86322	4.765
M2	0.63422	0.79648	0.94495	0.94314	0.056237	0.8429	4.623

Table 4: Successive annulus triples (ρ, κ, m'). A dash means that only the resulting inequality $|\mu| \le \kappa B + n^{-2}$ is retained.

Row	ρ	κ	m'
B	0.169	3.2333	0.178344
	0.39	2.3091	0.127367
	0.427	2.2504	0.124129
	0.43	2.2462	0.123897
	0.43	2.2462	–
A	0.206	2.9424	0.061721
	0.543	2.1268	0.044613
	0.558	2.1155	0.044376
	0.558	2.1155	–
G1	0.216	2.8802	0.11959
	0.498	2.1661	0.08994
	0.523	2.1432	0.088989
	0.523	2.1432	–
G2	0.197	3.0035	0.1404
	0.457	2.2109	0.10335
	0.486	2.1783	0.101826
	0.488	2.1762	0.101728
	0.488	2.1762	–
M0	0.096	4.4043	0.261894
	0.229	2.8073	0.166932
	0.281	2.5822	0.153547
	0.29	2.5515	0.151721
	0.291	2.5482	–
M1	0.125	3.7872	0.217054
	0.301	2.5165	0.144227
	0.348	2.3928	0.137138
	0.353	2.3817	0.136502
	0.353	2.3817	–
M2	0.144	3.5093	0.197354
	0.343	2.4043	0.135212
	0.385	2.318	0.130359
	0.389	2.3108	0.129954
	0.389	2.3108	–

The numerical statements used in the preceding proof are finite rational comparisons, with logarithms bounded by convergent positive series. The remaining comparisons are as follows, with $n_0 = 6251$, $N_0 = 6250$, $k_0 = N_0/n_0$, and $A_- = 1 - \alpha_0/n_0$:

$$2e < a_0^2, \quad 1 + 2e < d_0, \quad R_c^2 < (1-e)^2, \quad (7.11) \text{ for } j = 1, 2; \tag{7.26}$$

$$R_c^2 + \tfrac{22}{5}(\log 2 + (1 - R_c^2)/2) > 4, \quad \alpha_0/2 > \log(1/\delta_1), \quad e^{\ell} > 2/\delta_2; \tag{7.27}$$

$$T < 2a_0, \quad F_0 < q_c < m_c^2, \quad 2a_0 k_0 A_- R_c^2/s_1 > 22/5; \tag{7.28}$$

$$2R_c^2[k_0 A_-(a_0 - s_2)/s_1 + .9] > 22/5, \quad z^2 - zT + F_0 < .9R_c^2 \quad (z = 0, s_2); \tag{7.29}$$

$$N_0/20 > \log(200 n_0), \quad s_2 < 100\delta_2, \quad R_c > 1.001p, \quad N_0 \log(1.001) > \log(2p/\delta_2); \tag{7.30}$$

TABLE 5. Uniform bounds $Q < q_0$, $B < B_0$, $|\mu| < m_0$, and $|P(0)| > p_f^n$ furnished by the moment formulas at $n = 6251$.

Row	q_0	B_0	m_0	p_f
B	0.06001	0.0023642	0.005311	0.997625
A	0.28164	0.0009535	0.0020177	0.999036
G1	0.08531	0.0015932	0.0034151	0.998396
G2	0.07689	0.0018604	0.0040492	0.998129
M0	0.20395	0.0027331	0.0069651	0.997256
M1	0.14726	0.0025497	0.0060732	0.99744
M2	0.10888	0.0024586	0.0056819	0.997531

$$-\log R_c + \ell/n_0 < B_c, \quad 1/m_c < 2, \quad m_c/p < d_s, \quad (1-d_s)n_0 > \log(2n_0^2); \tag{7.31}$$

$$-n_0 \log(s_*/p) > \log n_0, \quad -n_0 \log(s_*/p) > 1, \quad (1-d_s)n_0 > 2; \tag{7.32}$$

$$\rho^2 + 2\rho m + q_c < s_*^2, \quad \rho < s_*, \quad \rho^{-1} - \rho < -\kappa \log \rho, \quad \kappa B_c + n_0^{-2} < m'; \tag{7.33}$$

$$\frac{4}{N_0 a_0^2} < \frac{7}{625}, \quad \frac{2}{N_0 a_0 (1-e)} < \frac{1}{270}, \quad w > 2\kappa, \quad w \log 2 > 3. \tag{7.34}$$

In (7.33), $m = m_c$ at the first stage and is the preceding m' thereafter; the last inequality is omitted for a dash. The caps in Table 5 are checked by substituting $a = a_0, r = e, n = n_0$ in (7.13)–(7.15), and by $\kappa B_0 + n_0^{-2} < m_0$, $1 - B_0 > p_f$. These conditions prove all the displayed estimates for every $n \geq n_0$: k, A_- increase, the moment errors decrease, and $n(s_*/p)^n$ and $2n^2 d_s^n$ decrease after their stated derivative bounds.

7.3. **Support and the normal of a boundary edge.** The following transfer is a finite identity between the original and critical empirical measures. It retains the normalization difference between n and N.

Lemma 7.5 (Moment support bound)**.** *Suppose $P(0) \neq 0$, $B \leq B_0$, $Q \leq q < 1$, and $|\mu| \leq m$. Let $0 < h < 1$ and put*

$$\Lambda = \frac{(1-q)^2}{1+q+2m}.$$

If

$$(N\Lambda + 1)(1-h)^2 > 2(1-h) + 2nB_0, \tag{7.35}$$

then the convex hull of the original zeros contains $\overline{D(0,h)}$ in its interior.

Proof. Rotate a proposed contrary support direction to the real axis, so that $\operatorname{Re} z_i \leq h$ for all i. Define

$$C_o(z) = \prod_i (1 - z_i z), \quad C_c(z) = \prod_j (1 - \xi_j z), \quad f(z) = \frac{1}{n} \sum_i \frac{1}{1 - z_i z}, \quad g(z) = \frac{1}{N} \sum_j \frac{1}{1 - \xi_j z}.$$

The exact identities $C_c = C_o - zC_o'/n$ and $C_c/C_o = f$ give

$$N(g - f) = f - 1 - zf'/f. \tag{7.36}$$

The support assumption, and Gauss–Lucas for the critical points, make every term well defined at $z = 1$. Also

$$|f'(1)| \leq \frac{1}{n} \sum_i |1 - z_i|^{-2}, \qquad \operatorname{Re} f(1) \geq (1-h) \frac{1}{n} \sum_i |1 - z_i|^{-2}.$$

Consequently $f(1)\neq 0$ and $|f'(1)/f(1)|\le (1-h)^{-1}$. Set $p_o=2\operatorname{Re} f(1)-1$, $p_c=2\operatorname{Re} g(1)-1$. Then

$$p_o=\frac{1}{n}\sum_i \frac{1-|z_i|^2}{|1-z_i|^2}\le \frac{2B_0}{(1-h)^2},\qquad Np_c=np_o-1-2\operatorname{Re}(f'(1)/f(1)).$$

Here we used $1-|z_i|^2\le -2\log|z_i|$. For the other measure, Cauchy–Schwarz gives

$$p_c\left(\frac{1}{N}\sum_j |1-\xi_j|^2\right)\ge \left(\frac{1}{N}\sum_j \sqrt{1-|\xi_j|^2}\right)^2\ge (1-Q)^2.$$

Its denominator equals $1+Q-2\operatorname{Re}\mu$, whence $p_c\ge\Lambda$. Combining these estimates contradicts (7.35). Since every direction was allowed, all support values exceed h. □

We use the inverse-log quantities of Lemma 6.1. Suppose, at the actual rightmost zero b,

$$H>H_0>0,\quad E_b/H<V,\quad d^{-1}<M. \tag{7.37}$$

For $R>0$ define

$$\lambda=\frac{VR}{1-MR},\qquad e_R=V\left(\frac{R^2}{2}+\frac{MR^3}{3(1-MR)}\right),\qquad \widehat{r}=R-e_R, \tag{7.38}$$

$$B_2=\frac{V}{(1-MR)^2(1-\lambda)^3},\qquad p_-=(1+\lambda)^{-1},\qquad p_+=(1-\lambda)^{-1}. \tag{7.39}$$

Choose $0<\nu<1$, $W>0$, and $e_W>e^{-W}$, and put

$$\epsilon=1.001B_2/H_0,\qquad g=1.001\nu B_2\widehat{r}.$$

The sufficient comparisons are

$$MR<1,\quad 0<\lambda<1,\quad \widehat{r}>0,\quad W^2<(1-\nu^2)(H_0\widehat{r})^2, \tag{7.40}$$

$$\epsilon(1+e_W)+ge_W<1-p_+e_W, \tag{7.41}$$

$$\epsilon(1+e_W)+ge_W^2<p_--e_W,\qquad \epsilon+ge_W<p_-. \tag{7.42}$$

They ensure that the analytic logarithm $\mathcal{L}(z)=\log(P'(z)/P'(b))$, fixed at b, is injective on $D(b,R)$ and has an inverse ϕ on $D(0,H\widehat{r})$. Moreover

$$|\phi(w)-b-w/S|<e_R,\qquad \frac{SP(\phi(w))}{P'(b)}=\int_0^w e^v A(v)\,dv,\quad A=S\phi', \tag{7.43}$$

with $\operatorname{Re} A>p_-$, $|A|<p_+$ and $|A'|<B_2/H$. On every rectangle with vertical sides $\operatorname{Re} w=\pm W$ and odd multiples of π as horizontal sides, contained in $|\operatorname{Im} w|\le \nu H\widehat{r}$, the complete side estimates and the homotopy $(1-s)+sA$ give the same zero count as e^w-1. In particular the number of original zeros in $D(b,R)$ is greater than

$$\nu H\widehat{r}/\pi-2>\nu(7/22)H\widehat{r}-2, \tag{7.44}$$

provided the last expression is positive. The loss two includes the choice of the outermost odd heights. When $\nu H_0\widehat{r}>4\pi$, the top adjacent strip and the bottom adjacent strip each contain one distinct simple original zero, with

$$|\operatorname{Re} w|<W,\qquad \operatorname{Im} w_+>\nu H\widehat{r}-4\pi,\qquad \operatorname{Im} w_-<-\nu H\widehat{r}+4\pi. \tag{7.45}$$

These statements concern zeros of P itself, through (7.43).

Lemma 7.6 (Localization of an edge normal)**.** *Suppose a is on a supporting edge of the original hull, the hull contains a positive centered disk, and $|b-a| \le e$. Suppose (7.37)–(7.42) hold, $\nu H_0 \widehat{r} > 4\pi$, and $|\operatorname{Im} S| < J_0$. Set $s = J_0/H_0 < 1$ and*

$$\begin{aligned} E_0 &= W/H_0 + s\nu\widehat{r} + e_R, \\ L_0 &= (1-s^2)(\nu\widehat{r} - 4\pi/H_0) - sW/H_0 - e_R. \end{aligned} \tag{7.46}$$

If $L_0 > e$, the outward normal $(\cos\theta, \sin\theta)$ of this edge satisfies

$$\cos\theta > 0, \qquad |\tan\theta| < K := \frac{E_0}{L_0 - e}. \tag{7.47}$$

In particular its distance from the origin is greater than $a/\sqrt{1+K^2}$.

Proof. Write $\sigma = S/H$. Since $\operatorname{Re} S > 0$, one has $|\operatorname{Im}\sigma| < s$ and $\operatorname{Re}\sigma > 1 - s^2$. The two zeros supplied by (7.45), and $w/S = \overline{\sigma} w/H$, satisfy

$$\operatorname{Re} z_\pm > x - E_0, \qquad \operatorname{Im} z_+ > y + L_0, \qquad \operatorname{Im} z_- < y - L_0.$$

These bounds follow by applying (7.43) on the strips. The positive support disk gives $a\cos\theta > 0$. Every original zero obeys

$$\cos\theta(\operatorname{Re} z - a) + \sin\theta \operatorname{Im} z \le 0.$$

Use z_+ when $\sin\theta \ge 0$ and z_- otherwise, and use $x - a \ge 0$, $|y| \le e$. This gives $|\sin\theta|(L_0 - e) < E_0 \cos\theta$, proving the lemma. □

Lemma 7.7 (Geometry of the whole edge and radial paths)**.** *Suppose the edge through a has distance at least $h > 0$ from the origin and its normal satisfies (7.47). Set*

$$c = \frac{h - K\sqrt{1-h^2}}{\sqrt{1+K^2}}.$$

Every point z of the edge has $\cos(\arg z) \ge c$ and

$$|z - \xi_j| > 1 - K - \sqrt{1-h^2}. \tag{7.48}$$

Furthermore, suppose $a \ge a_1$, $0 < \gamma < 1$, and $0 < r_0 < h$. If

$$1 - v^2 + 2t(vc - \gamma) - t^2(1-\gamma^2) > 0 \quad (v = a_1, 1;\ t = r_0, 1), \tag{7.49}$$

then the whole radial segment from $r_0 u/|u|$ to each endpoint u of the edge satisfies $|z - \xi_j| > \gamma|z|$.

Proof. If h_e is the distance of the line and θ is its normal angle, every point of its unit-disk chord satisfies $|\arg z - \theta| \le \arccos h_e \le \arccos h$. Combining this with $|\tan\theta| < K$ gives the bound c. In normal and tangent coordinates the chord has tangent coordinate in $[-\sqrt{1-h_e^2}, \sqrt{1-h_e^2}]$, while the tangent coordinate of a has modulus $h_e|\tan\theta|$. Therefore $|z - a| \le \sqrt{1-h^2} + K$, which proves (7.48) by (7.2). For $z = te^{i\arg u}$, the difference

$$(1-\gamma t)^2 - |a - z|^2 \ge 1 - a^2 + 2t(ac - \gamma) - t^2(1-\gamma^2)$$

is bounded below by a polynomial concave separately in a and t. Its minimum on $[a_1, 1] \times [r_0, 1]$ is therefore at a corner. Under (7.49), $|a - z| < 1 - \gamma t$, and (7.2) gives the claimed radial gap. □

7.4. **A potential estimate with an error proportional to the moment.** The following estimate retains a factor Q in the interpolation error.

Lemma 7.8 (Endpoint potential estimate)**.** *Let $0 < r_0 < h$, and let $C_0, v_0, \beta, J, s_0, g, e_0$ be positive parameters. Suppose an edge through a strict witness a has distance at least h from the origin, and all its points have critical distance greater than $\gamma > 0$. Suppose, at its two endpoints u, the radial paths from $r_0u/|u|$ to u have critical distance greater than $\gamma|z|$. Assume*

$$Q \le q < 1, \quad |\mu| \le m < \min(h, r_0), \quad |P(0)| > p_f^n, \qquad p_f > .99.$$

If $n \ge 1000$, the comparisons (7.52), (7.54), (7.57), and (7.59) hold, and $12+J/(1-Jq)+1/99 < 24$, then

$$-\log|P(0)| < 24Q. \tag{7.50}$$

Proof. Put $v = Q - |\mu|^2 = N^{-1}\sum_j |\xi_j - \mu|^2 \ge 0$. The exact centered resolvent identity is

$$\frac{1}{N}\sum_j \frac{1}{z-\xi_j} - \frac{1}{z-\mu} = \frac{1}{N}\sum_j \frac{(\xi_j-\mu)^2}{(z-\mu)^2(z-\xi_j)}. \tag{7.51}$$

On each outer radial path its modulus is at most $C_0 v$ if

$$\begin{gathered}\frac{1}{(r_0-m)^2\gamma r_0} < C_0, \qquad 1 - \frac{m}{r_0(r_0-m)} > v_0, \qquad v_0\frac{N}{n} - C_0 q > \beta, \\ \frac{C_0}{\beta^2(h-m)} < J, \qquad Jq < 1.\end{gathered} \tag{7.52}$$

The second comparison bounds from below the inward logarithmic speed of the centered radius $|z-\mu|$. Indeed, for $z = t\omega$,

$$\operatorname{Re}\frac{\omega}{t\omega-\mu} \ge \frac{1}{t} - \frac{m}{t(t-m)} \ge 1 - \frac{m}{r_0(r_0-m)}.$$

Let $q_P = P'/n$, $C_u = q_P(u)/(u-\mu)^N$, and $w_u = r_0u/|u|$. Integrate the logarithmic derivative in (7.51) inward from u a distance s. The centered monomial decays at least as e^{-Nv_0s}, and the logarithm of its ratio to q_P has modulus at most NC_0vs. Since $e^u - 1 \le ue^u$ for $u \ge 0$, the absolute difference integral is at most

$$|P'(u)|\int_0^\infty nC_0vse^{-\beta ns}\,ds = \frac{C_0v|P'(u)|}{\beta^2 n}.$$

Consequently

$$\begin{gathered}\int_{w_u}^u P'(z)\,dz = C_u\big((u-\mu)^n - (w_u-\mu)^n\big) + E_{\text{out}}, \\ |E_{\text{out}}| \le Jv|C_u(u-\mu)^n|.\end{gathered} \tag{7.53}$$

For the endpoint logarithm set $\zeta_j = (\xi_j-\mu)/(u-\mu)$. Their mean is zero. The parameter comparisons we use are

$$\frac{1+m}{h-m} < 1.1, \quad \log\frac{1+m}{\gamma} < 1.2, \quad \log\frac{2}{h-m} < 1.2, \quad \frac{4(1.2+1.1)}{(h-m)^2} < 12. \tag{7.54}$$

They imply $|\zeta_j| < 1.1$ and $\gamma/(1+m) < |1-\zeta_j| < 2/(h-m)$. For $|\zeta| \le 1/2$, the complete logarithmic series gives $|\log|1-\zeta| + \operatorname{Re}\zeta| \le |\zeta|^2$. For $|\zeta| > 1/2$, the corresponding bound is $4(1.2+1.1)|\zeta|^2$. Averaging, and using the zero mean, gives

$$|L(u) - \log|u-\mu|| \le 12v. \tag{7.55}$$

For the inner segment, write $\eta_j = \xi_j - \mu$ and, for fixed z, consider $f_z(t) = \prod_j (z - \mu - t\eta_j)$. Its first derivative at $t = 0$ vanishes. For every product missing two labels, AM–GM gives

$$\prod_{j \neq i,k} |z - \mu - t\eta_j| \leq e\big(|z-\mu|^2 + t^2 v\big)^{(N-2)/2},$$

because $(N/(N-2))^{(N-2)/2} \leq e$. Also $\sum_{i \neq k} |\eta_i \eta_k| \leq N^2 v$. Taylor's integral remainder therefore shows that

$$|q_P(z) - (z-\mu)^N| \leq 2N^2 v s_0^{N-2} \qquad (|z| \leq r_0),$$

provided $(r_0 + m)^2 + q < s_0^2$. The same argument applied to $\prod_j (1 - t\eta_j/(u-\mu))$ gives

$$|C_u| \leq e^{gn}, \qquad |C_u - 1| \leq 2N^2 v e^{gn}$$

under $q/[2(h-m)^2] < g$ and $e/[2(h-m)^2] < 2$. If $(r_0 + m)e^g < s_0$ and $e^g s_0^2 < 1$, it follows that

$$|q_P(z) - C_u(z-\mu)^N| \leq 4N^2 v s_0^{N-2} \qquad (|z| \leq r_0).$$

Combining its integral from 0 to w_u with (7.53) and $P(u) = 0$ gives

$$\begin{gathered} 0 = P(0) + C_u(u-\mu)^n(1+\theta_u) + E_u, \quad |\theta_u| \leq Jv, \\ |E_u| \leq 4n^3 v s_0^{n-3} + e^{gn}|\mu|^n. \end{gathered} \tag{7.56}$$

The condition $m < .006$ gives $|\mu|^n \leq Q(.006)^{n-2}$. The inner conditions are

$$\begin{aligned} &(r_0+m)^2 + q < s_0^2, \quad q/[2(h-m)^2] < g, \quad e < 4(h-m)^2, \\ &(r_0+m)e^g < s_0, \quad e^g s_0^2 < 1, \quad 4/s_0^3 < 6, \quad s_0/.99 < e_0, \quad m < .006, \\ &e^g(.006)/.99 < .01, \quad 6(1000)^3 e_0^{1000} + 10^6(.01)^{1000} < .01, \quad e_0(1001/1000)^3 < 1. \end{aligned} \tag{7.57}$$

They imply, for every $n \geq 1000$,

$$|E_u/P(0)| \leq Q(6n^3 e_0^n + 10^6(.01)^n) \leq Q/100. \tag{7.58}$$

Since $Jq < 1$ and $|E_u/P(0)| < 1$, we may take logarithms in (7.56). Using $\log|C_u(u-\mu)^n| = NL(u) + \log|u-\mu|$, we obtain

$$n\left|L(u) - \frac{1}{n}\log|P(0)|\right| \leq \left(12 + \frac{J}{1-Jq} + \frac{1}{99}\right) Q \leq 24Q.$$

The inequalities are strict when $Q > 0$.

The assumed inequality (7.59) implies $2h^2 - 1 > 0$. The edge has unit tangent $v_e = ie^{i\theta}$ and support coordinates $z = e^{i\theta}(h_e + it)$, with $h_e \geq h$ and $h_e^2 + t^2 = |z|^2 \leq 1$. Hence $h_e^2 - t^2 \geq 2h_e^2 - 1 > 0$, and

$$-\operatorname{Re}\frac{v_e^2}{z^2} = \frac{h_e^2 - t^2}{(h_e^2 + t^2)^2} \geq h_e^2 - t^2 \geq 2h_e^2 - 1 \geq 2h^2 - 1.$$

The identity

$$(z-\xi)^{-2} = z^{-2} + 2\xi z^{-3} + \frac{\xi^2(3z - 2\xi)}{z^3(z-\xi)^2}, \qquad 3z - 2\xi = z + 2(z-\xi),$$

therefore bounds the second derivative of L along the whole edge from below by

$$2h^2 - 1 - \frac{2m}{h^3} - q\left(\frac{1}{h^2\gamma^2} + \frac{2}{h^3\gamma}\right) > 0. \tag{7.59}$$

Thus L is convex on that edge. Its two endpoint estimates imply $nL(a) \leq \log|P(0)| + 24Q$. Since $L(a) > 0$, this is (7.50). The argument also covers $v = 0$. □

Lemma 7.9 (Annular cubic rigidity)**.** *Suppose all original zeros satisfy $\rho \le |z_i| \le 1$, where $0 < \rho \le 1$, and all critical points belong to $\{z : |z| \le 1,\ |z-1| \ge 1\}$. Suppose $C \ge 1$, $0 < q < 1$, $Q \le q$, and $|P(0)| \ge e^{-CQ}$. Then $Q = 0$, and $P(z) = z^n + c$ with $|c| = 1$, provided*

$$n \ge \left\lceil \max\left\{ \frac{256}{(1-q)^2},\ \frac{8Cq}{1-q},\ C\left(\frac{8}{\rho} + \frac{16}{\rho^2} + \frac{6}{\rho^3}\right) + 11 \right\} \right\rceil. \tag{7.60}$$

Proof. For $z = x + iy$ and $s = |z|^2$ the identity

$$\operatorname{Re}(-z - z^2 - z^3/4) - s/4 = (1+x)(3s/4 - x - x^2)$$

has nonnegative right side on the stated lens. Indeed $-1 \le x \le 1/2$, $s \ge x^2$, and $s \ge 2x$; use the first lower bound when $x \le 0$ and the second when $x \ge 0$. Writing $t_k = N^{-1}\sum_j \xi_j^k$, we deduce

$$Q \le 4|t_1| + 4|t_2| + |t_3|. \tag{7.61}$$

Assume $Q > 0$ and put $a_j = P^{(j)}(0)/P(0)$, $j = 1, 2, 3$. Differentiate the product for P' and apply AM–GM to every remaining subproduct. Since $(N/(N-j+1))^{(N-j+1)/2} \le e^{(j-1)/2} < 3$,

$$|a_j| \le 3e^{Cq} n^3 Q^{(n-3)/2}.$$

Set $\delta = 1 - q$. The first two conditions in (7.60) imply

$$3e^{Cq} n^4 q^{(n-5)/2} < 1.$$

$\log n \le \sqrt{n}$ and $\log q \le -\delta$ give

$$\begin{aligned} \log 3 + Cq + 4\log n + \tfrac{n-5}{2}\log q &\le \log 3 + Cq - \delta n/4 + 5\delta/2 \\ &\le \log 3 - \delta n/8 + 5\delta/2 \\ &\le \log 3 - 32/\delta + 5\delta/2 < 0. \end{aligned}$$

It follows that $|a_j| \le Q/n$.

Let $R_k = \sum_i z_i^{-k}$ and $s_k = n^{-1}\sum_i z_i^k$. The exact identities

$$R_1 = -a_1, \quad R_2 = a_1^2 - a_2, \quad R_3 = -(a_3 - 3a_1a_2 + 2a_1^3)/2$$

give $|R_k| \le 2Q/n$ for $k = 1, 2, 3$. For an original modulus $s \in [\rho, 1]$, integrate the derivative of $e^{kt} - e^{-kt}$ from 0 to $t = -\log s$ to obtain

$$|z_i^k - \overline{z_i^{-k}}| = s^{-k} - s^k \le 2k\rho^{-k}(-\log s).$$

Sum over all original zeros. The radial budget $\sum_i -\log|z_i| \le CQ$ implies

$$|s_k| \le A_k Q/n, \qquad A_k = 2kC\rho^{-k} + 1, \quad k = 1, 2, 3. \tag{7.62}$$

Here $n \ge 256$ justifies replacing the term $2/n$ by one.

Newton's identities for P and P'/n give

$$t_1 = s_1, \qquad t_2 = \frac{(n-2)s_2 + s_1^2}{n-1}, \qquad t_3 = \frac{(n-3)s_3 + 3s_1s_2 - s_1^3}{n-1}.$$

Using $|s_1| \le 1$, (7.61), and (7.62), we obtain

$$Q \le \left(B_1 + 9 + \frac{E_1}{n-1}\right)\frac{Q}{n},$$

where

$$B_1 = C(8/\rho + 16/\rho^2 + 6/\rho^3), \qquad E_1 = 10C/\rho + 12C/\rho^2 + 8.$$

But $B_1 + 10 - E_1 = C(-2/\rho + 4/\rho^2 + 6/\rho^3) + 2 > 0$. Since $n \ge B_1 + 11$, the coefficient in parentheses is less than $B_1 + 10 < n$, a contradiction. Thus $Q = 0$. Now $P' = nz^{n-1}$, and the product hypothesis and the unit-disk roots force the constant term of P to have modulus one. □

Proposition 7.10 (Boundary and small displacement ranges)**.** *For every $n \geq 6251$, a hull point satisfying (7.2) is impossible in each of the following cases:*

	a	r
B	$[.975, 1]$	*arbitrary*
A	$[.85, 1]$	$[0, .02]$
G1	$[.96, 1]$	$[0, .04]$
G2	$[.965, 1]$	$[0, .045]$.

(7.63)

The latter three conclusions hold for any choice of actual rightmost zero satisfying the indicated displacement bound.

Proof. We first treat a boundary witness $a < 1$ which is not an original zero. The normalization and moment proof above gives Table 5. Apply Lemma 7.5 using the weak support radii in Table 6. For row A, repeat the support test and the actual-parameter moment inequality in the following order:

lower a	new upper Q	next support radius
.9305	.13831	.9454
.9454	.11036	.9478
.9478	.10582	.9481
.9481	.10525	–.

(7.64)

Each new moment bound is obtained from (7.15) by increasing only the lower bound for its explicit a^2 term. The error $\mathcal{E}(.02; .85, N)$ is retained throughout, and the caps for $B, |\mu|$ remain unchanged. Put $a_1 = .9481$, $q = .10525$ in row A; in the other three rows put $a_1 = a_0$, $q = q_0$.

At b the finite resolvent expansion

$$\frac{S}{N} = \frac{1}{b} + \frac{\mu}{b^2} + \frac{1}{N}\sum_j \frac{\xi_j^2}{b^2(b - \xi_j)}$$

gives

$$H > Nc_f, \quad E_b/H < V, \quad d^{-1} < M, \qquad c_f = 1 - \frac{m}{a_1} - \frac{q}{a_1(1-e)}, \quad V = \frac{1}{c_f(1-e)^2}, \quad M = \frac{1}{1-e}. \tag{7.65}$$

The listed parameters give $c_f > 0$. Since $1 - m/|b| - q/(|b|d) > c_f$ and $|b| \leq 1$,

$$H \geq \frac{N}{|b|}\left(1 - \frac{m}{|b|} - \frac{q}{|b|d}\right) > Nc_f.$$

Use $H_0 = N_0 c_f$, $\nu = .99$, $W = 4$ and the radii R in Table 6. All comparisons (7.40)–(7.42) hold, as does $\nu H_0 \widehat{r} > 4\pi$. Equation (7.24) supplies $J_0 = 8/a_0$, with the original a_0 also in row A. Lemma 7.6 consequently gives K.

For B, G1, and G2 the direct comparisons $a_0^2 > h^2(1 + K^2)$ show that this particular edge has distance greater than the listed h. Row A uses its support radius .9481. All four rows satisfy $1 - K - \sqrt{1 - h^2} > \gamma$ and the four corner inequalities (7.49) with $r_0 = .9$. Lemma 7.7 therefore supplies the gaps on the whole edge and the endpoint radial paths.

The endpoint parameters in Table 7 are positive, and $r_0 = .9 < h$ in all four rows. All the hypotheses of Lemma 7.8 hold. In particular the Hessian bounds and endpoint coefficients in Table 8 are strictly positive and strictly less than 24, respectively. We obtain $-\log|P(0)| < 24Q$. The additional annulus radii ρ_f in the table follow from

$$\rho_f^2 + 2\rho_f m + q < s_f^2, \qquad \rho_f < s_f, \qquad s_f/p_f < s_*/p,$$

and the already proved whole-degree bound $n(s_*/p)^n < 1$. Thus every original zero satisfies $|z_i| > \rho_f$.

Every critical point belongs to the lens in Lemma 7.9, since

$$|\xi - 1|^2 - |\xi - a|^2 = (1 - a)(1 + a - 2\,\mathrm{Re}\,\xi) > 0.$$

The three bounds in (7.60), with $C = 24$, $\rho = \rho_f$, and the final q, are all less than 900. Lemma 7.9 gives $Q = 0$. All critical points then equal zero, contradicting (7.2) for $a < 1$.

The weak support test already excludes a hull of dimension less than two. A boundary vertex is an original zero and was excluded by the energy inequality; every other boundary point lies on a genuine edge as used above. If $a = 1$, strict convexity of $\overline{\mathbb{D}}$ makes 1 an original zero, again impossible. Finally, for an interior witness move outward on its real ray to the boundary point A. Since $a \le A \le x$,

$$|A - \xi|^2 - |a - \xi|^2 = (A - a)(A + a - 2\,\mathrm{Re}\,\xi) \ge 0, \qquad |b - A| \le |b - a|.$$

The strict hole and each applicable displacement hypothesis persist. The boundary result therefore proves all of (7.63). □

Table 6. Support and geometric parameters for the four exclusion rows.

Row	weak h	final h	R	γ	ρ_f	s_f	final q
B	0.9239	0.95027	0.113	0.455	0.955	0.993	0.06001
A	0.9305	0.9481	0.061	0.583	0.934	0.993	.10525
G1	0.9351	0.94709	0.092	0.51	0.943	0.993	0.08531
G2	0.9308	0.94828	0.1	0.49	0.947	0.993	0.07689

Table 7. Endpoint parameters, with $r_0 = .9$ in every row.

Row	C_0	v_0	β	J	s_0	g	e_0
B	3.056	0.9932	0.8086	4.952	0.93797	0.03371	0.947455
A	2.369	0.9973	0.7468	4.495	0.95869	0.0589	0.968384
G1	2.716	0.9955	0.7626	4.954	0.94956	0.048	0.959162
G2	2.83	0.9947	0.7759	4.984	0.94572	0.04323	0.955283

Table 8. Outward numerical bounds for the boundary geometry. The comparisons themselves use rational intervals.

Row	upper bound for K	lower Hessian bound	upper endpoint coefficient
B	.229122900	.165248328	19.055896748
A	.092635534	.024898554	20.541111328
G1	.164970033	.026449841	20.590324175
G2	.187982215	.064807027	20.090774931

The first support comparison and each feedback comparison check both (7.35) at n_0 and its slope

$$\Lambda(1 - h)^2 - 2B_0 > 0.$$

The four inverse applications check (7.40)–(7.42) for both signs of the strip, as well as $s < 1$, $\nu\widehat{r} - 4\pi/H_0 > 0$, and $L_0 > e$. The geometric comparisons are $h > .9$, $1 - K - \sqrt{1 - h^2} > \gamma$, $c > 0$, and the four instances of (7.49). The endpoint checks are exactly (7.52), (7.54), (7.57), the stricter lower bound .01 in (7.59), and $12 + J/(1 - Jq) + 1/99 < 24$. The final annulus and the three

inequalities in (7.60) are checked as in the proof. These tests, together with (7.26)–(7.34), form the boundary certificate.

As n increases, the fixed moment caps remain valid, while the field Nc_f increases. Thus the inverse corner inequality improves and ϵ decreases. In (7.46), E_0 decreases and L_0 increases, so K decreases. The geometric inequalities therefore persist. The outer rate $v_0N/n - C_0q$ increases, and the inner errors have the explicit decreasing consecutive-degree ratios in (7.57). This proves the boundary conclusions on the whole infinite degree range, including all stated endpoints.

7.5. **A direct entrance for the compact range.**

Lemma 7.11 (Crescent entrance)**.** *Under* (7.2)*, if* $a \ge .1$ *and* $n \ge 6251$*, then*

$$D := Hd > \frac{5N}{14}\frac{c}{1-c+c^2} > 117.17, \qquad c = x/2. \tag{7.66}$$

Thus Propositions 4.11 and 5.6 apply.

Proof. A closest point to $b = x + iy$, $x \ge a$, $y \ge 0$, in the closed crescent $\{|\xi| \le 1,\ |\xi - a| \ge 1\}$ is $a/2 + i\sqrt{1 - a^2/4}$. Indeed a minimizer must be on one of its two circular boundary arcs. On $|\xi| = 1$, maximizing $\mathrm{Re}(\bar{b}\xi)$ over $\mathrm{Re}\,\xi \le a/2$ gives the stated endpoint, because the direction of b lies to its right. On $|\xi - a| = 1$, the same argument maximizes $\mathrm{Re}((\bar{b} - a)(\xi - a))$ over $\mathrm{Re}(\xi - a) \le -a/2$; $b - a$ has nonnegative real and imaginary parts, giving the same endpoint. Since the strict hole excludes that endpoint,

$$d^2 > 1 + x^2 + y^2 - ax - 2y\sqrt{1 - a^2/4}.$$

The right side decreases in a for $0 < a \le x \le 1$, since its derivative is $-x + ay/[2\sqrt{1 - a^2/4}] < 0$. Set $c = x/2$, $v = \sqrt{1 - c^2} - y \ge 0$, and $G = \sqrt{c^2 + v^2}$. Comparing at $a = x$ gives $d > G$. Also, if $p = 1 - |b|^2$, then

$$p = -3c^2 + 2v\sqrt{1 - c^2} - v^2 \le 2v - 3c^2 - v^2.$$

The exact radial identity, obtained by summing $2\,\mathrm{Re}(b/(b - \xi)) + (1 - |b|^2)/|b - \xi|^2 = 1 + (1 - |\xi|^2)/|b - \xi|^2$, is

$$N = 2\,\mathrm{Re}(bS) + (1 - |b|^2)E_b - \sum_j \frac{1 - |\xi_j|^2}{|b - \xi_j|^2}. \tag{7.67}$$

Furthermore $\mathrm{Re}\,S > cE_b$, because $a \le x$. Using $|b| \le 1 - p/2$ in (7.67), we obtain

$$\frac{N}{H} \le 2 + p(c^{-1} - 1), \qquad Hd > \frac{NcG}{2c + (1 - c)p}.$$

The second inequality is strict because $d > G$. By Cauchy–Schwarz and the identity $(3c^2 + v^2)^2 - 8c^2(c^2 + v^2) = (v^2 - c^2)^2$,

$$\begin{aligned}\frac{2c + (1 - c)p}{G} &\le \frac{2c + 2(1 - c)v}{G} - (1 - c)\frac{3c^2 + v^2}{G} \\ &\le 2\sqrt{2 - 2c + c^2} - 2\sqrt{2}\,c(1 - c).\end{aligned}$$

For $.05 \le c \le 1/2$, $\sqrt{2 - 2c + c^2} < 7/5 < \sqrt{2}$. This proves the first inequality in (7.66). The function $c/(1 - c + c^2)$ increases on that interval. Its value at $c = .05$, with $N = 6250$, proves the second. □

Proposition 7.12 (The bridge interval)**.** *For every* $n \ge 6251$*, no hull witness satisfies* (7.2) *with* $.1 \le a \le .11$.

Proof. Lemma 7.11 gives $D > 100$. Throughout the argument, C and γ are the quantities in (7.5). The logarithmic form of Lemma 7.2 is

$$-\log \mathcal{B}_P(\gamma) + CE_b/2 < \log(8/C). \tag{7.68}$$

The range $|b| \le .2$. Here $d > .8$ and Proposition 4.11 gives $E_b d/H < 9/4$. In particular $V = E_b/H < 9/(4 \cdot .8)$ and $H > N(.8)/(9/4)$, using $E_b \ge N$. Apply (7.40)–(7.42) with $R = .1$, $\nu = .99$, $W = 6$, $e_W = 1/400$, $M = 1/.8$, and $H_0 = N_0(.8)/(9/4)$. The image radius is $\widehat{r} = 379/4480$, and the count (7.44) is greater than $1831/32$ at the floor. Since $|\gamma| \le |b|$, each counted zero has factor at most

$$\frac{R + .11}{1 - (.2 + R)(.2)} < \frac{1}{4}.$$

As $e < 4$, its logarithmic weight exceeds one. The count divided by n increases, and exceeds $1/120$ for all $n \ge 6251$.

The actual-zero variance inequality (3.11) gives $d^2 \le 1 - (1 - |b|^2)/n$, and therefore $d < 1 - .47/n$. Since $d + r > 1$, we have $r > .4/n$. Since $a \ge .1$ and $u = \min(r^2, 1/2) > (.4/n)^2$, the monotonicity of $u(1-u)$ on $[0, 1/2]$ gives

$$C \ge a^2 u(1-u) > (.1)^2 \left(\frac{.4}{n}\right)^2 \left(1 - \left(\frac{.4}{n}\right)^2\right) > \frac{1}{1000n^2}.$$

Hence

$$\log(8/C) < 10 + 2\log n < n/120 < -\log \mathcal{B}_P(\gamma),$$

a contradiction to (7.68). For the middle inequality, at $n = 6251$ one has $\log n < 9$ and $n/120 > 28$; its difference increases since $1/120 > 2/n$.

The range $.2 < |b| \le .8$. Now $r > .09$, so $C > C_m = (.1)^2(.09)^2(1 - (.09)^2)$. The field conclusion of Proposition 5.6, with $D > 100$ from Lemma 7.11, gives

$$H > \frac{N - (1899/20)/(.2)}{2(1 + .11)}.$$

Use $d > .2$, $V = 9/(4 \cdot .2)$, $M = 1/.2$, $R = .04$, and the same ν, W, e_W. The image radius is $.0295$, the count at the floor exceeds $26254139/1184000$, and every counted factor is at most

$$\frac{.04 + .11}{1 - (.8 + .04)(.8)} < .46.$$

The comparison $e^{.75} < 1/.46$ gives a product budget exceeding $16.63 > 12$, while $e^{12} > 8/C_m$. This again contradicts (7.68); its count increases with N.

The range $.8 < |b| \le .94$. Here $r > .69$, so $u \ge .4761$, $1/2 \le t \le .5239$, and

$$C > C_u = (.1)^2(.4761)(.5239) = 24942879/10^{10}.$$

Retain only the factor of the original zero b itself. Its numerator is $ta \le .057629$, while the real part of its denominator is $1 - |b|^2 + tax \ge 1 - .94^2 + .5(.1)^2 = .1214$. Thus that factor is at most $57629/121400 < .475$, and its logarithmic weight exceeds $.74$. Since

$$.74 + 6250C_u/2 > 8.08, \qquad e^{8.08} > 8/C_u,$$

the product factor and the energy term for the same C contradict (7.68).

The range $|b| > .94$. Now $r > .83$, hence $u = t = 1/2$. Put $a_* = .03$, $x_* = .1$, and $s = 3a_*/4$. Concavity of $a(x - 3a/4)$ on $[a_*, x]$, and $x > 3a_*$, give

$$C = a(x - 3a/4) \ge a_*(x - s).$$

In fact the value at $a = x$ exceeds the value at $a = a_*$ because $(x - a_*)(x - 3a_*) > 0$. The comparison $e^{8.15} > 8/[a_*(x_* - s)]$ and (7.68) imply $E_b/N < k/(x - s)$, where $k = 2(8.15)/(Na_*)$. Also $H^2 \le NE_b$ by Cauchy–Schwarz. The phase conclusion of Proposition 5.6 is $|\operatorname{Im} S| < (21/4)E_b/H$;

the half-plane bound $E_b/H < 2/x$ follows directly from $\operatorname{Re} S > (x/2)E_b$. Use these estimates in the exact identity (7.67). Writing $p_{\max} = 1 - .94^2$, we find the necessary inequality

$$1 < 2x\sqrt{\frac{k}{x-s}} + \frac{p_{\max}k}{x-s} + \frac{4(21/4)}{Nx}. \tag{7.69}$$

Under $v = \sqrt{x-s}$ the right side is a positive linear combination of $v + s/v$, v^{-2} and $(v^2+s)^{-1}$. Each is convex throughout the interval in question: for the last term this follows from $3v^2 > s$. Its maximum is therefore at $x = x_*$ or $x = 1$. At each of those endpoints, with $N = N_0$, the exact comparisons

$$R_x := 1 - \frac{p_{\max}k}{x-s} - \frac{21}{Nx} > 0, \qquad R_x^2 > \frac{4x^2k}{x-s}$$

make the right side of (7.69) less than one. Every term decreases with N, which proves the contradiction in the whole high-modulus range. The four cases include their common endpoints and exhaust every possible b. □

7.6. **Nested original-zero counts in the middle range.**

Proposition 7.13 (The compact range)**.** *For every $n \ge 6251$, no hull witness satisfies (7.2) with $.1 \le a \le .975$.*

Proof. We prove the part $.11 \le a \le .975$ by the closed cover in Table 9; Proposition 7.12 supplies the remaining interval. A row specifies $A \le a \le B$ and $r_-(n) \le r \le r_+$. All numbers in the tables are exact rationals.

We first construct a lower bound $H(n) = \alpha(n-1) - \beta$ for H, together with constants V, M as in (7.37). There are two constructions.

For a trace row put

$$M = \frac{1}{1-r_+}, \qquad c = \frac{A}{2(1+r_+)^2}.$$

Lemma 7.11 gives $D > 100$. Since $1 - |b|^2 \le 2d$, (7.67) and $E_b d/H < 9/4$ give $H > 2N/13$ and $D > 2N(1-r_+)/13$. Set

$$D_f = \frac{2N_0(1-r_+)}{13} - 10^{-6}. \tag{7.70}$$

For the six trace rows $D_f \ge 201$. The Fine signed-moment inequality (5.14), derived from Lemma 5.3, on its domain $E_b/H^2 \le 7/625$, $(Hd)^{-1} \le 1/200$, applies here: $E_b/H^2 < (9/4)/201 < 7/625$. For its ratio consequence, set $v_j = [H(b-\xi_j)]^{-1}$, $q_b = E_b/H^2$, $\mu_b = (Hd)^{-1}$, and $y_b = E_b d/H = q_b/\mu_b$. Then

$$q_b - 2\mu_b \le -\operatorname{Re}\sum_j v_j^2 < (9/4)q_b^2 + 13q_b\mu_b,$$

because $0 \le \operatorname{Re} v_j \le \mu_b$ and $\sum_j \operatorname{Re} v_j \le 1$. Consequently

$$(9/4)y_b^2 - (D-13)y_b + 2D > 0. \tag{7.71}$$

The quadratic is strictly decreasing on $[2, 9/4]$ for $D \ge 201$. The values κ in Table 10 satisfy

$$2 < \kappa < 9/4, \qquad (9/4)\kappa^2 - (D_f - 13)\kappa + 2D_f < 0. \tag{7.72}$$

As its value at $\kappa > 2$ decreases with D, and $y_b < 9/4$ is already known, (7.71) implies $y_b < \kappa$. We may therefore take

$$V = \min(2/A, \kappa/(1-r_+)), \qquad H_{\text{init}} = \max(N_0A/2, 2N_0/13).$$

The comparisons $V/H_{\text{init}} < 7/625$ and $M/H_{\text{init}} < 1/270$ justify the signed estimate

$$\operatorname{Re}\sum_j (b-\xi_j)^{-2} > -(9/4)V^2 - 13VM.$$

Each real reciprocal coordinate $X_j = \operatorname{Re}(b-\xi_j)^{-1}$ belongs to (c, M):

$$2(a-\operatorname{Re}\xi) - a|a-\xi|^2 = a(1-|\xi|^2) + (1-a^2)(a-2\operatorname{Re}\xi) > 0,$$

and $|b-\xi| < (1+r_+)|a-\xi|$. Using $X_j^2 \le (c+M)X_j - cM$ and $E_b \ge N$ in $E_b = 2\sum_j X_j^2 - \operatorname{Re}\sum_j (b-\xi_j)^{-2}$ gives

$$H > \alpha N - \beta, \quad \alpha = \frac{1+2cM}{2(c+M)}, \qquad \beta = \frac{(9/4)V^2 + 13VM}{2(c+M)}. \tag{7.73}$$

For a moment row take (ℓ, w, κ) from M0, M1, or M2 as indicated in Table 9. Define, at the fixed floor,

$$\begin{aligned} e_r &= \frac{8r_+}{AN_0} + \frac{r_+^2}{2(1-r_+)(1-2r_+)}, \\ q_m &= 1 - A^2 + w(\ell/N_0 + e_r) + 2/n_0^2, \qquad m_m = \kappa(\ell/n_0 + e_r) + 1/n_0^2. \end{aligned} \tag{7.74}$$

These are upper bounds for $Q, |\mu|$ throughout the row and for every larger degree, by Proposition 7.4. Set

$$\alpha = 1 - \frac{m_m}{A} - \frac{q_m}{A(1-r_+)}, \quad \beta = 0, \quad V = \frac{1}{\alpha(1-r_+)^2}, \quad M = \frac{1}{1-r_+}. \tag{7.75}$$

The finite comparisons give $\alpha > 0$; the resolvent estimate (7.65) gives $H > \alpha N$.

For each row and each indicated integer j, use $R_j = j/1000$ and the parameters ν_j, W_j specified in Tables 12 and 13. All intermediate integers between the displayed endpoints are included. Use (7.38)–(7.39) with this row's V, M, and set $H_0 = \alpha N_0 - \beta$. Every one of the comparisons (7.40)–(7.42) holds, together with $\nu_j H_0 \widehat{r}_j > 22/7$. Equation (7.44) consequently gives, for the actual number $\mathcal{N}_j$ of original zeros in $D(b, R_j)$,

$$\mathcal{N}_j > k_j(n) := \nu_j(7/22)\widehat{r}_j(\alpha(n-1) - \beta) - 2 > 0. \tag{7.76}$$

We estimate their Blaschke factors at γ from (7.5). In every middle row $r^2 \le r_+^2 < 1/2$, so $t = 1 - r^2$. For a counted zero $z = b + v$, rightmostness gives $\operatorname{Re} v \le 0$. It follows that

$$|z-\gamma|^2 = |v + ta|^2 \le R_j^2 + B^2.$$

Substitution of $C = ta[2x - (1+t)a]$ gives the denominator identity

$$\operatorname{Re}(\bar{b}\gamma) = \frac{C(1+r^2)}{2(1-r^2)} + r^2 + \frac{a^2r^2(1-r^2)}{2}. \tag{7.77}$$

Also $\operatorname{Re}(\overline{v}\gamma) = (x-ta)\operatorname{Re} v + y \operatorname{Im} v \le r_+ R_j$; here $x - ta \ge 0$. Thus, with

$$\Delta_j = \frac{C_*(1+r_+^2)}{2(1-r_+^2)} + (1 + B^2/2)r_+^2 + R_j r_+, \tag{7.78}$$

we have $\operatorname{Re}(\overline{z}\gamma) < \Delta_j$. The row comparisons give $0 < \Delta_j < 1$ and hence

$$\left|\frac{\gamma - z}{1 - \overline{z}\gamma}\right| \le \frac{\sqrt{B^2 + R_j^2}}{1 - \Delta_j} \le \eta_j < 1. \tag{7.79}$$

Here η_j is the smallest multiple of 10^{-12} whose square is at least $(B^2 + R_j^2)/(1-\Delta_j)^2$.

Put $t_j = (1-\eta_j)/(1+\eta_j)$ and define the rational weights by

$$w_j = 10^{-12}\left\lfloor 10^{12}\left(2\sum_{k=0}^{23}\frac{t_j^{2k+1}}{2k+1}\right)\right\rfloor. \tag{7.80}$$

All omitted terms of the logarithmic series are positive, so $0 < w_j < -\log\eta_j$. For each row the w_j decrease and the $k_j(n_0)$ increase with j. The physical disks are nested, so finite Abel summation gives

$$\begin{aligned} -\log\mathcal{B}_P(\gamma) &> w_J\mathcal{N}_J + \sum_{j<J}(w_j - w_{j+1})\mathcal{N}_j \\ &> w_J k_J(n) + \sum_{j<J}(w_j - w_{j+1})k_j(n) = \sum_j w_j(k_j(n) - k_{j-1}(n)) =: \mathcal{A}(n), \end{aligned} \tag{7.81}$$

where $k_0 = 0$ and the indices are understood in their successive order within a row. The first inequality assigns to each counted zero the largest available weight. The second uses nonnegative coefficients and lower bounds on nested disk counts.

Set

$$C_-(n) = A^2 r_-(n)^2(1 - r_-(n)^2).$$

Because $r^2 < 1/2$, $C \geq C_-(n)$. For a fixed radial lower endpoint retain the energy bonus $(n-1)C_-/2$ in (7.68); for a shrinking lower endpoint $r_-(n) = c/n$ discard that positive term. The targets T_0 in Table 11 satisfy

$$e^{T_0} > \frac{8}{C_-(n_0)}, \qquad \mathcal{A}(n_0) + \text{bonus} > T_0. \tag{7.82}$$

Put $\lambda_j = \nu_j(7/22)\widehat{r}_j$ and

$$W_* = \sum_j w_j(\lambda_j - \lambda_{j-1}), \qquad \lambda_0 = 0.$$

The exact telescoping identity is

$$\mathcal{A}(n) = W_*(\alpha(n-1) - \beta) - 2w_{\text{first}}.$$

The comparisons verify that $\mathcal{A}(n) + \text{bonus} = A_* n - B_*$ with $A_*, B_* > 0$. For fixed r_- the left side increases, while $\log(8/C_-)$ is constant, so (7.82) contradicts (7.68) for every $n \geq n_0$. For $r_-(n) = c/n$, $C_-(n) \geq C_-(n_0)(n_0/n)^2$, while $\mathcal{A}(n)/n$ increases. As $T_0 > 2$ in these four rows,

$$\log\frac{8}{C_-(n)} < T_0 + 2\log(n/n_0) \leq T_0(n/n_0) < \mathcal{A}(n),$$

again a contradiction on the entire degree range. The field entrances and every inverse contour condition also improve with n, since $H(n)$ increases and the other contour constants are fixed.

It remains to verify that the closed cells just excluded leave no gaps. For each interval in a, its final far radius r_f is listed below:

A	.11	.2	.45	.75	.85	.9	.93	.95	.96	.965
r_f	.65	.27	.115	.069	.06	.057	.055	.054	.053	.053.

(7.83)

Each satisfies $r_f^2 < 1/2$ and $A^2 r_f^2(1 - r_f^2) > C_*$. Lemma 7.2 therefore excludes every $r \geq r_f$, including $r^2 \geq 1/2$. For the four shrinking lower endpoints, let $\overline{B} = B + .001$. If $r \leq c/n$, then $|b| < \overline{B}$, and (3.11) gives

$$d^2 < 1 - (1 - \overline{B}^2)/n, \qquad d < 1 - \frac{1-\overline{B}^2}{2n} < 1 - c/n.$$

Here the four exact comparisons are $B + c/n_0 < \overline{B}$ and $c < (1-\overline{B}^2)/2$. This contradicts $d + r > 1$. For $a \geq .85$ the interval $r \leq .02$ is excluded by row A of Proposition 7.10. For $a \geq .96, r \leq .04$,

and $a \geq .965, r \leq .045$, use rows G1 and G2, respectively. All witness and radial endpoints are shared by closed exclusions. Together with Proposition 7.12, this proves Proposition 7.13. □

TABLE 9. The thirteen closed middle cells. Every integer in a displayed j interval is used, with $R_j = j/1000$.

Row	A	B	$r_-(n)$	r_+	j interval	number	field
Ia	0.11	0.2	$0.45/n$	0.1	5–266	262	Trace
Ib	0.11	0.2	0.1	0.65	6–91	86	Trace
II	0.2	0.45	$0.35/n$	0.27	5–212	208	Trace
III	0.45	0.75	$0.2/n$	0.115	5–262	258	Trace
IV	0.75	0.85	$0.12/n$	0.069	5–278	274	Trace
V	0.85	0.9	0.02	0.06	5–281	277	Trace
VI	0.9	0.93	0.02	0.057	5–299	295	M0
VII	0.93	0.95	0.02	0.055	5–244	240	M1
VIIIa1	0.95	0.96	0.02	0.025	5–248	244	M2
VIIIa2	0.95	0.96	0.025	0.035	5–236	232	M2
VIIIb	0.95	0.96	0.035	0.054	5–211	207	M2
IXa	0.96	0.965	0.04	0.053	5–193	189	M2
IXb	0.965	0.975	0.045	0.053	5–150	146	M2

TABLE 10. Trace-row ratio parameters. In each row D_f is given by (7.70), so that the same lower bound is used in (7.72).

Row	Ia	Ib	II	III	IV	V
κ	2.04150406	2.11136288	2.0514852	2.04222636	2.04008696	2.0396935

TABLE 11. Targets and rational lower bounds for the final compact margins.

Row	T_0	$\mathcal{A}(n_0) + \text{bonus} - T_0 >$
Ia	25.574	193.4195
Ib	11.111	8.6177
II	24.88	44.1758
III	24.378	16.8682
IV	24.378	3.1044
V	10.23	7.9635
VI	10.116	5.4864
VII	10.051	.5545
VIIIa1	10.008	.4089
VIIIa2	9.562	.7142
VIIIb	8.89	1.4735
IXa	8.602	1.6317
IXb	8.356	.4620

The next two tables specify every contour choice by intervals of integers. The notation $[u, v] : k$ assigns the value k to every $j = u, u+1, \ldots, v$; a single integer denotes a singleton interval. Assignments in a row are disjoint and cover exactly its j interval in Table 9.

Table 12: The exact values of $10^4\nu_j$.

Row	Assignments
Ia	$5:9975,\quad 6:9982,\quad 7:9987,\quad 8:9989,\quad 9:9991$ $10:9993,\quad [11,12]:9994,\quad [13,14]:9995,\quad [15,17]:9996,\quad [18,24]:9997$ $[25,257]:9998,\quad [258,264]:9997,\quad [265,266]:9996$
Ib	$6:9891,\quad 7:9919,\quad 8:9937,\quad 9:9950,\quad 10:9959$ $11:9965,\quad 12:9971,\quad 13:9975,\quad 14:9978,\quad 15:9980$ $16:9982,\quad 17:9984,\quad 18:9986,\quad 19:9987,\quad 20:9988$ $21:9989,\quad [22,23]:9990,\quad [24,25]:9991,\quad [26,27]:9992,\quad [28,30]:9993$ $[31,34]:9994,\quad [35,40]:9995,\quad [41,48]:9996,\quad [49,52]:9990,\quad [53,57]:9991$ $[58,64]:9992,\quad [65,68]:9993,\quad [69,70]:9986,\quad [71,75]:9987,\quad [76,78]:9988$ $[79,81]:9980,\quad [82,84]:9981,\quad 85:9971,\quad [86,88]:9972,\quad [89,90]:9961$ $91:9932$
II	$5:9967,\quad 6:9976,\quad 7:9982,\quad 8:9986,\quad 9:9989$ $10:9990,\quad 11:9992,\quad 12:9993,\quad [13,14]:9994,\quad [15,16]:9995$ $[17,20]:9996,\quad [21,29]:9997,\quad [30,188]:9998,\quad [189,206]:9997,\quad [207,211]:9996$ $212:9994$
III	$5:9980,\quad 6:9986,\quad 7:9989,\quad 8:9991,\quad 9:9993$ $10:9994,\quad [11,12]:9995,\quad [13,15]:9996,\quad [16,21]:9997,\quad [22,253]:9998$ $[254,262]:9997$
IV	$5:9984,\quad 6:9989,\quad 7:9991,\quad 8:9993,\quad 9:9994$ $10:9995,\quad [11,13]:9996,\quad [14,19]:9997,\quad [20,276]:9998,\quad [277,278]:9997$
V	$5:9985,\quad 6:9989,\quad 7:9992,\quad 8:9993,\quad 9:9994$ $10:9995,\quad [11,13]:9996,\quad [14,18]:9997,\quad [19,279]:9998,\quad [280,281]:9997$
VI	$5:9989,\quad 6:9992,\quad 7:9994,\quad 8:9995,\quad [9,10]:9996$ $[11,15]:9997,\quad [16,299]:9998$
VII	$5:9991,\quad 6:9993,\quad 7:9995,\quad [8,9]:9996,\quad [10,13]:9997$ $[14,244]:9998$
VIIIa1	$5:9992,\quad 6:9994,\quad 7:9995,\quad [8,9]:9996,\quad [10,12]:9997$ $[13,248]:9998$
VIIIa2	$5:9992,\quad 6:9994,\quad 7:9995,\quad [8,9]:9996,\quad [10,12]:9997$ $[13,236]:9998$
VIIIb	$5:9992,\quad 6:9994,\quad 7:9995,\quad [8,9]:9996,\quad [10,13]:9997$ $[14,211]:9998$
IXa	$5:9992,\quad 6:9994,\quad 7:9995,\quad 8:9996,\quad [9,12]:9997$ $[13,193]:9998$
IXb	$5:9992,\quad 6:9994,\quad 7:9995,\quad 8:9996,\quad [9,12]:9997$ $[13,150]:9998$

Table 13: The exact values of W_j.

Row	Assignments
Ia	$[5,129]:1,\quad [130,184]:2,\quad [185,212]:3,\quad [213,233]:4,\quad [234,248]:5$ $[249,257]:6,\quad [258,264]:8,\quad [265,266]:10$
Ib	$[6,48]:1,\quad [49,68]:2,\quad [69,78]:3,\quad [79,84]:4,\quad [85,88]:5$

Row	Assignments
	$[89,90]:6,\quad 91:8$
II	$[5,104]:1,\quad [105,149]:2,\quad [150,171]:3,\quad [172,188]:4,\quad [189,199]:5$ $[200,206]:6,\quad [207,211]:8,\quad 212:10$
III	$[5,127]:1,\quad [128,181]:2,\quad [182,209]:3,\quad [210,229]:4,\quad [230,244]:5$ $[245,253]:6,\quad [254,261]:8,\quad 262:10$
IV	$[5,134]:1,\quad [135,191]:2,\quad [192,220]:3,\quad [221,242]:4,\quad [243,257]:5$ $[258,268]:6,\quad [269,276]:8,\quad [277,278]:10$
V	$[5,135]:1,\quad [136,193]:2,\quad [194,222]:3,\quad [223,244]:4,\quad [245,260]:5$ $[261,270]:6,\quad [271,279]:8,\quad [280,281]:10$
VI	$[5,181]:1,\quad [182,253]:2,\quad [254,289]:3,\quad [290,299]:4$
VII	$[5,195]:1,\quad [196,244]:2$
VIIIa1	$[5,217]:1,\quad [218,248]:2$
VIIIa2	$[5,213]:1,\quad [214,236]:2$
VIIIb	$[5,204]:1,\quad [205,211]:2$
IXa	$[5,193]:1$
IXb	$[5,150]:1$

7.7. **Finite arithmetic verification.** For the boundary calculations, all rational expressions in Tables 2–7 and (7.64) are evaluated with interval endpoints in $\mathbb{Q}$. The following rules make the arithmetic finite and reproducible. For $0 \le x \le 4$, put $S_{96}(x) = \sum_{j=0}^{96} x^j/j!$. Then

$$S_{96}(x) < e^x < S_{96}(x) + \frac{x^{97}}{97!}\frac{1}{1-x/98} \quad (x>0);$$

negative arguments are handled by reciprocal intervals. For logarithms, first multiply the argument by a power of two to place it in $[1,2)$. If $t=(x-1)/(x+1)$ in that interval, then

$$2\sum_{j=0}^{95}\frac{t^{2j+1}}{2j+1} \le \log x \le 2\sum_{j=0}^{95}\frac{t^{2j+1}}{2j+1} + \frac{2t^{193}}{193(1-t^2)}.$$

Apply the same rule to $\log 2$ when undoing the power-of-two change. Square roots are enclosed between consecutive multiples of 10^{-56}, using integer square roots after multiplying the radicand by 10^{112}. For π, use

$$\pi = 16\arctan(1/5) - 4\arctan(1/239)$$

with 100 alternating terms for each arctangent and the first omitted term as its remainder bound. These rules suffice to verify every strict comparison in the boundary certificate. Integer powers, even those with exponent 1000, are evaluated exactly.

For the middle contours, only rational arithmetic, integer square roots, and positive exponential and logarithmic sums are needed. The choices for e_W in (7.41)–(7.42) are as follows:

W	$10^{12}e_W$	W	$10^{12}e_W$
1	367879441172	5	6737947000
2	135335283237	6	2478752177
3	49787068368	8	335462628
4	18315638889	10	45399930

The positive sums $S_{100}(W) = \sum_{j=0}^{100} W^j/j!$ satisfy $S_{100}(W) > 1/e_W$, and hence $e_W > e^{-W}$. Likewise $S_{100}(T_0) > 8/C_-(n_0)$ certifies the first comparison in (7.82); all other terms in the final budget are rational.

An explicit verification consists of the following finite steps. For each of the thirteen rows, form V, M, α, β from (7.73) or (7.75) and verify all its preceding field conditions. For each integer j in its listed interval, read ν_j, W_j from the two assignment tables, form (7.38)–(7.39), and verify (7.40)–(7.42), $\nu_j H_0 \widehat{r}_j > 22/7$, and $k_j(n_0) > 0$. Next form (7.78) and the outward radical (7.79); verify $0 < \Delta_j < 1$, $0 < \eta_j < 1$, and

$$(\eta_j - 10^{-12})^2 < \frac{B^2 + R_j^2}{(1-\Delta_j)^2} \le \eta_j^2.$$

Compute the weight by (7.80), and check its positivity, its nonincrease with j, and the strict increase of $k_j(n_0)$. Finally verify the two equivalent forms of the Abel sum in (7.81), the positive slope and intercept loss A_*, B_*, and the strict margins in Table 11. There are exactly 2918 such contours, all specified by the printed tables.

The two bridge contours use $(\nu, W, e_W) = (.99, 6, 1/400)$ with the parameters given in Proposition 7.12. Their remaining checks are (7.40)–(7.42), the displayed factor and count bounds, and the scalar comparisons in its four cases. Upper bounds for e^x at $x = 1, .75, .74$ follow from

$$e^x < \sum_{j=0}^{24} \frac{x^j}{j!} + \frac{x^{25}}{25!} \frac{1}{1 - x/26};$$

all the other bridge exponential comparisons use S_{100} from below. The high-modulus radical tests are the two positive-side squared comparisons following (7.69). Thus the entire compact argument reduces to the displayed finite integer and rational operations, followed by the explicit monotonicity arguments for $n \ge 6251$.

8. The origin region

In this section we exclude a rightmost zero close to the upper endpoint of the unit disk. The estimates apply down to arbitrarily small positive scales.

Proposition 8.1 (Origin region). *Let P have degree $n \ge 6251$, with all its zeros in $\overline{\mathbb{D}}$. Suppose that $a > 0$ and*

$$|a - \xi| > 1 \qquad (P'(\xi) = 0).$$

Let $b = x + iy$ be a rightmost zero of P, where $x \ge a$ and $y \ge 0$. Then

$$\max(x, 1 - y) > \rho_* := \frac{30501}{500000}. \tag{8.1}$$

All zeros and critical points in this statement, and in the sums below, are counted with their algebraic multiplicities.

8.1. **Normalization and radial identities.** Suppose, to the contrary, that

$$\rho = \max(x, 1-y) \le \rho_*, \qquad F(u) = P(i + \rho u), \qquad \beta = X - iT = \frac{b - i}{\rho}, \qquad \alpha = \frac{a}{\rho}.$$

Then

$$0 < \alpha \le X \le 1, \qquad 0 < T \le 1, \qquad \max(X, T) = 1. \tag{8.2}$$

Here $T > 0$: a point $x + iy \in \overline{\mathbb{D}}$ with $x > 0$ cannot have $y = 1$. Every zero u of F satisfies

$$\operatorname{Re} u \le X, \qquad |i + \rho u| \le 1. \tag{8.3}$$

Gauss–Lucas and the strict hole give, for each critical point $v = r - it$,

$$\frac{\rho}{2}|v|^2 \le t < \frac{\rho}{2}|v - \alpha|^2, \qquad r < \frac{\alpha}{2}. \tag{8.4}$$

The root β is simple, since its real part is strictly larger than that of every critical point. Put

$$N = n-1, \quad N_0 = 6250, \quad n_0 = 6251, \quad c_\rho = \rho/2, \quad I(u) = -\operatorname{Im} u - \frac{\rho}{2}|u|^2,$$
$$S = \frac{F''(\beta)}{F'(\beta)} = \sum_v \frac{1}{\beta - v}, \quad h = |S|, \quad Q = \sum_v \frac{1}{|\beta - v|^2}, \quad d = \min_v |\beta - v|, \tag{8.5}$$
$$y_0 = \frac{Q}{h}, \quad q = \frac{Q}{h^2}, \quad \mu = \frac{1}{dh}, \quad D = dh, \quad \tau = \frac{N\rho}{2}.$$

We use y_0 for the moment ratio, to distinguish it from the physical coordinate y of b. The original and critical insets I are nonnegative. Moreover,

$$\operatorname{Re} S > 0, \quad h > 0, \quad Q < \frac{2\operatorname{Re} S}{X}, \quad h^2 \le NQ, \quad \frac{h}{n} < y_0. \tag{8.6}$$

Indeed, $X - r > X/2$ in every summand of $\operatorname{Re} S$; the penultimate inequality is Cauchy–Schwarz. Also, by (8.4),

$$I(v) < \rho\alpha(\alpha/2 - r) \le \rho X(X/2 - r). \tag{8.7}$$

For the last comparison subtract the two quadratic expressions: their difference is $(X - \alpha)((X + \alpha)/2 - r) \ge 0$.

For $b = i + \rho\beta$ and any $z \ne \beta$,

$$\operatorname{Re} \frac{b}{\beta - z} = \frac{\rho}{2} + \frac{I(z) - I(\beta)}{|\beta - z|^2}. \tag{8.8}$$

Summing over the critical points, and writing $E_c = \sum_v I(v)/|\beta - v|^2$, gives

$$\tau = \rho X \operatorname{Re} S - (1 - \rho T) \operatorname{Im} S - E_c + I(\beta)Q. \tag{8.9}$$

Since $E_c \ge 0$, $|b| \le 1$, and $I(\beta) < T$, it follows that

$$\tau < h(1 + Ty_0). \tag{8.10}$$

On the other hand (8.7) gives

$$E_c < \rho(X \operatorname{Re} S - X^2 Q/2).$$

Substitution into (8.9), retaining the term $-E_c$, yields

$$T(1 - \rho T/2)Q < \tau + (1 - \rho T) \operatorname{Im} S. \tag{8.11}$$

Thus, if $|\operatorname{Im} S| < J$,

$$\tau < \rho h + J + Ty_0 h, \tag{8.12}$$
$$\left(\frac{h}{n}\right)^2 < \left(\frac{h}{N}\right)^2 < \frac{\rho/2 + J/N}{T(1 - \rho T/2)}. \tag{8.13}$$

At zero inset we have the following bound.

Lemma 8.2 (A boundary source)**.** *If $X = 1$ and $I(\beta) = 0$, then $h > N/2$.*

Proof. Now $|b| = 1$. At a simple zero the original-root factorization gives $S = 2\sum_{u \ne \beta}(\beta - u)^{-1}$. If $z = i + \rho u$ is another physical zero, then

$$2\operatorname{Re} \frac{b}{b - z} - 1 = \frac{1 - |z|^2}{|b - z|^2} \ge 0.$$

Consequently

$$\operatorname{Re}(bS) = 2\rho \sum_{z \ne b} \operatorname{Re} \frac{b}{b - z} \ge N\rho.$$

The critical-root version of (8.8) says $\operatorname{Re}(bS) = N\rho/2 + E_c$. Hence

$$N\rho/2 \le E_c < \rho(\operatorname{Re} S - Q/2), \qquad h \ge \operatorname{Re} S > N/2 + Q/2 > N/2.$$

□

8.2. **A finite Cayley–Hermite–Biehler inequality.** The identities (8.6)–(8.13) and the next two source lemmas hold for every finite degree $n \geq 2$, $0 < \rho < 1$, $0 < \alpha \leq X \leq 1$, and $0 < T \leq 1$ satisfying (8.3)–(8.4) at an actual simple rightmost zero. They also apply at the larger scales in the next section.

Lemma 8.3 (Finite polar source estimate)**.** *For real s suppose that*

$$\operatorname{Re} \frac{s}{1 + ic_\rho s} > X. \tag{8.14}$$

Define

$$\begin{aligned} W(s) &= \sum_{F(u)=0} \frac{I(u)}{|s - (1 + ic_\rho s)u|^2}, \qquad C(s) = \sum_{F'(v)=0} \frac{I(v)}{|s - (1 + ic_\rho s)v|^2}, \\ E(s) &= \sum_{F(u)=0} \frac{I(u)}{|s - u|^2}, \qquad \theta = \frac{n\rho}{2(1 + c_\rho^2 s^2)}. \end{aligned} \tag{8.15}$$

Then

$$W \geq E, \qquad \theta W \leq (C + \rho)(W + \theta). \tag{8.16}$$

In particular, if an upper bound k for $C + \rho$ satisfies $k < \theta$, then $W \leq k/(1 - k/\theta)$.

Proof. The maps

$$w = \frac{u}{1 - ic_\rho u}, \qquad u = \frac{w}{1 + ic_\rho w}, \qquad d_p(w) = 1 + ic_\rho w$$

send the disk in (8.3) to the closed lower half-plane, because

$$-\operatorname{Im} w = \frac{I(u)}{|1 - ic_\rho u|^2}.$$

Consider the finite polynomials

$$G(w) = d_p(w)^n F(w/d_p(w)), \qquad J_p(w) = d_p(w)G'(w) - nic_\rho G(w) = d_p(w)^{n-1} F'(w/d_p(w)). \tag{8.17}$$

A root $u = -i/c_\rho$ makes its transformed linear factor constant. It has $I(u) = 0$ and contributes zero to the sums in (8.15). All other labels and multiplicities are preserved. The coefficient n in (8.17) is the original degree, even if G has smaller degree. Both G and J_p have their finite roots in the closed lower half-plane.

By (8.14), $G(s) \neq 0$. Put

$$g = G'/G, \qquad a(w) = -nic_\rho/d_p(w), \qquad H_s(w) = G'(w) + a(s)G(w).$$

Direct subtraction in the Cayley map proves

$$-\operatorname{Im} g(s) = W, \quad -\operatorname{Im}(J_p'/J_p)(s) = C, \quad -\operatorname{Im} a(s) = \theta, \qquad |g(s) + a(s)| \geq W + \theta > 0.$$

In particular, $H_s(s)$ and $J_p(s)$ are nonzero. Differentiating the polar identity before freezing a gives

$$\frac{H_s'}{H_s}(s) = \frac{J_p'}{J_p}(s) - \frac{ic_\rho}{d_p(s)} - \frac{a'(s)}{g(s) + a(s)}. \tag{8.18}$$

Here $|a'(s)| = c_\rho\theta$ and $|ic_\rho/d_p(s)| \leq c_\rho$. The last two terms therefore each have modulus at most c_ρ, and

$$-\operatorname{Im}(H_s'/H_s)(s) \leq C + \rho. \tag{8.19}$$

The real-rootedness needed below follows from a finite argument. If a degree m polynomial has all roots strictly below the real axis, a continuous argument of its value on the real axis decreases strictly through a total angle $m\pi$: its derivative is the negative sum of the positive Poisson kernels

of the roots. After any fixed rotation, each crossing of an integer multiple of π is a real zero of its imaginary part. These crossings account for the degree of that imaginary part, which is m, or $m-1$ when its leading coefficient vanishes. Hence that imaginary part is real-rooted. Moving roots a small distance downward and taking a coefficient limit proves the assertion for roots on the boundary as well, allowing repeated roots and degree drop. A nonzero coefficient limit of real-rooted polynomials is real-rooted, with roots allowed to escape to infinity.

Rotate G so that $H_s(s) > 0$ is real, and write $G = U + iV$. Let $p = \operatorname{Re} a(s)$ and

$$B = V' + pV - \theta U = \operatorname{Im} H_s.$$

At the selected point s,

$$B(s) = 0, \qquad H_s(s)V(s) = (W+\theta)|G(s)|^2 > 0, \qquad U'V - UV' = W|G|^2.$$

In particular V is not identically zero. By the preceding argument it is real-rooted, so

$$V'^2 - VV'' \geq 0.$$

This last inequality follows directly by logarithmic differentiation away from the zeros, and by continuity at them. The exact Wronskian identity

$$BV' - B'V = \theta(U'V - UV') + V'^2 - VV''$$

now gives

$$\theta W \leq [-\operatorname{Im}(H_s'/H_s)(s)](W+\theta) \leq (C+\rho)(W+\theta).$$

Finally, for $u = r - it$,

$$|s - (1 + ic_\rho s)u|^2 = |s-u|^2 - \rho s^2 t + \frac{\rho^2 s^2}{4}|u|^2 \leq |s-u|^2, \tag{8.20}$$

since $t \geq \rho|u|^2/2$. Termwise comparison gives $W \geq E$. Rearranging with $k < \theta$ gives the solved form. □

We will also use the unsolved form. If $W \geq b_0 > 0$, then

$$C + \rho \geq \frac{\theta b_0}{\theta + b_0}. \tag{8.21}$$

The function $uv/(u+v)$ is increasing in each positive argument. Thus (8.21) remains useful without an upper bound on h or a positive solved denominator.

8.3. **The critical-label comparison.** For $s \geq \alpha$, the critical domain yields the exact identity

$$\begin{aligned} &|s - (1 + ic_\rho s)v|^2 - (1 - \rho^2 s^2/2)|s-v|^2 \\ &\quad = \rho s^2 \left(\frac{\rho}{2}|v-\alpha|^2 - t\right) + \frac{\rho^2 s^2}{2}(s-\alpha)(s+\alpha-2r) + \frac{\rho^2 s^2}{4}|v|^2 > 0. \end{aligned} \tag{8.22}$$

All three terms have their stated sign on the whole unbounded critical domain. In particular, if $\rho \leq r_+$ and $s \leq s_+$, with $r_+^2 s_+^2/2 < 1$, set

$$f = \frac{1}{1 - r_+^2 s_+^2/2}, \qquad D_p = 1 + (r_+ s_+/2)^2. \tag{8.23}$$

Then

$$C < f C_{\text{real}}, \qquad C_{\text{real}} = \sum_v \frac{I(v)}{|s-v|^2}, \qquad \theta \geq \frac{n\rho}{2D_p}.$$

Lemma 8.4 (A negative square-moment term)**.** *For $s = X + \delta$ with $\delta \ge 0$, put*

$$\kappa_X(T,\delta) = \frac{X^2}{2} - \frac{X\max(0, T^2 - X\delta - \delta^2)^2}{4(T^2+\delta^2)(X/2+\delta)}. \tag{8.24}$$

Then

$$C_{\mathrm{real}} < \rho\{X \operatorname{Re} S - \kappa_X(T,\delta)Q\}. \tag{8.25}$$

For $X = 1$, the coefficient decreases with T, and is nonnegative for $0 < T \le 1$, $\delta \ge 0$. For $T = 1$ and $\delta \in \{3/10, 1/2, 71/100\}$ there is also the bound

$$C_{\mathrm{real}} < m\rho \operatorname{Re} S \quad (0 < X \le m), \qquad m = (1-\delta)^2. \tag{8.26}$$

Proof. Let $A = X - r > X/2$, $B = A + \delta$, $b_0 = X/2 + \delta$, and $L = T^2 + \delta^2$. Use (8.7) and clear the positive denominators of one summand in (8.25). The coefficient available after division by X is

$$\frac{X}{2} + (A - X/2)\frac{2\delta A + \delta^2 + 2Tt - T^2}{(A+\delta)^2 + t^2}.$$

If the second numerator is nonnegative there is no loss. Otherwise, with $z_0 = 2\delta A + \delta^2 - T^2 < 0$,

$$\frac{z_0 + 2Tt}{B^2 + t^2} - \frac{z_0}{B^2} = \frac{2TtB^2 - z_0 t^2}{B^2(B^2+t^2)} \ge 0.$$

It suffices to take $t = 0$. The possible loss is

$$\frac{(B - b_0)(L - 2\delta B)}{B^2}.$$

As a function of $w = 1/B \in [0, 1/b_0]$ this is $(1 - b_0 w)(Lw - 2\delta)$. Its maximum is $\max(0, L - 2\delta b_0)^2/(4Lb_0)$, by completing the square; when $L - 2\delta b_0 \le 0$ its maximum is zero at $w = 1/b_0$. This proves the stated coefficient. For $X = 1$ the subtracted term increases with T^2 on its active interval. Its numerator is at most $T^4 \le T^2 \le (1 + 2\delta)(T^2 + \delta^2)$, which also proves nonnegativity of κ_1.

For (8.26), put $T = 1$. The critical inset divided by ρ is less than $XA - X^2/2$. If $A \ge m$ and $X \le m$, this is at most $mA - m^2/2$. The required comparison then reduces to a quadratic in A whose discriminant is $[(1-\delta)^2 - m][(1+\delta)^2 + m] = 0$; its remaining t and t^2 coefficients are nonnegative. If $A \le m$, use $XA - X^2/2 \le A^2/2$. The constant term after clearing denominators is

$$P(A) = 2m(A+\delta)^2 - A(A^2+1).$$

Its derivative is $-3A^2 + 4mA + 4m\delta - 1$. In each of the three specified cases, $16m^2 + 12(4m\delta - 1) < 0$, so P decreases on the real line. Furthermore $P(m) = m\delta^4 > 0$. The other terms are nonnegative ($2At$ and $(2m - A)t^2$ after the same division), proving the assertion. □

For a closed parameter cell suppose that $C_{\mathrm{real}} < \rho h A$, where $0 < A \le \overline{A}$, and that $h \ge H$, $h/n \le U$. With the constants in (8.23), if

$$B = 2D_p(f\overline{A}U + 1/n_0) < 1, \tag{8.27}$$

then Lemma 8.3 gives

$$\frac{W}{h} < \frac{r_+(f\overline{A} + 1/H)}{1 - B}. \tag{8.28}$$

If a ceiling $h < H_+$ has previously been proved, we may instead use $B = 2D_p(f\overline{A}H_+ + 1)/n_0$. A source lower bound $W \ge b_0$ gives, without either ceiling,

$$\rho h A > K(\rho), \qquad K(\rho) = \frac{1}{f}\left\{\frac{n_0\rho b_0}{n_0\rho + 2D_p b_0} - \rho\right\}. \tag{8.29}$$

The expression in braces is concave in ρ, since its second derivative is $-4n_0^2D_pb_0^2/(n_0\rho+2D_pb_0)^3<0$. It increases with n_0 when the original degree is retained. Thus the smaller endpoint value on a closed scale interval is a valid lower bound for the entire interval. In each application $K>0$, so $A>0$ and $h>K/(\rho A)$.

8.4. **Moving sources and a uniform crescent gap.** When $X=1$ and $I=I(\beta)>0$, set

$$D_0=1-c_\rho(T+I),\qquad s=\frac{1+I}{D_0}. \tag{8.30}$$

The contribution of the actual zero β is exactly

$$\frac{I}{|s-(1+ic_\rho s)\beta|^2}=\frac{D_0}{2I}\ge\frac{(1-\rho)^2}{2T}. \tag{8.31}$$

Indeed $D_0=|1-ic_\rho\beta|^2$, and expansion gives $D_0|s-(1+ic_\rho s)\beta|^2=2I^2$. Moreover $(1-\rho)^2\le D_0<1$, $1<s<2/(1-\rho)^2<3$, and

$$D_0^2(s-1-c_\rho^2s^2)=I\{1+2c_\rho(1-T)+c_\rho^2(T^2-4T-1)+2c_\rho^3(1+T^2)\}>0. \tag{8.32}$$

The bracket is at least $1-4c_\rho^2$: after subtraction the remaining terms are $2c_\rho(1-T)+c_\rho^2(T-1)(T-3)+2c_\rho^3(1+T^2)$. Thus the test satisfies (8.14).

On a cell $\rho\le r_+$, $T\le t_+$ put

$$b_m=\frac{(1-r_+)^2}{2t_+},\quad f_m=(1-9r_+^2/2)^{-1},\quad D_m=1+9r_+^2/4,\quad H_m=\frac{n_0b_m/(n_0r_++2D_mb_m)-1}{f_m}. \tag{8.33}$$

Since $C<f_m\rho h$, the unsolved inequality (8.21) proves $h>H_m$. At zero inset it remains valid with $h\ge H_m$. At $X=1$ and $I(\beta)=0$,

$$T=\frac{\rho}{2}(1+T^2)\le\rho<1.$$

To justify that endpoint, multiply every physical zero and the physical witness by $1-\epsilon$. For a sufficiently small positive ϵ all the finitely many strict holes persist, and $T'<1$ in the $X=1$ normalization, where

$$\rho'=(1-\epsilon)\rho,\quad T'=T+\frac{\epsilon}{(1-\epsilon)\rho},\quad \beta'-v'=\beta-v,\quad S'=S,\quad h'=h. \tag{8.34}$$

The scaled source has positive inset. Applying the formula with a depth endpoint tending down to t_+ and taking the rational limit gives $h\ge H_m$.

We also need a distance bound which keeps the actual depth. The lower intersection of the hole circle with $r=\alpha/2$ has depth

$$\eta=\frac{1-\sqrt{1-\rho^2\alpha^2/4}}{\rho}.$$

The inequality

$$\frac{2}{125}-\rho_*\left(\frac14+\frac{1}{15625}\right)>0 \tag{8.35}$$

gives $0\le\eta<\alpha^2/125$ by squaring two positive sides. To bound the distance on the critical crescent, write $u=r-\alpha\le-\alpha/2$ on the lower hole arc. The scalar product with the vector $(X-\alpha,T-1/\rho)$ is

$$(X-\alpha)u+(1/\rho-T)\sqrt{\rho^{-2}-u^2}.$$

Its derivative is positive for $u \le 0$. Hence the nearest permitted arc point has the largest real coordinate. The upper arc and exterior radial segments cannot be nearer; if $T < \eta$, the horizontal separation alone suffices. We obtain

$$X = 1: \quad d^2 > \frac{1}{4} + \max(T - 1/125, 0)^2, \tag{8.36}$$

$$T = 1: \quad d^2 > G(X) := 1 + \frac{117}{500}X^2. \tag{8.37}$$

For the second line, the stronger lower bound is $(X-\alpha/2)^2+(1-\alpha^2/125)^2 \ge X^2/4+(1-X^2/125)^2 = G(X) + X^4/15625$.

8.5. **From inverse contours to source moments.** We use the inverse and symmetric contour estimates of Lemmas 6.1 and 6.2, and prove two refinements for one-sided strips.

Suppose $Q/h \le Y$, $1/d \le m$, and $h \ge H$. For $0 < mR < 1$ put

$$\begin{aligned} \lambda &= \frac{YR}{1-mR}, \qquad e_R = Y\left\{\sum_{j=2}^{16} \frac{m^{j-2}R^j}{j} + \frac{m^{15}R^{17}}{17(1-mR)}\right\}, \\ \widehat{R} &= R - e_R, \qquad B_R = \frac{Y}{(1-mR)^2(1-\lambda)^3}. \end{aligned} \tag{8.38}$$

The simpler upper bound $e_R = Y\{R^2/2 + mR^3/[3(1-mR)]\}$ is also permissible and will be used for the base range. These bound the complete logarithmic series of the actual critical product; the last displayed term includes every omitted order. For example

$$\frac{1}{S}\log\frac{F'(\beta+z)}{F'(\beta)} = z + E_R(z), \qquad |E_R(z)| \le e_R \quad (|z| \le R),$$

and the derivative differs from 1 by at most λ. If $\lambda < 1$ and $\widehat{R} > 0$, this logarithm is injective and its image contains the disk of radius $\widehat{R}$. In the unscaled coordinate $w = \log(F'(\beta+z)/F'(\beta))$, the inverse position $\phi(w) = \beta + z$ is defined on $|w| < h\widehat{R}$, with $\phi(0) = \beta$ and $F'(\phi(w))/F'(\beta) = e^w$. Set $A(w) = S\phi'(w)$. The normalized primitive $SF(\phi(w))/F'(\beta)$ has derivative $e^w A(w)$, and

$$\operatorname{Re} A \ge (1+\lambda)^{-1}, \quad |A| \le (1-\lambda)^{-1}, \quad |A'| \le B_R/h.$$

Choose $0 < \nu < 1$, $W_0 > 0$, and rational bounds

$$\begin{gathered} B_R/H < \epsilon_0, \quad \nu B_R\widehat{R} < g_0, \quad 0 < p_0 \le (1+\lambda)^{-1} < 1 < (1-\lambda)^{-1} \le a_0, \\ e_0 > e^{-W_0}, \qquad W_0^2 < (1-\nu^2)(H\widehat{R})^2, \\ \epsilon_0(1+e_0) + g_0e_0 < 1 - a_0e_0, \qquad \epsilon_0(1+e_0) + g_0e_0^2 < p_0 - e_0. \end{gathered} \tag{8.39}$$

For an odd-π contour require

$$\epsilon_0 + g_0e_0 < p_0, \qquad \nu H\widehat{R} > \pi. \tag{8.40}$$

For a quarter-period contour replace these requirements by

$$\epsilon_0 + g_0e_0 < \frac{p_0}{1+p_0+a_0}, \qquad \nu H\widehat{R} > \pi/2. \tag{8.41}$$

These are conditions on the containing disk, all corners, both vertical sides, both horizontal sides, and the amplitude homotopy. The resulting number L_R of actual original roots in $|u-\beta| < R$ obeys

$$\frac{L_R}{h} > \frac{\nu\widehat{R}}{\pi} - \frac{\ell_0}{h}, \qquad \ell_0 = 2 \text{ in (8.40)}, \quad \ell_0 = 3/2 \text{ in (8.41)}. \tag{8.42}$$

The trimmed odd height is the largest odd multiple of π not exceeding the available height, including equality. The full side estimates already cover such an endpoint. A one-sided odd-π strip between the fractions γ and ν of the available height, $0 < \gamma < \nu$, gives

$$\frac{L_{\gamma,\nu}}{h} > \frac{(\nu-\gamma)\widehat{R}}{2\pi} - \frac{2}{h}. \tag{8.43}$$

Lemma 8.5 (One-sided inverse strips)**.** *Under* (8.39) *and* (8.40)*, the upper strip* $\gamma h\widehat{R} < \operatorname{Im} w < \nu h\widehat{R}$ *and its reflected lower strip each have the count* (8.43)*. They exclude the source* β*. Writing* $S/h = c_0 + is_0$ *with* $c_0 > 0$*, each upper-strip root lies above* β *by more than*

$$c_0\gamma\widehat{R} - e_R - |s_0|W_0/h; \tag{8.44}$$

each lower-strip root lies below it by more than the same quantity.

Proof. Put $\mathcal{J}(w) = SF(\phi(w))/F'(\beta)$. The estimates preceding (8.39) give

$$A(0) = 1, \quad \operatorname{Re} A \ge p_0, \quad |A| \le a_0, \quad |A'| < \epsilon_0, \quad \mathcal{J}(w) = \int_0^w e^z A(z)\,dz.$$

All corners of the containing rectangle lie in the inverse disk. Integration by parts along the entire path $0 \longrightarrow -W_0 \longrightarrow -W_0 + i\eta \longrightarrow x + i\eta$ gives

$$\mathcal{J}(w) = e^w A(w) - 1 - \mathcal{E}(w), \qquad |\mathcal{E}(w)| < \epsilon_0(1+e^x) + g_0 e^{-W_0}.$$

The vertical segment contributes at most $(B_R/h)|\eta|e^{-W_0} \le \nu B_R\widehat{R}e^{-W_0}$; this gives the $g_0e^{-W_0}$ term. On the left side the modulus of $e^wA(w) - 1$ is at least $1 - a_0e^{-W_0}$, and on the right side it is at least $p_0e^{W_0} - 1$. The two inequalities in the last line of (8.39) control these whole sides, respectively. On every horizontal line $\eta = (2k+1)\pi$ the modulus is at least $1 + p_0e^x$. It dominates the error, because

$$\epsilon_0(1+e^x) + g_0e^{-W_0} \le (\epsilon_0 + g_0e^{-W_0})(1+e^x) < p_0(1+e^x) \le 1 + p_0e^x.$$

Thus Rouché applies on every subrectangle bounded by two such horizontal lines. The homotopy $A_t = 1 + t(A-1)$ preserves $\operatorname{Re} A_t \ge p_0$ and $|A_t| \le a_0$, so $e^wA_t(w) - 1$ stays nonzero on this boundary. The zero count is therefore exactly that of $e^w - 1$.

For the upper strip take the smallest odd multiple L of π at least $\gamma h\widehat{R}$, and the largest odd multiple U at most $\nu h\widehat{R}$. If $L < U$, the subrectangle contains exactly $(U-L)/(2\pi)$ roots, and

$$\frac{U-L}{2\pi} > \frac{(\nu-\gamma)h\widehat{R}}{2\pi} - 2.$$

If the right side is negative, the claimed lower count is automatic; if it is nonnegative, the strip width is at least 4π and $L < U$ necessarily. The same argument applies to $-U, -L$. The boundaries contain no zeros, including when an endpoint is exactly an odd multiple of π. The inverse is injective and its image is critical-free, so all counted roots are distinct simple actual roots. Since $\gamma > 0$, neither strip contains $w = 0$. Finally

$$|\phi(w) - \beta - w/S| \le e_R, \qquad \operatorname{Im}(w/S) = \frac{c_0 \operatorname{Im} w - s_0 \operatorname{Re} w}{h},$$

which proves (8.44) and its lower-strip counterpart. □

Lemma 8.6 (Real widths and an upper root)**.** *If* $|\operatorname{Im} S| < J$*, the roots counted in* (8.42) *satisfy*

$$|\operatorname{Re}(u-\beta)| \le \ell_R := \min\left(R, \frac{W_0}{H} + \frac{J\nu\widehat{R}}{H} + e_R\right). \tag{8.45}$$

For a row satisfying (8.40), a positive upper strip also supplies an actual root above β by more than

$$\left(1-\frac{J^2}{H^2}\right)\nu\widehat{R}-e_R-\frac{4\pi}{H}-\frac{JW_0}{H^2}, \tag{8.46}$$

whenever this expression is positive. If it exceeds the depth T, the original disk condition is contradicted.

Proof. In the counting rectangle $|\operatorname{Re} w|\le W_0$, $|\operatorname{Im} w|\le \nu h\widehat{R}$. Since

$$\operatorname{Re}(w/S)=\frac{c_0\operatorname{Re} w+s_0\operatorname{Im} w}{h},\qquad |s_0|<J/h,$$

the inverse error bound gives (8.45). For the upper root let U be the largest odd multiple of π not exceeding $\nu h\widehat{R}$. The adjacent rectangle $U-2\pi<\operatorname{Im} w<U$ contains exactly one actual root by the complete boundary argument of Lemma 8.5. When (8.46) is positive, $\nu H\widehat{R}>4\pi$ and $J/H<1$, so that this rectangle has positive height above zero. Its root has $\operatorname{Im} w>\nu h\widehat{R}-4\pi$. Using $c_0=\sqrt{1-s_0^2}\ge 1-(J/H)^2$, $c_0\le 1$, and $h\ge H$ in the imaginary inverse displacement gives exactly (8.46). Every original zero has nonpositive imaginary part in the normalized coordinates, which proves the last assertion. □

We now convert these counts into lower bounds for $E(s)$. Suppose $T\in[t_-,t_+]$, $\rho\le r_+$, and $s=X+\delta$. A counted region with radius R and real width ℓ has $|\operatorname{Re} u|\le 1$ in every application below. For $X=1$ this follows from rightmostness and $\ell<2$; for $T=1$ it follows from $R<1$ and $0<X\le 1$. Its whole-region lower weight is

$$\omega(R,\ell;\delta,t_-,t_+,r_+)=\max\left(0,\min_{t\in\{\max(0,t_--R),\,t_++R\}}\frac{t-r_+(1+t^2)/2}{(\delta+\ell)^2+t^2}\right). \tag{8.47}$$

The derivative numerator of the quotient, writing $A=(\delta+\ell)^2>0$, is $A-t^2+r_+(1-A)t$. Its two roots have opposite signs, and its positive root is a maximum. Thus the minimum on a nonnegative interval is at an endpoint. If that minimum is negative we instead use $I(u)\ge 0$.

Take increasing radii R_j; replace their widths by running maxima so the regions are nested. Let s_j be any increasing rational lower bounds for $\nu_j\widehat{R}_j/\pi$, and replace the weights by the maximum of all later weights. This gives valid $w_1\ge\cdots\ge w_k\ge 0$ on the nested regions. The source root has a separate lower weight b_0. Subtracting that root once from each odd-π count, and applying finite Abel summation, gives

$$\frac{E}{h}>\sum_{j=1}^{k}w_j(s_j-s_{j-1})+\frac{b_0-3w_1}{h},\qquad s_0=0. \tag{8.48}$$

Indeed, if M_j are the actual cumulative counts excluding β, then $M_j>hs_j-3$ and $\sum w\ge w_kM_k+\sum_{j<k}(w_j-w_{j+1})M_j$. The coefficients of the cumulative counts are nonnegative, so the bound remains valid when an initial hs_j-3 is negative. If $b_0-3w_1<0$, replace its denominator in (8.48) by H. If it is nonnegative and $h<600$, replace it by 600; if no upper field is known, discard the nonnegative term. This accounts for both signs of the source compensation. For a one-sided strip the corresponding calculation is simpler: if its roots have weight at least $a>0$ and β has weight at least b_0, then, since they are distinct from β,

$$\frac{E}{h}>a\frac{(\nu-\gamma)\widehat{R}}{2\pi}+\frac{b_0-2a}{h}. \tag{8.49}$$

Thus $b_0\ge 2a$ makes the last term nonnegative.

For quarter-period counts, one may instead use the cumulative lower bounds $k_j = \nu_j \widehat{R}_j/\pi - 3/(2H)$ directly, when they and their successive increments are positive. Then

$$\frac{E}{h} > \sum_j w_j(k_j - k_{j-1}), \qquad k_0 = 0. \tag{8.50}$$

8.6. **The base range** $0 < \rho \le 1/100$**.** All constants in this subsection use $r_b = 1/100$. Set

$$f_m = \frac{1001}{1000}, \qquad f_X = f_T = \frac{10001}{10000}, D_b = 1 + 9r_b^2/4. \tag{8.51}$$

These dominate the quadratic polar factors for, respectively, the moving tests $s < 3$, the fixed test $s = 101/100$, and the fixed test $s = X + 3/10 \le 13/10$. The fixed inverse-Cayley conditions are $(r_b(101/100)/2)^2 < 1/100$ and $(r_b(13/10)/2)^2 < 3/10$. Lemma 8.4 gives $C_{\mathrm{real}} < \rho h$ for the X tests, and $C_{\mathrm{real}} < \rho \max(X, 49/100)h$ for the T test.

There are 55 inverse rows in this range. The columns in Table 14 are

$$(Y, m, R, H, \nu, W_0, \epsilon_0, g_0, p_0, a_0, e_0).$$

The simpler e_R following (8.38) is used. Here $(W_j, g_j, e_j) = (2, 1/5, 1/7)$ for $6 \le j \le 9$, and

TABLE 14. All inverse rows for the base range. Each j-row occurs for every integer $6 \le j \le 30$.

row	Y	m	R	H	ν	W_0	ϵ_0	g_0	p_0	a_0	e_0
X_1	2	2	3/20	600	99/100	4	1/25	3	7/10	7/4	1/50
X_D	2	2	13/100	600	99/100	4	1/25	3	7/10	7/4	1/50
$X_{450,j}$	2	2	$j/200$	450	23/25	W_j	11/100	g_j	7/10	7/4	e_j
$X_{200,j}$	2	2	$j/200$	200	9/10	W_j	11/100	g_j	7/10	7/4	e_j
X_{78}	2	2	3/20	78	97/100	11/5	3/10	3	7/10	7/4	1/9
T_{150}	25/11	100/99	3/20	150	9/10	4	1/6	3	7/10	7/4	1/50
T_{103}	25/11	100/99	3/20	103	9/10	4	1/6	3	7/10	7/4	1/50

$(4, 3, 1/50)$ for $10 \le j \le 30$. Only the two j-families use quarter-period contours; all other rows use odd-π contours. Every row satisfies (8.39) and its appropriate horizontal and height condition. These are rational inequalities apart from e^{-W_0}; the latter is bounded by the reciprocal of the positive Taylor sum through degree 48. Thus Table 14 and the displayed formulae specify the finite verification.

8.6.1. *The base case* $X = 1$*: the first ceiling.* Suppose $h \ge 600$. Then $d > 1/2$ and $y_0 < 2$, so $q < 2/600 < 7/625$ and $\mu < 2/600 < 1/200$. Proposition 5.2 therefore gives

$$|\operatorname{Im} S| < 4y_0 < 8, \qquad |\operatorname{Im}(S/h)| < 1/70, \qquad \operatorname{Re}(S/h) > 999/1000. \tag{8.52}$$

The upper one-sided strip of row X_1, with $\gamma = 4/5$, has positive count, since

$$\frac{(99/100 - 4/5)\widehat{R}_1}{2(22/7)} - \frac{2}{600} > 0.$$

Its actual root lies above β by more than $(4/5)(999/1000)\widehat{R}_1 - e_{R_1} - 4/(70 \cdot 600) > 67/1000$. Thus $T > 67/1000$.

First let $67/1000 \le T \le 3/10$. Equation (8.13), with $J = 9$, gives $u = h/n < 311/1000 =: U$ for every $N \ge 6250$. At $s = 101/100$,

$$\kappa_1(T, 1/100) \ge \kappa = \frac{29591}{63600} > 0.$$

As $y_0 > u$, put $a(u) = f_X(1-\kappa u)$ and $B(u) = 2D_b(a(u)u + 1/n_0)$. On $0 \le u \le U$ the factors $1-\kappa u$, $1-2\kappa u$, and $1-B(u)$ are positive. The derivative numerator of $(a(u)+1/600)/(1-B(u))$ is

$$-f_X\kappa(1-2D_b/n_0) + 2D_b\{f_X^2(1-\kappa u)^2 + (f_X/600)(1-2\kappa u)\}.$$

It has a positive lower bound obtained by replacing the first term by $-f_X\kappa$ and u by U in the remaining terms. Therefore the solved source estimate is valid throughout this interval and

$$W/h < r_b\frac{a(U)+1/600}{1-B(U)} < 1/50. \tag{8.53}$$

Use the lower one-sided strip of X_D, with $\gamma = 7/25$, and set

$$\Delta = \frac{7}{25}\frac{999}{1000}\widehat{R}_D - e_{R_D} - \frac{4}{70\cdot 600} - \frac{3481}{5000}r_b > 0,$$
$$a_D(T) = \frac{T+\Delta}{(T+157/1200)^2}, \qquad b_D(T) = \frac{T - r_b(1+T^2)/2}{T^2+1/10000}.$$

These are full strip and source weights. The strip depth estimate comes from (8.52); its roots have $|u| < \sqrt{1.09} + .13 < 59/50$ and $|s-u| < \sqrt{T^2+.0001} + .13 \le T + 157/1200$. The polynomial

$$[T - r_b(1+T^2)/2](T+157/1200)^2 - 2(T+\Delta)(T^2+1/10000)$$

has all its Bernstein coefficients positive on $[67/1000, 3/10]$. Hence $b_D(T) > 2a_D(T)$ there. Adding β to the strip count in (8.43) compensates the loss $2/h$ and yields

$$\frac{E}{h} > \frac{(99/100 - 7/25)\widehat{R}_D}{2(22/7)}a_D(T) > 1/50.$$

For the final comparison it suffices to check the two endpoints: the only positive stationary point of a_D is a maximum. This contradicts (8.53).

If $3/10 \le T \le 1$, equation (8.13) gives $h/n < 147/1000$. Its source denominator is less than $3/10$, and

$$\frac{W}{h} < \frac{r_b(f_X + 1/600)}{1 - 2D_b(f_X(147/1000) + 1/n_0)} < 2/125.$$

The symmetric X_1 count has lower weight

$$\frac{T - 3/20 - 5r_b/4}{(T+901/6000)^2}.$$

Indeed $|u| < \sqrt{2} + 3/20 < \sqrt{5/2}$ and the displayed denominator bounds the complete disk. Multiplication by $(99/100)(7/22)\widehat{R}_1 - 2/600$ gives a lower bound exceeding $1/50$ at both endpoints, and therefore on the whole interval, again by its single-maximum derivative. Thus $h \ge 600$ is impossible.

8.6.2. *The base case $X = 1$: the remaining fields.* We now have $h < 600$. Lemma 8.2 excludes $I(\beta) = 0$, since $N/2 \ge 3125$. Apply the moving-source estimate at this positive inset. The following table gives entire-domain denominator and depth bounds:

known ceiling $h < H_+$	$(C+\rho)/\theta < B_+$	$T > t_+$
600	193/1000	13/200
450	29/200	9/100
200	13/200	21/100

(8.54)

For each row the two exact checks are

$$\frac{2D_b(H_+f_m+1)}{6251} < B_+, \qquad \frac{(1-r_b)^2}{2t_+} > \frac{r_b(H_+f_m+1)}{1-B_+}.$$

Thus they hold for every positive $\rho \le r_b$.

Suppose first $h \ge 450$ and use all 25 rows $X_{450,j}$. For each radius, apply (8.47) with $\ell = R$ and $\delta = 1/100$ on each of the two closed depth intervals $[13/200, 1/5]$ and $[1/5, 1]$. Use one of these depth intervals for the entire nested family. The cumulative bounds

$$k_j = \nu_j(7/22)\widehat{R}_j - \frac{3}{2H}$$

and their successive increments are positive, and the resulting weights decrease. Formula (8.50) applies. After excluding this field, repeat with $H = 200$ and depth intervals $[9/100, 1/5]$, $[1/5, 1]$. The exact inequalities, rounded outwards here to shorter rational numbers, are

H	$[t_-, t_+]$	E/h exceeds	W/h is less than
450	$[.065, .2]$	.042	.0125
450	$[.2, 1]$	.029	.0125
200	$[.09, .2]$	.041	.0118
200	$[.2, 1]$	.0245	.0118

(8.55)

The source upper in each row is exactly $r_b(f_X + 1/H)/(1 - B_+)$ with the previously established B_+. These comparisons prove successively $h < 450$, $T > .09$, $h < 200$, and $T > .21$.

On $[.21, 1]$ the source weight is at least

$$b_X = \frac{9900}{10001}.$$

To check the whole interval, the numerator difference after clearing the positive denominators factors as $(1-T)(AT-B)$, where $A = 1-(r_b/2)(1-1/10000)$, $B = (r_b/2)(1-1/10000)+1/10000$, and $A(.21) > B > 0$. Let $a_{78} = \omega(3/20, 3/20; 1/100, .21, 1, r_b)$. If $h \ge 78$, the row X_{78} and source compensation give

$$\frac{E}{h} > a_{78}\frac{97}{100}\frac{7}{22}\widehat{R}_{78} + \frac{\min(0, b_X - 3a_{78})}{78} > .01176, \qquad \frac{W}{h} < \frac{r_b(f_X + 1/78)}{1 - .065} < .01084.$$

Thus $h < 78$. Finally,

$$W < \frac{r_b(78f_X + 1)}{1 - .065} < .846 < b_X \le E,$$

which excludes the entire base case $X = 1$.

8.6.3. *The base case* $T = 1$. A direct physical comparison gives $d > 99/100$. In fact, with $d_0 = 99/100$,

$$(1 - d_0\rho)^2 - |b - a|^2 \ge \rho\{2(1 - d_0) - (2 - d_0^2)\rho\} > 0,$$

and $|a - \xi| > 1$ then implies $|b - \xi| > d_0\rho$. If $h \ge 150$, this first gives $dh > 100$, so Proposition 4.11 yields $y_0 < 25/11$. The bounds $q < 25/(11 \cdot 150) < 9/400$ and $\mu < 1/(.99 \cdot 150) < 1/100$ permit Proposition 5.2, giving $|\operatorname{Im} S| < (23/5)y_0 < 11$. Equation (8.13) implies $h/n < 21/250$. The row T_{150} then gives

$$\frac{E}{h} > .01599 > .01211 > \frac{r_b(f_T + 1/150)}{1 - 2D_b(f_T(21/250) + 1/n_0)} > \frac{W}{h}.$$

Thus $h < 150$, whence $(C + \rho)/\theta < 1/20$.

If $h \ge 103$, the bound $103(.99) > 100$ permits Proposition 4.11. The row T_{103} gives

$$\frac{E}{h} > .01145 > .01063 > \frac{r_b(f_T + 1/103)}{.95} > \frac{W}{h}.$$

Hence $h < 103$. The source β itself always has weight at least $b_T = 99/109$. For $X \le .77$, the uniform small-X comparison gives

$$W < \frac{r_b(103f_T(.77) + 1)}{.95} < .846 < b_T,$$

a contradiction.

It remains to consider $X > .77$. If $h \geq 79$, (8.6) and $d > .99$ give

$$q < \frac{2}{.77 \cdot 79} < \frac{33}{1000}, \qquad \mu < \frac{1}{.99 \cdot 79} < \frac{8}{625}. \tag{8.56}$$

The quartic construction of Proposition 5.2 supplies two distinct simple actual roots at distance less than $15/(2h)$ from β. For this construction, at $q_b = 33/1000$, $\mu_b = 8/625$, $\nu_b = q_b\mu_b$, use

$$(Z_0, M_0, E_0, P_0, Z_{\mathrm{out}}, L_0, R_0, \eta_0) = (719/100, 7/25, 71/50, 350, 15/2, 32/25, 3/10, 11/50).$$

The center and path majorants of that construction are bounded by

$$\begin{aligned} Z(q) &= 63/10 + 21q + 133q^2 + 1030q^3 + 9000q^4, \\ M(q) &= (63/10)q + 45q^2 + 400q^3 + 3400q^4 + 86000q^5 \\ &\quad + 320000q^6 + 1700000q^7 + 9100000q^8 + 39000000q^9. \end{aligned}$$

They satisfy $Z(q_b) < Z_0$, $M(q_b) < M_0$, $\mu_b Z_{\mathrm{out}} < 1$, $Z_0 + R_0 < Z_{\mathrm{out}}$, and

$$\begin{gathered} \exp\left(M_0 + \frac{\nu_b Z_0^3}{3(1-\mu_b Z_0)}\right) < E_0, \qquad \frac{Z_0^4 E_0}{12(1-\mu_b Z_0)} < P_0, \\ E_0(132200q_b^5 + P_0\nu_b) < (1-\eta_0)R_0, \qquad 1 + q_b Z_{\mathrm{out}} + \frac{\nu_b Z_{\mathrm{out}}^2}{1-\mu_b Z_{\mathrm{out}}} < L_0, \\ \frac{e^{L_0R_0} - 1 - L_0R_0}{L_0} < \eta_0 R_0. \end{gathered} \tag{8.57}$$

The complete Gaussian residual is $132200q_b^5$, on its valid domain $q_b < 1/20$; the integrated actual cubic logarithmic tail retains the factors $1/3$ and $1/4$ in (8.57). The last two residual bounds give strict Rouché inequalities on both entire circles. Those circles avoid the source and are disjoint because

$$6 - 21q_b - 133q_b^2 - 1030q_b^3 - 9000q_b^4 > R_0, \quad 12(1 - q_b - 17q_b^2 - 137q_b^3 - 313q_b^4) > 2R_0.$$

Rouché's theorem therefore gives the two distinct simple zeros of F.

Since $15/(2 \cdot 79) < 19/200$, each companion has weight at least

$$\omega(19/200, 19/200; 3/10, 1, 1, r_b) = \frac{8672039}{10840400}.$$

Adding the source once gives $E \geq b_T + 2(8672039/10840400) > r_b(103f_T + 1)/.95 > W$. Thus $h < 79$, and the remaining source comparison $W < r_b(79f_T + 1)/.95 < .843 < b_T \leq E$ is again impossible. This proves Proposition 8.1 for all $0 < \rho \leq 1/100$.

8.7. **The higher strip with $T = 1$: entering the local estimates.** Assume throughout this subsection that

$$1/100 \leq \rho \leq \rho_*, \quad T = 1, \quad 0 < X \leq 1. \tag{8.58}$$

The closed lower endpoint overlaps the base range. By (8.37), $d > \sqrt{G(X)} > 1$. For a test $s = X + \delta$ write

$$\kappa_\delta(X) = \kappa_X(1, \delta), \qquad A_\delta = X - \kappa_\delta(X)y_0, \quad b_\delta(X) = \frac{1 - \rho_*(1 + X^2)/2}{1 + \delta^2}.$$

On the active interval $1 - X\delta - \delta^2 \geq 0$ the coefficient factors as

$$\kappa_\delta(X) = \frac{X[X - (1-\delta)^2][X + (1+\delta)^2]}{2(1+\delta^2)(X + 2\delta)}; \tag{8.59}$$

outside it $\kappa_\delta(X) = X^2/2$. With $D_\delta(X) = 1 + \rho_*^2(X+\delta)^2/4$ and $f_\delta(X) = (1-\rho_*^2(X+\delta)^2/2)^{-1}$, define

$$K_\delta(r,X) = \frac{1}{f_\delta(X)}\left\{\frac{n_0 r b_\delta(X)}{n_0 r + 2D_\delta(X)b_\delta(X)} - r\right\}. \tag{8.60}$$

Equation (8.29) gives $\rho h A_\delta > \min_{r\in\{1/100,\rho_*\}} K_\delta(r,X)$. All tests used here have $1/200 \le \delta \le 1$; the inverse-Cayley condition follows from $\rho_*^2 < 1/200$.

For $\delta = 1/2$, replacing the source and polar factors by their $X = 1$ bounds gives

$$K_h = \min_{r\in\{1/100,\rho_*\}} K_{1/2}(r,1) > .6843, \qquad \rho h A > K_h, \tag{8.61}$$

where $A = 1/4$ if $X \le 1/4$, by (8.26), and $A = X - \kappa_{1/2}(X)y_0$ otherwise. Here

$$\kappa_{1/2}(X) = \frac{X(4X-1)(4X+9)}{40(X+1)}.$$

If $X \le 2/5$, (8.61) gives $D > h > K_h/((2/5)\rho_*) > 25$. For $2/5 \le X \le 1$, the one-variable inequality

$$4(227/200)\kappa_{1/2} - (X+\kappa_{1/2})^2 > 0 \tag{8.62}$$

implies $A(1+y_0) < 227/200$ by completion of the square in y_0. Multiplication of (8.61) and (8.10) then gives $h^2 > K_h N/[2(227/200)] > 25^2$. Thus Proposition 4.10 is applicable in all cases, and

$$y_0 < 3/d < 3/\sqrt{G(X)} \le 3. \tag{8.63}$$

The same moment ratio sharpens the joint bound to

$$\frac{A(1+y_0)}{G(X)} \le \frac{21}{20}. \tag{8.64}$$

Here are whole-interval certificates for this assertion. For $X \le 1/4$ use $(1/4)(1+3) = 1$. Put $Y(X) = 3(1 - 11X^2/100)$. Squaring positive sides gives $Y(X) > 3/\sqrt{G(X)}$ on $0 < X \le .49$; after removal of X^2 the required polynomial is strictly positive. The ratio $\kappa_{1/2}(X)/X$ increases, and seven times its value at .41 is less than one. Hence on $[1/4, .41]$ the product $(X - \kappa_{1/2}y_0)(1+y_0)$ increases up to its allowed endpoint, and the certificate is

$$(21/20)G(X) - (X - \kappa_{1/2}Y(X))(1+Y(X)) > 0.$$

On $[.41, 1]$ completion of the square gives the certificate $4(21/20)G(X)\kappa_{1/2} - (X+\kappa_{1/2})^2 > 0$. Consequently

$$D^2 > \frac{K_h N}{2(21/20)} > 45^2. \tag{8.65}$$

Write $S/h = c_0 + is_0$, $c_0 > 0$, $c_0^2 + s_0^2 = 1$. The universal envelope in Proposition 4.1 is

$$q < \frac{21}{5}\mu^2 + \frac{11}{2}\mu c_0 + \frac{c_0^2}{4} - \frac{s_0^2}{2}. \tag{8.66}$$

If $|s_0| \ge .7$, then $c_0 < 18/25$, and the right side is at most

$$\frac{21/5}{45^2} + \frac{(11/2)(18/25)}{45} + \frac{1-.7^2}{4} - \frac{.7^2}{2} < 0.$$

Thus $|\operatorname{Im} S| < .7h$.

Now take $\delta = 3/5$. On $[0,1]$ the coefficient is always active. The rational functions $K_{3/5}(1/100, X) - K_{3/5}(\rho_*, X)$ and $K_{3/5}(\rho_*, X)$ are positive. Write

$$K_{.6}(X) = K_{3/5}(\rho_*, X), \quad A_{.6} = X - \kappa_{3/5}(X)y_0, \quad R_{.6}(y_0) = .7 + .3\rho_* + (199/200)y_0.$$

Equation (8.9) gives $N\rho/2 < hR_{.6}(y_0)$. Combining this with $\rho h A_{.6} > K_{.6}(X)$ gives

$$h^2 > \frac{NK_{.6}(X)}{2A_{.6}R_{.6}(y_0)}, \qquad D^2 > \frac{NK_{.6}(X)G(X)}{2A_{.6}R_{.6}(y_0)}.$$

The two rectangle inequalities are

$$N_0K_{.6}(X)G(X) - 2\cdot 51^2 A_{.6}R_{.6}(y_0) > 0, \qquad 0 \le X \le 1, 0 \le y_0 \le 3, \tag{8.67}$$

$$N_0K_{.6}(X) - 2\cdot 45^2 A_{.6}R_{.6}(y_0) > 0, \qquad 0 \le X \le 1, 0 \le y_0 \le 3. \tag{8.68}$$

They prove $D > 51$ and $h > 45$. An exact finite positivity procedure for these two rational quadratics and all the preceding one-variable certificates is given below. If $|s_0| \ge 2/3$, then $c_0 < 3/4$, and

$$\frac{21/5}{51^2} + \frac{(11/2)(3/4)}{51} + \frac{5}{36} - \frac{2}{9} < 0.$$

Therefore

$$h > 45, \quad D > 51, \quad |\operatorname{Im} S| < (2/3)h. \tag{8.69}$$

8.7.1. *Entry to the larger companion rectangle.* Set $\delta = .71$ and $K(X) = K_\delta(\rho_*, X)$. The same endpoint comparison proves that $K(X)$ is the smaller scale endpoint; also $K(X) \ge K_* := K(1) > 0$. Using (8.69) in the exact radial identity gives

$$\frac{N\rho}{2} < hR, \qquad R = \rho X + \frac{2}{3}(1-\rho) + \left(1 - \frac{\rho}{2}(1+X^2)\right) y_0.$$

Consequently the same source gives

$$q^2 < \frac{2y_0^2[X - \kappa_\delta(X)y_0]R}{NK(X)}. \tag{8.70}$$

For $X \le .49$, R decreases with ρ. At $\rho = .01$ put $\beta_0 = 33/50 + X/100$, $I_0 = 199/200 - X^2/200$. If $r(X) = \kappa_\delta(X)/X$, the derivative in y_0 of $y_0^2(X - \kappa_\delta y_0)(\beta_0 + I_0 y_0)$, divided by Xy_0, is

$$2\beta_0 + 3(I_0 - r\beta_0)y_0 - 4rI_0 y_0^2.$$

The ratio r increases: after positive denominators are removed its derivative numerator is $\delta^4 + 6\delta^2 + 4\delta X + X^2 + 1 > 0$. Let r_+ be its value at .49, and take the endpoint extrema of β_0, I_0 on $[0, .49]$. The last display is bounded below by

$$2\beta_- + 3(I_- - r_+\beta_+)y_0 - 4r_+ I_+ y_0^2.$$

This concave quadratic is positive at 0 and 3. It follows that the permitted y_0 may be replaced by $Y(X)$ from above. The entire interval is then certified by

$$N_0K(X) - \frac{2}{(.057)^2}Y(X)^2[X - \kappa_\delta(X)Y(X)](\beta_0 + I_0 Y(X)) > 0 \quad (0 \le X \le .49). \tag{8.71}$$

For $X \ge .49$ the active interval has $\kappa_\delta(X) \ge (12/25)X^2$; its inactive value $X^2/2$ is stronger. Completion of the square gives $y_0[X - \kappa_\delta(X)y_0] \le 25/48$. Also (8.63) gives $y_0 < 73/25$. Thus the right side of (8.70) is at most

$$\frac{2(25/48)(73/25)\{2/3 + \rho_*/3 + (199/200)(73/25)\}}{N_0K_*} < (.057)^2.$$

We have proved

$$q < 57/1000, \qquad \mu < 1/51. \tag{8.72}$$

Proposition 5.4 applies in the form (5.21). Its two distinct actual roots $u_\pm$ have normalized positions $w_\pm = S(u_\pm - \beta)$ satisfying

$$|w_\pm| < 8917/1000, \quad \left|\frac{w_+ + w_-}{2}\right| < 2109/1000, \quad |\operatorname{Im} S| < (37/5)y_0. \tag{8.73}$$

The mean estimate uses the even part of the two centers at the same moment parameters, as specified in that proposition.

8.8. **The higher $T = 1$ strip: two field ceilings.** If $h \ge 105$, then $D > 100$ by (8.37). Propositions 4.11 and 5.2 give

$$y_0 < 9/4, \quad q < 9/400, \quad \mu < 1/100, \quad |\operatorname{Im} S| < (23/5)y_0. \tag{8.74}$$

For a closed cell $\rho \in [p, r]$, $y_0 \in [l, u]$, take $J = (23/5)u$ and a rational $H \ge 105$ with either $H = 105$ or

$$H \le \frac{N_0 p/2 - J}{r + u}. \tag{8.75}$$

Then $h \ge H$. Put

$$B_r = \frac{r/2 + J/N_0}{1 - r/2}. \tag{8.76}$$

A positive rational U is a valid bound for h/n if at least one of $U \ge u$, $U^2 \ge B_r$, or $Ul \ge B_r$ holds; the last is used only for $l > 0$. These follow respectively from (8.6), (8.13), and (8.11).

Retain the same-X distance information. A cap $0 < X_+ \le 1$ is valid if $X_+ = 1$ or

$$l^2 G(X_+) \ge (9/4)^2. \tag{8.77}$$

For the chosen δ, certify

$$\overline{A} - X + \kappa_\delta(X)v > 0 \quad (0 \le X \le X_+, \quad v \in \{l, u\}). \tag{8.78}$$

Split the X interval at $(1 - \delta^2)/\delta$ to use the correct active or inactive formula. Since the expression is affine in v, this verifies the whole (X, y_0) cell. The constants in (8.23) use $s_+ = 1 + \delta$ and $r_+ = r$. After checking (8.27), (8.28) is the source upper bound.

For each contour use (8.38) with $Y = u$, $m = 1$, and set

$$\epsilon_0 = \frac{1001 B_R}{1000 H}, \quad g_0 = \frac{1001 \nu B_R \widehat{R}}{1000}, \quad p_0 = (1 + \lambda)^{-1}, \quad a_0 = (1 - \lambda)^{-1}, \quad e_0 = \left(\sum_{j=0}^{80} \frac{W_0^j}{j!} \right)^{-1}. \tag{8.79}$$

Verify (8.39) and (8.40), using $\pi < 22/7$. Use (8.45), rounded upwards, and the weights (8.47) with $t_- = t_+ = 1$. A valid source weight is $b_0 = (1 - r)/(1 + \delta^2)$. The source lower bound is

$$\sum_j w_j (s_j - s_{j-1}) + \frac{\min(0, b_0 - 3w_1)}{H}. \tag{8.80}$$

The source upper is (8.28). Table 15 specifies the two full covers; the rational choices of cells and contours are in the electronic appendix. On every cell of the first cover, (8.80) strictly exceeds that upper. The coverage and arithmetic procedures are specified in §8.12. This proves $h < 105$.

Suppose next that $90 \le h < 105$. Since $D > 90$, Proposition 5.6 gives, with $z = Qd/h$,

$$z < 8/3, \quad |\operatorname{Im} S| < (21/4)y_0, \quad 3z^2 - (D - 91/10)z + 2D > 0.$$

At $D = 90$ this quadratic is negative at both $z = 8/3$ and $z = 49/20$. These values decrease with D, since $2 - z < 0$. The prior $z < 8/3$ excludes the larger root branch, so

$$z < 49/20, \qquad y_0 < 49/20. \tag{8.81}$$

Use the nested-count procedure with $H \ge 90$, $J = (21/4)u$ and $49/20$ in place of $9/4$ in (8.77). It excludes 26 closed cells of the second cover. The remaining nine cells use the already proved $h < 105$ and the following pair estimate.

Lemma 8.7 (Two actual sources). *Suppose $T = 1$, $0 < \rho \le r < 1$, $R > 0$, and $0 < m < 1$. Suppose two distinct actual zeros other than β have displacements $v_\pm = u_\pm - \beta$ satisfying $|v_\pm| < R$, $|(v_+ + v_-)/2| < m$. Suppose also that their individual insets belong to $[3/4, 6/5]$. For a test $s = X + \delta$ with $\delta > 0$, put*

$$I_\Sigma = 2(1-r)(1-m) - rR^2, \qquad D_\Sigma = 2(1+\delta^2) + 4\sqrt{1+\delta^2}\, m + 2R^2.$$

Assume $I_\Sigma > 0$. Then

$$E > \frac{1-r}{1+\delta^2} + \frac{(I_\Sigma/2 + 49/50)^2}{D_\Sigma}. \tag{8.82}$$

Proof. Rightmostness gives $\operatorname{Re} v_\pm \le 0$. Expanding the inset,

$$I(\beta + v) = I(\beta) - (1-\rho)\operatorname{Im} v - \rho X \operatorname{Re} v - \frac{\rho}{2}|v|^2.$$

Since $I(\beta) \ge 1 - \rho$ and $|v_+ + v_-| < 2m$,

$$I(u_+) + I(u_-) > 2(1-\rho)(1-m) - \rho R^2 \ge 2(1-r)(1-m) - rR^2 = I_\Sigma > 0.$$

The second inequality uses $2(1-m) + R^2 > 0$. The squared distances satisfy

$$\begin{aligned} |s-u_+|^2 + |s-u_-|^2 &= 2(1+\delta^2) - 2\operatorname{Re}\big((\delta + i)\overline{v_+ + v_-}\big) + |v_+|^2 + |v_-|^2 \\ &< D_\Sigma. \end{aligned}$$

On $[3/4, 6/5]$ one has $\sqrt{t} \ge t/2 + 49/100$. Indeed $t - (t/2 + 49/100)^2$ is concave and positive at both endpoints. Cauchy–Schwarz gives

$$\frac{I(u_+)}{|s-u_+|^2} + \frac{I(u_-)}{|s-u_-|^2} \ge \frac{(\sqrt{I(u_+)} + \sqrt{I(u_-)})^2}{|s-u_+|^2 + |s-u_-|^2} > \frac{(I_\Sigma/2 + 49/50)^2}{D_\Sigma}.$$

Adding the contribution of β proves (8.82). □

By (8.81), the field $h \ge 90$ gives $q < (49/20)/90 < .05$, $\mu < 1/90 < .02$. Proposition 5.4 therefore supplies

$$R = 17/180, \qquad m = 19/900.$$

Here $R > 0$, $0 < m < 1$, and, for each of the nine cells,

$$I_\Sigma \ge 2(1-\rho_*)(1-m) - \rho_* R^2 = \frac{367561059}{200000000} > 0.$$

The individual insets lie in $[3/4, 6/5]$, since

$$(1-r)(1-R) - rR^2/2 > 3/4, \qquad 1 + R < 6/5. \tag{8.83}$$

For each pair cell verify (8.78) and $B_{105} = 2D_p(105 f\overline{A} + 1)/n_0 < 1$. Its source upper is

$$W < \frac{r(105 f\overline{A} + 1)}{1 - B_{105}}. \tag{8.84}$$

The lower bound (8.82) exceeds (8.84) in all nine cells. This proves $h < 90$.

TABLE 15. Closed covers for the higher $T = 1$ strip. Each cover uses the three scale slabs $[.01, .025]$, $[.025, .045]$, and $[.045, .061002]$.

field assumption	moment interval	closed cells	count cells	pair cells	contours
$h \ge 105$	$[0, 9/4]$	28	28	0	1950
$90 \le h < 105$	$[0, 49/20]$	35	26	9	1881

8.9. **The higher $T = 1$ strip: the terminal pairs.** For $\delta = .71$ use $A = (1-\delta)^2$ when $X \le (1-\delta)^2$, and $A = X - \kappa_\delta(X)y_0$ otherwise. The source coefficient satisfies

$$y_0 A \le 11/20. \tag{8.85}$$

Here is the whole-domain verification. Put $m_0 = (1-\delta)^2$. For $X \le m_0$, use $3m_0 < 11/20$. On $[m_0, .35]$ six times the endpoint value of κ_δ/X is less than one, so $y_0(X - \kappa_\delta y_0)$ increases up to $y_0 = 3$; certify $11/20 - 3(X - 3\kappa_\delta) > 0$ on this full interval. Above .35, the active coefficient satisfies $\kappa_\delta \ge (5/11)X^2$, and the inactive coefficient is stronger. Completion of the square then gives (8.85).

If $|\operatorname{Im} S| < J_0 y_0$, equation (8.9) implies

$$\rho(N - 2h) < 2y_0(h + J_0).$$

For $h_L \le h \le h_U < N_0/2$, we have $N - 2h > 0$, and hence

$$\rho h A < \frac{2y_0 A\, h(h + J_0)}{N - 2h} \le \frac{2(11/20)h_U(h_U + J_0)}{N_0 - 2h_U}.$$

Use $f = (1 - \rho_*^2(1.71)^2/2)^{-1}$, $D_p = 1 + (\rho_* 1.71/2)^2$, and $B_U = 2D_p(fh_U + 1)/n_0 < 1$. The resulting finite source upper is

$$W < \frac{f\, 2(11/20)h_U(h_U + J_0)/(N_0 - 2h_U) + \rho_*}{1 - B_U}. \tag{8.86}$$

The following two intervals cover $45 < h < 90$, including their seam:

field interval	companion estimate	Z	M	J_0
$(45, 70]$	(5.21)	8917/1000	2109/1000	37/5
$[70, 90)$	5.4	17/2	19/10	7

(8.87)

The second row applies because $q < 3/70 < .05$ and $\mu < 1/70 < .02$; the first applies by (8.72). In each row put $R = Z/h_L$, $m = M/h_L$. In both rows $R > 0$, $0 < m < 1$, and the inset gates (8.83) hold with $r = \rho_*$. The corresponding values of I_Σ are

$$I_\Sigma = \begin{cases} \dfrac{201103346909179}{112500000000000} > 0, & (45, 70], \\ \dfrac{715839963}{392000000} > 0, & [70, 90). \end{cases}$$

Lemma 8.7, using $\sqrt{1 + .71^2} < 123/100$, gives the lower

$$\frac{1 - \rho_*}{1 + .71^2} + \frac{\{(1 - \rho_*)(1 - m) - \rho_* R^2/2 + 49/50\}^2}{2(1 + .71^2) + 4(123/100)m + 2R^2}. \tag{8.88}$$

It exceeds (8.86) by more than .6165 in the first row and more than .0527 in the second. These rational comparisons hold throughout the stated field intervals. They contradict $W \ge E$ and exclude (8.58).

8.10. **The higher strip with $X = 1$: a lower field bound.** Assume now

$$1/100 \le \rho \le \rho_*, \quad X = 1, \quad 0 < T \le 1. \tag{8.89}$$

In this case $d > 1/2$, $0 < y_0 < 2$, and $y_0 < 2\operatorname{Re}(S/h)$ by (8.6). We first prove

$$h > 50. \tag{8.90}$$

If $T \le 1/8$, apply (8.33) with $r_+ = \rho_*$, $t_+ = 1/8$. The exact expression H_m exceeds 54, so $h > 50$ at positive inset. At zero inset, $\rho = 2T/(1 + T^2)$ and $\rho_* < 2(1/8)/(1 + (1/8)^2)$ imply $T < 1/8$ strictly. The finite homothety (8.34) can therefore be chosen with $T' < 1/8$. It gives the same strict bound for the unchanged h.

Suppose $1/8 \le T \le 1$ and, for contradiction, $h \le 50$. The unphased radial identity gives $N\rho/2 < h(1+Ty_0) < 50(1+2T)$, hence

$$\rho < 300/N \le R_* = 6/125.$$

Use $\delta = 3/10$ and the polar constants

$$f = (1 - R_*^2(1.3)^2/2)^{-1}, \quad D_p = 1 + (R_*1.3/2)^2, B_{50} = 2D_p(50f+1)/n_0 < 1.$$

The source has the whole-interval lower weight

$$b_0 = \min_{t\in\{1/8,1\}} \frac{t - R_*(1+t^2)/2}{t^2 + .09} > 0.$$

Write $S/h = c_0 + is_0$, $c_0 > 0$, and set $k = \kappa_1(1,.3) = 13719/34880$. The critical comparison gives $C < f\rho h(c_0 - ky_0)$, and $c_0 - ky_0 > 0$ follows from $C \ge 0$ and its strictness. Under $h \le 50$, the denominator $1 - B_{50}$ is positive, so the source inequality gives

$$\rho h(c_0 - ky_0) > K := \frac{(1-B_{50})b_0 - R_*}{f} = \frac{1279084413230726}{1580047607421875} > 0. \tag{8.91}$$

Put $v_0 = y_0/c_0 \in (0,2)$, so $c_0 - ky_0 = c_0(1 - kv_0)$. Equation (8.9), using $I(\beta) \le T \le 1$, gives

$$\frac{N\rho}{2} < h\{c_0(R_* + v_0) + \sqrt{1 - c_0^2}\}.$$

Multiply by (8.91). The product in the denominator is the Rayleigh form, at the unit vector $(c_0, \sqrt{1-c_0^2})$, of

$$(1 - kv_0)\begin{pmatrix} R_* + v_0 & 1/2 \\ 1/2 & 0 \end{pmatrix}.$$

Its largest eigenvalue is less than $M = 4/5$ for every $0 \le v_0 \le 2$. Indeed the determinant of M times the identity minus this matrix is $a_2v_0^2 + a_1v_0 + a_0$, where

$$a_2 = Mk - k^2/4 > 0, \quad a_1 = -M(1 - kR_*) + k/2, \quad a_0 = M^2 - MR_* - 1/4, \qquad 4a_2a_0 - a_1^2 > 0.$$

This quadratic is positive on the whole real line; the other principal minor is $M > 0$. Also $1 - 2k > 0$. Thus

$$h^2 > \frac{KN_0}{2M} = \frac{639542206615363}{202246093750} > 2500,$$

contradicting $h \le 50$. This proves (8.90). Consequently $q < 2/h < .04$ and $\mu < 2/h < .04$. Proposition 5.4 gives

$$|\operatorname{Im} S| < (15/2)y_0. \tag{8.92}$$

The homogeneous version includes all positive q in this rectangle: the radius is $(45/4)q$, and the center bound $Z = 7.706829806638$ satisfies $Z + .45 < (13/5)\pi$.

8.11. **The higher $X = 1$ strip: field bounds on closed cells.** For a cell

$$\rho \in [p,r], \qquad y_0 \in [l,u], \qquad T \in [t_-, t_+], \tag{8.93}$$

use a phase bound $J = \gamma u$, with $\gamma = 4$ on a high-field branch and $\gamma = 15/2$ on the low-field branch. For a test δ, put

$$k = \kappa_1(t_+, \delta), \quad \overline{A} = 1 - kl, b_0 = \omega(0,0;\delta,t_-,t_+,r),$$

and use (8.23) with $s_+ = 1 + \delta$. Where $b_0 > 0$ the direct source gives

$$\rho h(1 - ky_0) > K, \qquad K = \frac{1}{f}\min_{z\in\{p,r\}}\left\{\frac{n_0 z b_0}{n_0 z + 2D_p b_0} - z\right\} > 0. \tag{8.94}$$

The cell conditions include positivity of $\overline{A}$, the polar denominator, and K. Set

$$G_0 = \max_{l\le y_0\le u}(1-ky_0)(1+t_+y_0), \qquad G_1 = \max_{l\le y_0\le u} y_0(1-ky_0).$$

Each is a concave quadratic maximum, evaluated at both endpoints and its vertex when the vertex belongs to the closed interval. Equations (8.94) and (8.10) imply

$$h > \frac{K}{r\overline{A}}, \qquad h^2 > \frac{KN_0}{2G_0}. \tag{8.95}$$

For the low-field phase (8.92), the exact radial identity also gives $\rho(N-2h) < 2y_0(Th+15/2)$. Multiplying by the same positive factor $h(1-ky_0)$ gives

$$K(N-2h) < 2y_0(1-ky_0)h(Th+15/2), \qquad h < N/2.$$

Thus a rational $H < N_0/2$ is a valid strict lower field if

$$K(N_0-2H) \ge 2G_1H(t_+H+15/2). \tag{8.96}$$

Indeed a contrary $h \le H$ makes its left side no smaller, and its right side no larger, than those in (8.96).

Besides these source bounds, every cell has the lower field bounds

$$\frac{N_0p}{2(1+t_+u)}, \qquad \frac{N_0p/2-J}{r+t_+u}, \qquad H_m \tag{8.97}$$

from (8.10), (8.12), and (8.33). At zero inset the last is non-strict, which suffices for a choice $H \le H_m$. The moving test supplies H_m; the final moment calculation includes β once.

For every inverse row choose a rational $m > 0$ satisfying

$$m^2\{1/4+\max(t_- - 1/125, 0)^2\} \ge 1. \tag{8.98}$$

By (8.36) this bounds $1/d$ on the whole cell. Use $Y = u + 10^{-9}$ in (8.38), and use (8.79). The rational enlargement of u ensures the required strict derivative inequalities also at a closed moment endpoint. All rows satisfy (8.39) and (8.40). Widths, slopes and depths are computed by (8.45), (8.42), and (8.46). The physical nesting conditions are $R_j < R_{j+1} < 2$ and increasing slopes; running maxima of the widths remain at most the corresponding radii. The regions therefore have $|\operatorname{Re} u| \le 1$ as required by (8.47).

If a depth lower bound exceeds t_+, the cell is impossible. This positive depth condition forces the height budget above 4π, so Lemma 8.6 supplies such a root. Otherwise compute the nested source lower in (8.48). For an unbounded high-field cell use the correction $\min(0, b_0 - 3w_1)/H$; for a low-field cell use

$$\begin{cases} (b_0-3w_1)/H, & b_0-3w_1 < 0, \\ (b_0-3w_1)/600, & b_0-3w_1 \ge 0. \end{cases} \tag{8.99}$$

The latter form is used only after the ceiling $h < 600$ is proved.

8.11.1. *The high-field cover.* Suppose $h \ge 600$. The bounds $q < 2/600 < 7/625$ and $\mu < 2/600 < 1/200$ permit Proposition 5.2, giving $|\operatorname{Im} S| < 4y_0$. On each cell take $J = 4u$ and H below at least one of 600 and the three lower fields in (8.97). For $t_- > 0$ put

$$B_r = \frac{r/2 + J/N_0}{t_-(1-rt_-/2)}. \tag{8.100}$$

A positive U bounds h/n if $U \ge u$, $U^2 \ge B_r$, or $Ul \ge B_r$ for $l > 0$. Apply (8.28) with $\overline{A} = 1 - \kappa_1(t_+, \delta)l$. The high cover has 61 closed cells covering $[0,2]\times[0,1]$ in (y_0, T) for the entire scale interval $[.01, .061002]$. Each cell has either an actual-root depth contradiction, or a strict source lower bound greater than (8.28). Cells with $t_- = 0$ use the depth alternative. It follows that $h < 600$.

8.11.2. *The low-field cover.* Use (8.90) and (8.92). The six closed scale slabs are

$$[.01,.013],\ [.013,.017],\ [.017,.023],\ [.023,.033],\ [.033,.045],\ [.045,.061002]. \tag{8.101}$$

On each cell take $J=(15/2)u$ and choose $H>J$ below at least one of: 50; a bound in (8.97); a bound in (8.95); or a value certified by (8.96). All source tests used for these choices satisfy $1/200\le\delta\le 3/4$ and $(r(1+\delta)/2)^2<\delta$. If $H\ge 600$, the cell contradicts the preceding ceiling. Otherwise it has either a depth contradiction or a nested moment contradiction. For the latter the upper denominator is

$$B=\frac{2D_p(600f\overline{A}+1)}{n_0}<1. \tag{8.102}$$

Thus every low-field source upper uses the bound $h<600$.

The six slabs have respectively $36,38,41,38,41,46$ closed cells. Together with the 61 high cells this makes 301 cells. Their three contradiction types are 196 moment comparisons, 57 actual root depth comparisons, and 48 field comparisons. There are 14891 complete inverse contours. With the rational rounding conventions below, the smallest moment reserve is

$$\frac{84739835370681964317289}{240500000000000000000000000}>.0003523485. \tag{8.103}$$

Every external endpoint and every internal depth, moment, and scale seam belongs to the closed cover. This excludes (8.89).

8.12. **Finite verification and the entire degree ray.** We finish by specifying the finite verification behind the interval comparisons above. The electronic appendix contains the exact rational parameter files and executable verifiers. The files specify cells, tests, radii and trims; the following procedure recomputes their bounds.

(i) Every integer, fraction and finite decimal is read as an exact rational number. The base rows are precisely Table 14. The higher T rows are the two files `T6251_H105_PROPOSAL.json` and `T6251_H90_PROPOSAL.json`; each record specifies $(p,r,l,u,H,\delta,\overline{A},U,X_+,J)$ and either the list (R,ν,W_0) or the pair alternative. The higher X rows are `X6251_high_0.01_0.06 1002.json` and the six `X6251_MOVING_low_` files indexed by the consecutive endpoints of (8.101). Each record specifies (l,u,t_-,t_+,H), its contradiction type, its source tests, and its contour choices (R,ν,W_0,m).

(ii) First verify the declared parameter domains, the inverse-Cayley condition, every local entrance, and the asserted lower field using the formulae preceding the relevant cover. For T cells verify (8.77), (8.78), and the radial upper. For X cells verify (8.98) and every source positivity and quadratic maximum used for its field.

(iii) For every contour recompute $\lambda,e_R,\widehat{R},B_R$, then all of (8.39) and the appropriate horizontal and height inequalities. In the higher strips use (8.79). The displayed Taylor reciprocal is strictly greater than e^{-W_0}. Exponential upper bounds in the quartic gates use, for $0\le x<32$,

$$e^x\le\sum_{j=0}^{30}\frac{x^j}{j!}+\frac{x^{31}}{31!(1-x/32)}.$$

The geometric tail proves the inequality for every omitted order.

(iv) In the higher covers, round slopes and weights down, and real widths up, to the rational lattice $10^{-12}\mathbb{Z}$. Use $7/22<1/\pi$ for slopes and $\pi<22/7$ for depth losses. For every X row also verify $Hs_j>1$ after rounding its slope. Square roots needed for radial uppers or pair denominators are rounded up to $10^{-16}\mathbb{Z}$, checking the squared enclosure by integer arithmetic. Take running maxima of the widths and future maxima of the weights, verify their nesting, and compute the Abel sums exactly with the signed correction (8.99) when appropriate. The X source upper is rounded up to $10^{-12}\mathbb{Z}$ before subtraction.

(v) Each cell using nested counts must have $0 < B < 1$ and a positive lower-minus-upper reserve. Each depth cell must have its rounded lower depth strictly greater than t_+. Each low-field exclusion must have $H \geq 600$. Each pair cell must satisfy $0 < r < 1$, $R > 0$, $0 < m < 1$, $\delta > 0$, $I_\Sigma > 0$, both inset gates, both endpoint chord inequalities, its squared radical enclosure, and the strict comparison (8.82)–(8.84).

(vi) Covering is checked as an assertion about closed rectangles. For each X file, sort all depth endpoints. At every endpoint and one point in each intervening open interval, collect the $[l, u]$ intervals of all cells whose closed depth interval contains the point. Their sorted union must cover $[0, 2]$ without a gap. Membership is constant between adjacent depth endpoints, so this proves coverage of the whole rectangle. Check exact adjacency and the exterior endpoints of the six scale slabs. For each T cover, sort all moment endpoints and check that the active scale intervals cover $[.01, .061002]$ on every intervening open fiber; the closed cells include their endpoint fibers as well.

Here is a finite positivity method for all rational functions used in the T entrances, source coefficients and terminal bound. Reduce the function to a polynomial numerator and denominator over $\mathbb{Q}$. Verify nonzero endpoint values, use the Euclidean signed-remainder sequence (Sturm's rule) to show that neither has a root in the closed interval, and check that their signs agree at one endpoint. At an explicit vanishing factor such as X^2 or $1 - X$, remove that factor symbolically and check its endpoint separately. This proves every one-variable positivity statement in (8.62)–(8.71), (8.78), and (8.85) on its specified interval. The base source-compensation polynomial is instead expanded in the Bernstein basis of its specified interval; all its coefficients are positive. To make this test explicit, if $P(a + (b - a)t) = \sum_{j=0}^{d} c_j t^j$, its Bernstein coefficients are

$$b_k = \sum_{j=0}^{k} c_j \frac{\binom{k}{j}}{\binom{d}{j}}.$$

Nonnegative basis functions whose sum is one then prove positivity throughout $[a, b]$.

For the two rectangle inequalities (8.67)–(8.68), collect the exact coefficients as $a(X)y_0^2 + b(X)y_0 + c(X)$. Subdivide the closed X interval dyadically. On each leaf the same one-variable procedure verifies the positive values at $y_0 = 0, 3$ and one of the following sufficient conditions: $b > 0$; $6a + b < 0$; $a < 0$; or $a > 0$ and $4ac - b^2 > 0$. The first condition suffices because if $a \geq 0$ the quadratic is increasing, whereas if $a < 0$ its two positive endpoint values suffice by concavity. The second has the analogous decreasing or concave justification. The third and fourth follow from concavity and completion of the square, respectively. The coefficients a, b have a factor X; checking their quotients and $c(0) > 0$ handles the entire $X = 0$ edge. Both children of each subdivision are retained. The resulting sorted leaves have exact adjacent endpoints 0 and 1, proving the entire rectangle. This proves the two coupled source and matrix inequalities over their full rectangles.

These procedures verify the 55 base contours and all base inequalities; the higher T procedure verifies 63 closed cells, 3831 contours and 9 pair cells; the higher X procedure verifies the 301 closed cells and reserve (8.103). The least higher-T reserve among cells using nested counts exceeds .0001411.

Finally every argument extends from $N_0 = 6250$ to the entire ray $N \geq N_0$. All local companion, inverse-log and crescent estimates are independent of the degree. The direct source in (8.29) increases with n; radial lower fields increase with N; the terms J/N and $1/n$ in upper bounds decrease. The coupled inequality (8.96) strengthens with N, and the two terminal uppers decrease with N. The fixed ceilings and closed parameter covers therefore work for every $n \geq 6251$. The base range and the two higher strips exhaust (8.2), proving Proposition 8.1.

9. Small positive witnesses

Throughout this section $N \ge N_0 = 6250$, and

$$\rho_0 = \frac{30501}{500000}, \qquad A_0 = \frac{\rho_0}{100}.$$

We work in the strict-hole configuration of Lemma 2.2, with $0 < a \le 1/10$ and an actual simple rightmost zero $b = x + iy$, $x \ge a$, $y \ge 0$. Put $r = |b|$, $p_b = 1 - r^2$, $\tau = 1 - y$. Use d, S, H, Q as in (2.2), and D, q, μ as in (2.7).

9.1. **The first field bound.**

Lemma 9.1. *Let $F(u) = p(i + \rho u)$ be a normalized strict-hole configuration with an actual simple rightmost zero $\beta = X - i$, where $\alpha = a/\rho$, $0 < \alpha \le X \le 1$, $1/100 \le \rho \le 9/100$, and $n \ge 6251$. Let v_j list all zeros of F', counted with multiplicity. Write $h = |F''(\beta)/F'(\beta)|$ and $d_u = \min_j |\beta - v_j|$. Then $d_u > 1$ and $D = d_u h > 25$.*

Proof. Put

$$R = 9/100, \qquad Q_u = \sum_j |\beta - v_j|^{-2}, \qquad Y = Q_u/h.$$

The critical domain is $r_j < \alpha/2$ and $\rho|v_j|^2/2 \le t_j < \rho|v_j - \alpha|^2/2$ for $v_j = r_j - it_j$. Lemma 6.6, scaled back to these coordinates, gives

$$d_u^2 > (X - \alpha/2)^2 + (1 - \eta)^2, \qquad \eta = \frac{\rho\alpha^2}{4(1 + \sqrt{1 - \rho^2\alpha^2/4})} \le R\alpha^2/4.$$

Since $0 \le \eta \le R\alpha^2/4 < 1$, it follows that

$$d_u^2 > G(\alpha) := (X - \alpha/2)^2 + (1 - R\alpha^2/4)^2.$$

This polynomial decreases with α on $0 \le \alpha \le X$: its derivative is $-X + (1/2 - R)\alpha + (R^2/4)\alpha^3 < 0$. Hence $d_u^2 > 1 + (1/4 - R/2)X^2 + (R^2/16)X^4 > 1$.

Apply the finite polar estimate in Lemma 8.3 at $s = X + 1/2$. The inverse Cayley test is strictly to the right of the original zeros, since $(R(1 + 1/2)/2)^2 < 1/2$. The source root contributes at least $b_0 = 4(1 - R)/5$. By the full critical comparison (8.25) and (8.26),

$$C_{\mathrm{real}} < \rho h \mathcal{A}(X, Y), \qquad \mathcal{A}(X, Y) = \begin{cases} 1/4, & 0 < X \le 1/4, \\ X - \kappa(X)Y, & 1/4 \le X \le 1, \end{cases}$$

$$\kappa(X) = \frac{X(4X - 1)(4X + 9)}{40(X + 1)}.$$

Use $f = [1 - (9/8)R^2]^{-1}$ and $D_p = 1 + (3R/4)^2$ in (8.21). The function $uv/(u + v)$ is increasing in both variables, so

$$\rho h \mathcal{A}(X, Y) > K_0, \qquad K_0 = \min_{\rho \in \{1/100, R\}} \frac{1}{f}\left(\frac{6251\rho b_0}{6251\rho + 2D_p b_0} - \rho\right) = \frac{239012578698741}{379195048000000} > 0. \quad (9.1)$$

The expression minimized here is concave in ρ and increases with the degree, as shown by differentiating it in (8.29). Thus the bound holds throughout the scale interval for every $n \ge 6251$. The unphased radial inequality is $N\rho/2 < h(1 + Y)$.

If $X \le 7/25$, then $\mathcal{A} \le 7/25$, whence $h > K_0/[R(7/25)] > 25$. If $X \ge 7/25$, completion of the square gives $\mathcal{A}(X, Y)(1 + Y) < 31/10$ for every real Y. For an exact proof, put $N_k = X(4X - 1)(4X + 9)$ and $D_k = 40(X + 1)$. The degree-six polynomial

$$(62/5)N_k D_k - (XD_k + N_k)^2$$

has the following positive Bernstein coefficients on $[7/25, 1]$:

$$\frac{120508031}{244140625},\quad \frac{1411226329}{1953125},\quad \frac{185744659}{78125},\quad \frac{416205433}{78125},\quad \frac{30945547}{3125},\quad \frac{407837}{25},\quad 24527.$$

Multiplying the two inequalities for the same X, Y therefore gives $h^2 > K_0 N/(2(31/10)) > 625$. Since $d_u > 1$, both branches give $D > 25$. □

Lemma 9.2. *The implication $D > 25$ holds on each of the two regions*

$$\max(x, 1-y) > \rho_0, \qquad |b| \le 47/50.$$

Both regions also satisfy $D > 70$.

Proof. If $x \ge \rho_0$, Lemma 6.7 gives $D > 70$. We treat $x \le \rho_0$.

Return to physical coordinates. Lemma 6.6 gives $d > \tau$. If $1/100 \le \tau \le 9/100$ and $x \le \tau$, the preceding lemma applies with $\rho = \tau$, $X = x/\tau$, $\alpha = a/\tau$; its D is exactly the physical D. For $\tau \ge 9/100$, use the matrix envelope in Proposition 4.1. Writing $c = \operatorname{Re} S/H$, $s = |\operatorname{Im} S|/H$, the rational bound $\pi > 333/106$ gives $\gamma < 241/1000$, and its constants satisfy

$$Qd^2 < 21/5 + (11/2)Dc + (241/1000)D^2c^2 - D^2s^2/2.$$

Retaining the corresponding negative coefficient $741/1000$ and completing the square in the radial identity yields

$$N < \frac{2Dx}{d} + \frac{p_b}{d^2}\left(21/5 + (11/2)D + (241/1000)D^2\right) + \frac{y^2}{(741/1000)p_b}.$$

On the whole $x \le \rho_0$, $\tau \ge R = 9/100$ range,

$$x/d < \rho_0/R, \qquad p_b/d^2 < (2-R)/R, \qquad y^2/p_b \le (1-R)/(2R).$$

For the last bound, $x < \tau$ gives $p_b > 2\tau(1-\tau)$; the case $y = 0$ is immediate. Under $D \le 25$ the preceding upper bound for N is $640685569/102600 < 6250$, a contradiction.

Outside the origin box, the remaining small-x range has $\tau > \rho_0 > x$. For $r \le 47/50$ one has $\tau \ge .06$. If $.06 \le \tau \le \rho_0$, then

$$(.94)^2 < \rho_0^2 + (1-\rho_0)^2, \qquad x^2 \le (.94)^2 - (1-\tau)^2 < \tau^2,$$

because $\tau^2 + (1-\tau)^2$ decreases on this interval. Thus every case enters either the normalized source or the large-τ matrix argument. This proves $D > 25$ on both regions.

Proposition 4.10 therefore gives $Qd/H < 3$. Outside the origin box and with $x \le \rho_0$, the radial identity first gives $D > 6250/(2/\rho_0 + 96) > 48$. The matrix envelope excludes $|s| \ge 2/3$, since $c < 3/4$, $c^2 \le 5/9$ would give

$$\frac{21/5}{48^2} + \frac{(11/2)(3/4)}{48} + \frac{241}{1000}\frac{5}{9} - \frac{2}{9} < 0.$$

The phase-sensitive radial bound then gives

$$D > \frac{6250\rho_0}{2\rho_0 + (4/3)(1-\rho_0) + 3(2-\rho_0)} > 53.$$

On $r \le .94$ the initial bound is $D > 6250(.06)/8 = 46.875$. The matrix envelope excludes $|s| \ge .7$, using $c < .715$ and $c^2 \le .51$. Consequently

$$D > \frac{6250(.06)}{2\rho_0 + 1.4(.94) + 3(1.94)} > 51.$$

Both regions satisfy $q < .06$ and $\mu < .02$, so the first weight-eight row of Proposition 5.4 applies. Its phase bound $|\operatorname{Im} S| < 8Q/H$ and the radial identity give respectively

$$D > \frac{6250 - 48/\rho_0}{98} = \frac{27771875}{498183}, \qquad D > \frac{6250 - 48/.06}{2\rho_0/.06 + 97} = \frac{27250000}{495167}.$$

Both exceed 5000/91, so Proposition 5.8 applies. Its field bound with constant 210 gives

$$D > \frac{6250\rho_0 - 210}{11/5} = \frac{13701}{176} > 70$$

outside the origin box, and $D > [6250(.06) - 210]/(11/5) = 75 > 70$ on $r \le .94$. The geometric estimate covers the large-x branch. Every numerical comparison increases with N, so the full degree ray is included. □

9.2. **Witnesses below A_0.**

Proposition 9.3. *There is no strict hole with $0 < a \le A_0$ and $\max(x, 1-y) > \rho_0$.*

Proof. The crescent gives

$$d > \rho_0 - a/2 \ge d_0 := \rho_0 - A_0/2.$$

Lemma 9.2 first gives $D > 70$. Then (5.23), followed in order by Proposition 4.11, (5.24), and Proposition 5.7, gives the strict successive bounds

$$D > \frac{6250d_0 - 596/5}{2(1+A_0)} > 100, \qquad D > \frac{6250d_0 - 1899/20}{2(1+A_0)} > 140,$$

$$D > \frac{6250d_0 - 2637/32}{2(1+A_0)} > 148.$$

The last step uses $140 > 139$. With $\kappa = 11067/5000$, the exact inequality

$$(5/4)\kappa^2 - (148 - 23/2)\kappa + 296 < 0, \qquad 2 < \kappa < 9/4,$$

and the smaller quadratic branch in Proposition 5.7 give $Qd/H < \kappa$.

For $R = td$ define

$$\Lambda = \frac{\kappa t}{1-t}, \quad f = t - \kappa\left(t^2/2 + t^3/[3(1-t)]\right), \quad B = \frac{\kappa(1-t)}{[1-(1+\kappa)t]^3}.$$

Thus $H\widehat{R} = Df$, and the inverse lemma applies with $\epsilon = (1001/1000)B/148$, $g = (1001/1000)\nu Bf$, $p_* = (1+\Lambda)^{-1}$ and $A_* = (1-\Lambda)^{-1}$. The exact data `FINE6251_INTEGER_DATA.json` give forty-one triples (j, m, W), with $t = j/200$ and $\nu = m/10000$. For every row the finite predicates are

$$\begin{aligned} &0 < t, \nu, \Lambda < 1, \quad f > 0, \quad W^2 < (1-\nu^2)(148f)^2, \quad \nu(148)f > 22/7, \\ &\epsilon(1+e) + ge < 1 - A_* e, \quad \epsilon(1+e) + ge^2 < p_* - e, \quad \epsilon + ge < p_*. \end{aligned} \tag{9.2}$$

Here, for $W = 1, \dots, 10$, the rational bounds $e > e^{-W}$ are

$$1/2, 1/7, 1/20, 1/50, 1/140, 1/400, 1/1000, 1/2900, 1/8000, 1/22000.$$

Their validity follows from the positive exponential sum through degree one hundred. The data also have increasing t and increasing νf. Equations (9.2) are exactly the full inverse, corner, four-side and homotopy conditions of Lemma 6.1; the resulting cumulative count is $L_j > \nu_j(7/22)Df_j - 2$.

Put $\nu_0 f_0 = 0$ and

$$c = \frac{6250}{6251}\left\{\frac{7}{22}\sum_j (1-t_j)^2(\nu_j f_j - \nu_{j-1}f_{j-1}) - \frac{2}{148}\right\} > 0.$$

Lemmas 6.4 and 6.5, with $A = A_0$, give $d < Ja$, where $J = E_{A_0}/[(1-q_{A_0})c]$. Exact substitution of the forty-one rows gives $J + 1/2 < 77 < 100$. Thus $d < Ja < 100a - a/2 \le \rho_0 - a/2$, contrary to the crescent lower bound. The finite loss $2/148$ was charged once through positive cumulative summation. The factor $N/(N+1)$ was bounded below by $6250/6251$, so the proof holds for every $N \ge 6250$. □

9.3. **The low and middle modulus ranges.**

Proposition 9.4. *There is no strict hole with $A_0 \le a \le 1/10$ and $|b| \le 47/50$.*

Proof. Lemma 9.2 gives $D > 70$. Since $d > .06$, (5.23) gives $D > [6250(.06) - 596/5]/(11/5) > 100$. Thus Proposition 4.11 and (5.24) apply. The simultaneous crescent bound $d^2 > (.06)^2 + a^2/4$ gives $D > 139$: after positive squaring the required inequality

$$6250\sqrt{(.06)^2 + a^2/4} > 1899/20 + 278(1+a)$$

is exactly

$$9688341a^2 - (1036801/5)a + 613319/400 > 0.$$

Its minimum over the entire real line is $65688734375/155013456 > 0$. Consequently Proposition 5.7 supplies

$$Qd/H < 9/4, \qquad |\operatorname{Im} S| < 4Q/H, \qquad H > \frac{N - C_F/d}{2(1+a)}, \qquad C_F = 2637/32. \tag{9.3}$$

Put $r_a = |b - a|$. If $r_a = 0$, Proposition 3.1 contradicts the strict hole. Otherwise use the adaptive shift from Lemma 7.2:

$$u = \min(r_a^2, 1/2), \quad t = 1 - u, \quad \gamma = b - ta, \quad C = ta[2x - (1+t)a] \ge a^2 u(1-u) > 0.$$

It gives

$$-\log B_p(\gamma) + CQ/2 < \log(8/C), \qquad B_p(\gamma) = \prod_{p(z_i)=0} \left| \frac{\gamma - z_i}{1 - \overline{z_i}\gamma} \right|. \tag{9.4}$$

A zero product gives a contradiction. Every unselected factor is at most one.

The range $r \le .11$. Take $d_* = .89$, $V = (9/4)/.89$ and $H_* = 6250(.89)/(9/4)$, using Proposition 3.1 and the prior ratio. The complete inverse row $R = .1$, $W = 8$, $\nu = .99$ satisfies (6.1)–(6.2). Its count $K = \nu(7/22)H_*\widehat{R} - 2$ obeys $K/6251 > 1/100$. The count lower bound has the form $cN - 2$, so its quotient by $n = N + 1$ increases. Every counted root has adaptive factor at most

$$\frac{R + .1}{1 - (.11 + R)(.11)} < 1/4 < e^{-1}.$$

Hence $-\log B_p > n/100$. Proposition 3.2 gives

$$d \le \sqrt{1 - (1 - .11^2)/n} < 1 - .49/n.$$

If $r_a \le .4/n$, the strict hole instead gives $d > 1 - r_a \ge 1 - .4/n$. Thus $r_a > .4/n$, and monotonicity of $u(1-u)$ on $[0, 1/2]$ gives

$$C > A_0^2(.4/n)^2[1 - (.4/n)^2] > 1/(20000000n^2).$$

The last scalar inequality holds at $n = 6251$ and strengthens as n increases. Positive exponential sums give $e^{20} > 8(20000000)$ and $e^{10} > 6251$. Equation (9.4) would imply $n/100 < 20 + 2\log n$, which fails at 6251 and thereafter, since the derivative of the difference is $1/100 - 2/n > 0$.

The range $.11 < r \le .82$. Use

$$d_* = .18, \quad V = (9/4)/.18, \quad H_* = [6250 - C_F/.18]/2.2, \quad R = 11/250, \quad \nu = .99, \quad W = 6.$$

The full inverse predicates hold. If $z = b + v$ is counted, $\operatorname{Re} v \le 0$ by rightmostness. Therefore $|\gamma - z|^2 = |v + ta|^2 \le R^2 + .01$ and the denominator is at least $1 - (.82 + R)(.82) > 0$. The exact squared comparison bounds the factor by $3/8$. Since $e^{.98} < 8/3$, the count gives $-\log B_p > .98K > 21.5$. Also $r_a > .11 - a$ and

$$C > a^2(.11 - a)^2[1 - (.11 - a)^2] > 4 \cdot 10^{-9}.$$

Indeed the concave function $a(.11-a)$ exceeds $1/15000$ at both endpoints of $[A_0,.1]$, and the remaining factor exceeds .9. The exact positive sum $e^{21.5} > 8/(4\cdot 10^{-9})$ contradicts (9.4).

The range $.82 \le r \le .94$. Here $r_a^2 \ge (.82-.1)^2 > 1/2$, so the shift is $\gamma = b - a/2$ and $C = ax - 3a^2/4 \ge a^2/4$. We use twenty-three product rows, specified by rational inputs in `LOW_MIDDLE6251_PRODUCT_PROPOSAL.json`. For each row $B_L \le r \le B_U$, $a_L \le a \le a_U$, $x \le X$, choose the displayed d_*, V, H_* from that data, and nested physical radii R_j. Each full contour uses

$$\Lambda_j = \frac{VR_j}{1-R_j/d_*}, \quad \widehat{R}_j = R_j - V\left(\frac{R_j^2}{2} + \frac{R_j^3}{3(d_*-R_j)}\right), \quad B_j = \frac{V}{(1-R_j/d_*)^2(1-\Lambda_j)^3},$$

with $\epsilon_j = (1001/1000)B_j/H_*$ and $g_j = (1001/1000)\nu_j B_j \widehat{R}_j$. All predicates (6.1)–(6.2) are checked at H_*, together with $\nu_j H_* \widehat{R}_j > 22/7$. Lemma 6.9 supplies a whole-disk factor $f_j < 1$. The weights $w_j < -\log f_j$ are rational lower bounds from the positive atanh series after dyadic scaling into $[1/2,1)$. They decrease with radius. All count slopes increase, and positive Abel summation (6.5) gives the budget

$$\mathcal{B} = H_* \sum_j w_j(s_j - s_{j-1}) - 2w_1, \qquad s_j = \nu_j(7/22)\widehat{R}_j, \quad s_0 = 0.$$

For a positive whole-row lower bound $C_{\min}$ and a rational target T_*, the final predicates are

$$e^{T_*} > 8/C_{\min}, \qquad \mathcal{B} + 6250C_{\min}/2 > T_*. \tag{9.5}$$

The energy term follows from Proposition 3.1 and $C \ge C_{\min}$. Exponential bounds are finite positive sums or (5.6); radical enclosures are accepted by exact squares.

For $.82 \le r \le .925$ take $d_* = .075$, $V = (9/4)/d_*$, $H_* < [6250 - C_F/d_*]/[2(1+a_U)]$, $X = .925$, $C_{\min} = a_L^2/4$. The seven closed witness intervals have endpoints

$$A_0, .003, .01, .02, .04, .06, .08, .1.$$

The respective strict reserves in (9.5) exceed $3.03, 4.54, 4.41, 1.91, 1.45, 1.79, 2.99$.

For $.925 \le r \le .94$, $x \ge \rho_0$, take $X = .94$ and choose d_* below at least one of

$$.06, \quad \rho_0 - a_U/2, \quad \sqrt{(.06)^2 + a_L^2/4}.$$

Each displayed quantity is a crescent lower bound for d. Choose $V \ge \min((9/4)/d_*, 1/(\rho_0 - a_U/2))$ and $H_* < [6250 - C_F/d_*]/[2(1+a_U)]$. The lower bound $C_{\min}$ is the exact minimum of $a\max(a,\rho_0) - 3a^2/4$ on the full row. Below ρ_0 the function is concave; above it the function is $a^2/4$ and increases. The eleven intervals have endpoints

$$A_0, .003, .01, .02, .03, .04, .05, .06, .07, .08, .09, .1.$$

All eleven strict reserves exceed .067.

It remains to consider $.925 \le r \le .94$, $x \le \rho_0$. Put $Y = \tau Q/H$. If $Y \le 5/4$, the phase and radial identity give

$$H > \frac{N - 8(5/4)/.06}{2\rho_0 + (2-.06)(5/4)}. \tag{9.6}$$

This uses $\tau \ge .06$, $p_b/\tau \le 2 - \tau \le 1.94$, and $|\operatorname{Im} S| < 4Q/H$. Take $d_* = .06$, $V = (5/4)/.06$, an integer H_* below (9.6), $X = \rho_0$ and $C_{\min} = a_L^2/4$. The five intervals with endpoints $A_0, .003, .01, .02, .04, \rho_0$ have reserves exceeding $3.68, 5.43, 5.44, 3.01, 2.31$. These twenty-three rows comprise 1807 nested complete contours; together with the two earlier single contours, there are 1809.

Finally, in this small-x region $\tau < .078$ since $y^2 \ge .925^2 - \rho_0^2 > (1-.078)^2$. If $\tau \ge \rho_0$ then $x \le \tau$; if $.06 \le \tau \le \rho_0$ the seam used in Lemma 9.2 again gives $x < \tau$. The normalization $u = (z-i)/\tau$

therefore gives $T = 1$, $0 < \alpha = a/\tau \le X = x/\tau \le 1$, $d_u > 1$ and $Y = Q_u/h < 9/4$. The only branch not covered by the preceding product rows lies in

$$.06 \le \tau \le .078, \qquad 0 < \alpha \le X \le 1, \qquad 5/4 \le Y \le 9/4.$$

The bounds $D > 139$ and (9.3) supply the hypotheses of Lemma 10.2. In normalized coordinates the field bound reads $h > [N\tau - C_F]/[2(1+\tau)]$, since $d_u > 1$ and $a \le \tau$. The lemma excludes this entire normalized range.

All modulus and witness seams are closed. Each field lower bound is affine increasing in N, while every fixed count and energy lower bound strengthens along the degree ray. The only logarithmic degree budget, in the low range, was checked with its derivative. Thus every configuration in the proposition is covered. □

9.4. **The high modulus range.**

Proposition 9.5. *There is no strict hole with $A_0 \le a \le 1/10$, $|b| > .94$, outside the origin box* $\max(x, 1-y) \le \rho_0$.

Proof. If $x < \rho_0$, the exact comparisons

$$.94^2 > (.0433)^2 + (1-\rho_0)^2, \qquad .94^2 - \rho_0^2 > (1-.062)^2, \qquad .0433/.062 > .69$$

give $\rho_0 < \tau < .062$ and $.69 < x/\tau < 1$. The normalization $u = (z-i)/\tau$ therefore lies entirely in Lemma 10.3, with $T = 1$, $\rho_0 \le \tau \le .062$, $.69 \le X \le 1$. We may henceforth assume $x \ge \rho_0$.

Lemma 6.7 gives $D > 70$ geometrically. Proposition 5.6 then gives phase $K_0 = 21/4$ and the field (5.23). Since $|b - a| > .84$, the adaptive shift is $\gamma = b - a/2$, and (9.4) holds with $C = ax - 3a^2/4 > 0$. Put $q_N = Q/N$. Cauchy–Schwarz and the radial identity give

$$1 < 2x\sqrt{q_N} + p_b q_N + 2K_0 Q/(NH). \tag{9.7}$$

The first witness cap. Suppose $a \ge a_0$ and $x \ge x_L > 3a_0$. Concavity of $a \mapsto ax - 3a^2/4$ on $[a_0, x]$ gives $C \ge a_0(x - 3a_0/4)$: the other endpoint is $x^2/4$, which is larger when $x > 3a_0$. Choose L with $e^L > 8/[a_0(x_L - 3a_0/4)]$, and put $s = 3a_0/4$, $k = 2L/(N_0 a_0)$. The product inequality gives $q_N < k/(x-s)$. The reciprocal half-plane gives $Q/H < 2/x$. Thus (9.7) requires

$$1 < 2x\sqrt{\frac{k}{x-s}} + (1-B_L^2)\frac{k}{x-s} + \frac{4K_0}{N_0 x} \qquad (r \ge B_L). \tag{9.8}$$

This expression is convex as a function of $v = \sqrt{x-s}$: its terms are $2\sqrt{k}(v + s/v)$, $(1-B_L^2)k/v^2$, and $4K_0/[N_0(v^2+s)]$. The final term has nonnegative second derivative when $3v^2 \ge s$. It suffices to check the two endpoints of the full interval in x. For

$$B_L = .94, \quad x_L = \rho_0, \quad x_U = 1, \quad a_0 = .013, \quad L = 1879/200, \quad K_0 = 21/4,$$

both endpoint values are strictly below one. These are rational checks: the remainder after subtracting the last two terms is positive, and its square exceeds $4x^2 k/(x-s)$. The logarithmic comparison uses a positive exponential sum. We conclude $a < .013$.

The cap $a < .013$ gives $d > \rho_0 - .0065 = .054502$, so (5.23) gives $D > 100$. Thus Proposition 4.11 and (5.24) apply. For $r \ge .946$, Lemma 6.8, with $a_U = .013$, $x_0 = \rho_0$, $K_0 = 21/4$ and $p_U = 1 - .946^2$, gives

$$D > 321306321875/1995240393 > 160.$$

Proposition 5.7 therefore supplies phase four, field constant $C_F = 2637/32$, and the smaller ratio branch $Y(D)$ throughout that modulus range.

Eight successive Schur estimates. For a known witness cap $a \le a_U$ and modulus floor B_L, Lemma 6.8 with phase four gives a whole-region lower bound for D. Choose D_0 strictly below that value, then choose κ with

$$2 < \kappa < 9/4, \qquad (5/4)\kappa^2 - (D_0 - 23/2)\kappa + 2D_0 < 0.$$

The smaller branch implies $Qd/H < \kappa$. Use the complete Schur profiles as in (9.2), replacing 148 and 11067/5000 by D_0, κ. Put $\nu_0 f_0 = 0$. If

$$c = \frac{6250}{6251}\left\{\frac{7}{22}\sum_j (1-t_j)^2(\nu_j f_j - \nu_{j-1}f_{j-1}) - \frac{2}{D_0}\right\} > 0,$$

then, with $J = E_{a_U}/[(1-q_{a_U})c]$, Lemma 6.5 gives $d < Ja$ and hence $x < (J+1/2)a$. The complete sequence is

B_L	prior a_U	D_0	κ	M	upper bound for $J+1/2$	rows
.946	.013	162	2.1928	75	74.260	42
.955	.01108	198	2.1545	67	66.009	44
.97	.00779	301	2.0985	57	56.441	49
.98	.00572	442	2.0659	52	51.715	52
.99	.00430	781	2.0367	49	48.244	54
.995	.00296	1256	2.0226	47	46.493	56
.997	.00216	1651	2.0172	46	45.798	57
.998	.00173	1951	2.0145	46	45.515	57

All 411 triples (j, m, W) are rational integer data in `HIGH6251_INTEGER_DATA.json`, with $t = j/200$, $\nu = m/10000$. For each, the complete inverse, every side and corner, the amplitude homotopy, positive odd height, and positive Abel increment are verified by (9.2). Thus the conclusion of each row is $x < Ma$ on its whole domain.

After each row we improve the witness cap, in the same order. Suppose $a \ge a_0$ and $x < Ma$, and put $x_0 = Ma_0$, $\gamma_M = 1/M - 3/(4M^2)$. Concavity gives

$$C \ge \begin{cases} a_0(x - 3a_0/4), & \rho_0 \le x \le x_0, \\ \gamma_M x^2, & x \ge x_0. \end{cases} \tag{9.9}$$

The two expressions agree at x_0. For the first branch use $k = 2L/(N_0 a_0)$, $s = 3a_0/4$, and for the second use $k_q = 2L/(N_0\gamma_M)$. The cap $a < a_U$ gives the phase error $8/[N_0(x - a_U/2)]$ in (9.7). The first branch is convex in $\sqrt{x-s}$ when $3(x-s) > s - a_U/2$; the second branch requires

$$1 < 2\sqrt{k_q} + (1 - B_L^2)k_q/x^2 + \frac{8}{N_0(x - a_U/2)},$$

whose right side decreases with x. Hence positive squared endpoint checks at $x = \rho_0, x_0$, together with $e^L > 8/[a_0(\rho_0 - 3a_0/4)]$, exclude $a \ge a_0$. The rows are

B_L	M	new cap a_0	L
.946	75	.01108	9.528
.955	67	.00779	9.834
.97	57	.00572	10.115
.98	52	.00430	10.382
.99	49	.00296	10.738
.995	47	.00216	11.043
.997	46	.00173	11.26
.998	46	.00146	11.426

The prior cap of each Schur row is exactly the conclusion of the preceding cap row, and its modulus domain is contained in the preceding one. Thus these estimates form a finite ordered chain.

Root counts in physical disks. The closed initial domains are as follows. Each contains all x from ρ_0 to the stated upper endpoint, and all $a \in [A_0, a_U]$ compatible with the configuration.

modulus interval	a_U	M	upper x	closed leaves
$[.94, .946]$	.013	–	1	1
$[.946, .955]$	.01108	75	.831	1
$[.955, .97]$	.00779	67	.52193	1
$[.97, .98]$	.00572	57	.32604	1
$[.98, .99]$	.00430	52	.2236	2
$[.99, .995]$	.00296	49	.14504	3
$[.995, .997]$	.00216	47	.10152	4
$[.997, .998]$	.00173	46	.07958	4
$[.998, .9985]$	.00146	46	.06716	3

The data give eleven midpoint splits, each along the modulus or x coordinate, with both closed children retained. This gives twenty closed leaves. The binary rule regenerates the whole cover, including the boundaries of all initial domains and their common seams.

On a leaf $B_L \le r \le B_U$, $x_L \le x \le x_U$, put $d_* = x_L - a_U/2$, $V = 1/d_*$ and $p_{\max} = 1 - B_L^2$. The available field bounds at the degree floor are

$$H_{\text{hole}} = \frac{6250 - C_{\text{field}}/d_*}{2(1 + a_U)},$$
$$H_{\text{phase}} = \frac{6250 - 2K_0/d_*}{\max\{2x_L + p_{\max}/(x_L - a_U/2), 2x_U + p_{\max}/(x_U - a_U/2)\}}. \tag{9.10}$$

The denominator in the second bound is the maximum of a convex function on the full interval. In the first initial domain use $K_0 = 21/4$, $C_{\text{field}} = 1899/20$; in all others use $K_0 = 4$, $C_{\text{field}} = 2637/32$. All numerators are positive. Each leaf declares a field mode and an integer H_* strictly below the corresponding expression in (9.10). Its full physical radius R, $\nu = m/10000$ and integer W satisfy the inverse predicates with d_*, V, H_* as above. Thus the actual cumulative count is greater than $\nu(7/22)H_*\widehat{R} - 2$.

For its factor $f < 1$ the leaf verifies (6.8) with $P = 1 - B_U^2$, $T = a_U/(2P)$, $U = R/P$ and $L = 1 + 2Tx_U$. In the first initial domain take $C_{\min} = A_0(x_L - 3A_0/4)$. In every other initial domain use both lower constraints on a:

$$C_{\min} = \max\{A_0(x_L - 3A_0/4), \gamma_M x_L^2\}.$$

Every field in (9.10) has the form $(N - c)/F$ with $c \ge 0$. Equation (6.9) therefore gives $Q > H_*^2/6250$ throughout the full degree ray. For a rational $\ell < -\log f$ the complete product lower budget is

$$\ell\{\nu(7/22)H_*\widehat{R} - 2\} + \frac{C_{\min}H_*^2}{12500}. \tag{9.11}$$

Each leaf has a rational target T_* with $e^{T_*} > 8/C_{\min}$ and with (9.11) strictly greater than T_*. The logarithmic lower weight is checked by $\mathcal{E}_{24}(\ell) < 1/f$, and the target by the positive exponential sum through degree one hundred. All twenty whole-leaf reserves exceed .2558. The factor estimate holds for every original zero in each physical disk, and all remaining product factors are at most one.

The unit-circle endpoint. On $r \ge .9985$, the preceding chain gives $x < 46a$ and $a < .00146$. Hence $C \ge \gamma_{46}x^2$. Take $L = 11521/1000$; the exact positive exponential gate gives $e^L > 8/(\gamma_{46}\rho_0^2)$. The

product, phase and radial identities would require

$$1 < 2\sqrt{k_q} + (1 - .9985^2)k_q/x^2 + \frac{8}{6250(x - .00146/2)}, \qquad k_q = \frac{2L}{6250\gamma_{46}}.$$

The right side decreases for all $x \geq \rho_0$. At $x = \rho_0$ it is less than $.990541 < 1$; equivalently the positive remainder after the two rational terms has square exceeding $4k_q$ by more than .0158. This excludes the whole endpoint range, including $p_b = 0$ and $|b| = 1$.

The fixed input file contains all eight Schur profiles, the eight successive caps, the nine initial domains, all eleven exact splits, the twenty leaf witnesses and the endpoint parameters. `check_high6251_exact.py` reconstructs all the displayed rational predicates and every inherited-domain equality. The first fields and the subsequent same-gap bounds increase with N; the Schur factor uses the smallest $N/(N+1)$; and the counts and field-energy budgets strengthen by (6.9). Thus the proof covers every $N \geq 6250$. □

10. Two supplementary scale intervals

We prove the two normalized source estimates used at the joins in Section 9. Throughout this section we use the strict-hole configuration of Lemma 2.2, with an actual simple rightmost zero b. Assume $N = n - 1 \geq N_0 = 6250$, put $n_0 = 6251$, and write

$$F(u) = p(i + \rho u), \qquad \beta = X - i, \qquad 0 < \alpha = a/\rho \leq X \leq 1.$$

Thus $\rho = 1 - \operatorname{Im} b$ and $X = \operatorname{Re} b/\rho$. Use $S = F''(\beta)/F'(\beta)$, $h = |S|$, $Q = \sum_v |\beta - v|^{-2}$, $d = \min_v |\beta - v|$, $D = dh$, and $y_0 = Q/h$, where v runs over all zeros of F', counted with multiplicity. The physical quantities $H_{\mathrm{phys}} = |p''(b)/p'(b)|$ and $d_{\mathrm{phys}} = \min_{p'(\xi)=0} |b - \xi|$ satisfy

$$h = \rho H_{\mathrm{phys}}, \qquad d_{\mathrm{phys}} = \rho d.$$

Every critical label $v = r - it$ satisfies

$$r < \alpha/2, \qquad \rho|v|^2/2 \leq t < \rho|v - \alpha|^2/2. \tag{10.1}$$

The inset is $I(u) = -\operatorname{Im} u - \rho|u|^2/2$. In particular, $I(\beta) = 1 - \rho(1 + X^2)/2 \geq 1 - \rho$. The finite polar quantities W, C, E, θ are those in (8.15); their proof applies for every $0 < \rho < 1$ used below.

10.1. A closed-cell source criterion.

Suppose

$$\rho \in [p, r], \quad y_0 \in [l, u], \quad X \in [X_-, X_+] \subseteq [0, 1], \quad d > 1, \quad h \geq H, \quad |\operatorname{Im} S| < J.$$

Here $0 < p \leq r < 1$, $0 \leq l \leq u$, and $H, J > 0$. The radial identities (8.9) and (8.7), with $T = 1$, give

$$\frac{N\rho}{2} < \rho h + J + y_0 h, \tag{10.2}$$

$$(1 - \rho/2)Q < \frac{N\rho}{2} + J. \tag{10.3}$$

For the second inequality the negative critical-inset term is retained:

$$E_c < \rho X \operatorname{Re} S - \rho X^2 Q/2.$$

Its substitution cancels the X^2 terms in $I(\beta)Q$. Consequently, putting

$$B_r = \frac{r/2 + J/N_0}{1 - r/2},$$

one may take any positive rational U satisfying at least one of

$$U \geq u, \qquad U^2 \geq B_r, \qquad Ul \geq B_r \quad (l > 0) \tag{10.4}$$

as an upper bound for h/n. The first alternative follows from $h^2 \leq NQ$; the second uses the same inequality and (10.3); the third uses $(h/n)y_0 = Q/n < B_r$.

Choose $\delta>0$ with $r^2(1+\delta)^2/2<1$ and test at $s=X+\delta$. Define

$$\kappa_\delta(X)=\frac{X^2}{2}-\frac{X\max(0,1-X\delta-\delta^2)^2}{4(1+\delta^2)(X/2+\delta)}. \tag{10.5}$$

Assume the whole-cell inequalities

$$[r(1+\delta)/2]^2<\delta, \qquad X-\kappa_\delta(X)y_0<\overline{A} \tag{10.6}$$

with $\overline{A}>0$. The first places the inverse Cayley test strictly to the right of all original zeros, because $s/(1+(\rho s/2)^2)>X$ follows from $\delta>X(\rho s/2)^2$. The second and Lemma 8.4 give $C_{\mathrm{real}}<\rho h\overline{A}$. Set

$$f=\frac{1}{1-r^2(1+\delta)^2/2}, \quad D_p=1+[r(1+\delta)/2]^2, \quad B=2D_p(f\overline{A}U+1/n_0).$$

If $0<B<1$, the finite polar inequality yields

$$\frac{E}{h}\le\frac{W}{h}<\mathcal{U}:=\frac{r(f\overline{A}+1/H)}{1-B}. \tag{10.7}$$

Both finite corrections $1/H$ and $1/n_0$ occur in this expression.

For a lower bound for E/h, choose a rational radius $0<R<1$, a number $0<\nu<1$, and a positive integer W_0. Put

$$\lambda=\frac{uR}{1-R}, \qquad e_R=u\left\{\sum_{k=2}^{16}\frac{R^k}{k}+\frac{R^{17}}{17(1-R)}\right\}, \qquad \widehat{R}=R-e_R, \tag{10.8}$$

$$B_R=\frac{u}{(1-R)^2(1-\lambda)^3}, \qquad \epsilon=\frac{1001B_R}{1000H}, \qquad g=\frac{1001}{1000}\nu B_R\widehat{R}, \qquad p_0=\frac{1}{1+\lambda}, \quad a_0=\frac{1}{1-\lambda}.$$

The complete logarithmic tail appears in e_R. For $W_0\in\{1,2,3,4,5,6,8,10\}$ use $e_0=1/d_{W_0}$, where

W_0	1	2	3	4	5	6	8	10
d_{W_0}	2	7	20	50	148	403	2900	22000.

In each case $\sum_{k=0}^{69}W_0^k/k!>d_{W_0}$, so $e_0>e^{-W_0}$. Require

$$0<R<1, \quad 0<\lambda<1, \quad \widehat{R}>0, \quad W_0^2<(1-\nu^2)(H\widehat{R})^2, \tag{10.9}$$

$$\epsilon(1+e_0)+ge_0<1-a_0e_0, \qquad \epsilon(1+e_0)+ge_0^2<p_0-e_0, \tag{10.10}$$

$$\epsilon+ge_0<p_0, \qquad 0<p_0<1<a_0, \qquad \nu H\widehat{R}>22/7. \tag{10.11}$$

These are the entire-contour hypotheses (8.39)–(8.40). The inverse-log and amplitude comparisons therefore give an actual original-root count $L_R>h\nu\widehat{R}/\pi-2$. The phase bound confines these roots to

$$|u-\beta|<R, \qquad |\operatorname{Re}(u-\beta)|\le\ell_R:=\min(R,W_0/H+J\nu\widehat{R}/H+e_R). \tag{10.12}$$

This is a bound on the entire inverse rectangle.

Use increasing radii R_j, round the slopes $\nu_j(7/22)\widehat{R}_j$ down and the widths up to the 10^{-12} mesh, and replace the widths by running maxima. Write the resulting slopes and widths as s_j,ℓ_j. They are checked to satisfy $0<s_1<\cdots<s_k$, $R_1<\cdots<R_k<1$, and $0\le\ell_1\le\cdots\le\ell_k\le R_k$, with each $\ell_j\le R_j$. Rightmostness and $R_j<1$ give $|\operatorname{Re}u|\le1$ for the counted roots. Their weight in $E(s)$ is at least

$$\omega_j=\max\left(0,\min_{t\in\{1-R_j,\,1+R_j\}}\frac{t-r(1+t^2)/2}{(\delta+\ell_j)^2+t^2}\right). \tag{10.13}$$

This controls the whole depth interval: the derivative numerator of the displayed quotient is $A-t^2+r(1-A)t$, $A=(\delta+\ell_j)^2>0$. It has one positive root, which is a maximum. If an endpoint lower bound is negative, the true inset is still nonnegative. Round ω_j down to the same mesh and put $w_j=\max_{m\ge j}\omega_m$. The nested regions make these nonincreasing weights valid.

The actual source β has weight at least $b_0 = (1-r)/(1+\delta^2)$. Formula (8.48) gives

$$\frac{E}{h} > \mathcal{L} := \sum_{j=1}^{k} w_j(s_j - s_{j-1}) + \frac{\min(0, b_0 - 3w_1)}{H}, \qquad s_0 = 0. \tag{10.14}$$

The source is removed exactly once from each cumulative count. The finite loss three is consequently retained. If the source compensation is positive, discarding it remains valid without an upper bound on h. In particular, the first count lower bound need not itself be positive.

Lemma 10.1. *Under the preceding whole-cell hypotheses, a collection satisfying (10.9)–(10.11), the stated nesting and rounding conditions, and $\mathcal{L} > \mathcal{U}$, excludes the entire closed cell.*

Proof. Equations (10.7) and (10.14) are contradictory. □

Equation (10.6) has a polynomial verification. The dependence on y_0 is affine, so it suffices to check $y_0 = l, u$. On the active interval $X < (1-\delta^2)/\delta$,

$$\kappa_\delta(X) = \frac{X[X-(1-\delta)^2][X+(1+\delta)^2]}{2(1+\delta^2)(X+2\delta)}. \tag{10.15}$$

On the other interval it is $X^2/2$. Clear the positive denominators in $\overline{A} - X + \kappa_\delta(X)l$ and $\overline{A} - X + \kappa_\delta(X)u$ separately on those two closed pieces. The rational Bernstein rule in Section 12, with exact dyadic subdivisions, proves positivity. The two closed pieces include their common endpoint, and both y_0 endpoints are checked for either sign of κ_δ.

10.2. **The moment interval up to $\rho = 39/500$.**

Lemma 10.2. *There is no strict-hole configuration in the normalized range*

$$\frac{3}{50} \le \rho \le \frac{39}{500}, \qquad 0 < \alpha \le X \le 1, \qquad \frac{5}{4} \le y_0 \le \frac{9}{4}$$

satisfying

$$D \ge 139, \quad Qd/h < 9/4, \quad |\operatorname{Im} S| < 4y_0, \quad H_{\text{phys}} > \frac{N - (2637/32)/d_{\text{phys}}}{2(1+a)}. \tag{10.16}$$

Proof. The source criterion requires $d > 1$. The lower intersection of the two critical-boundary circles at $r = \alpha/2$ has depth

$$\eta = \frac{1 - \sqrt{1 - \rho^2\alpha^2/4}}{\rho} < \frac{\rho\alpha^2}{4}.$$

The closest-point comparison gives $d^2 > (X - \alpha/2)^2 + (1-\eta)^2$. To see why it controls the whole crescent, on the lower hole arc put $u = r - \alpha \le -\alpha/2$. The relevant scalar product is

$$(X-\alpha)u + (1/\rho - 1)\sqrt{\rho^{-2} - u^2}.$$

Its derivative is positive for $u \le 0$. The largest allowed real coordinate therefore gives the nearest permitted arc point. The upper arc and the outward radial parts are farther away. Consequently

$$d^2 > 1 + \frac{X^2}{4} - \frac{\rho\alpha^2}{2} \ge 1 + (1/4 - \rho/2)X^2 > 1,$$

since $1/4 - (39/500)/2 > 0$.

Using $d > 1$, $a = \rho\alpha \le \rho$, and $N\rho > 2637/32$, the last hypothesis in (10.16) yields

$$h > \frac{N\rho - 2637/32}{2(1+\rho)}. \tag{10.17}$$

The bound $D \ge 139$ is an input hypothesis; the table below gives lower bounds $h \ge H$, starting with $H = 137$.

Use the five scale intervals

$$[.06,.064],\ [.064,.068],\ [.068,.072],\ [.072,.075],\ [.075,.078]$$

and the four moment intervals

$$Y_1=[5/4,3/2],\quad Y_2=[3/2,7/4],\quad Y_3=[7/4,2],\quad Y_4=[2,9/4].$$

Every pair is a closed cell. On the jth moment interval put $J=4u$ and use the following exact parameters:

j	1	2	3	4
δ	1/2	11/20	3/5	13/20
$10^6\overline{A}$	428583	358161	304796	262971
J	6	7	8	9

Take $X_-=0$, $X_+=1$ in every cell. The lower fields and upper ratios U are:

$[\rho_-,\rho_+]$	H	$10^{12}U_1$	$10^{12}U_2$	$10^{12}U_3$	$10^{12}U_4$
[.060, .064]	137	27239669422	22809917356	19645808737	17272727273
[.064, .068]	148	28952380953	24237405107	20869565218	18343685301
[.068, .072]	159	30672199171	25670816045	22098399526	19419087137
[.072, .075]	170	31966753247	26749783550	23023376624	20228571429
[.075, .078]	179	33265348596	27832119321	23951241267	21040582727

For each cell $[p,r]\times[l,u]$, its H is at most one of

$$\frac{N_0p-2637/32}{2(1+r)},\qquad \frac{N_0p/2-J}{r+u}.$$

The first comparison uses (10.17) and the second uses (10.2). Thus $h\ge H$ in each cell. The displayed U satisfy (10.4). Every polynomial comparison in (10.6) is verified by (10.15).

The fixed contour lists have, in the order of this table,

55	49	42	37
56	50	44	39
56	50	45	40
58	50	45	40
58	51	47	42

members, for a total of 954. Their exact triples (R,ν,W_0) are the rows with family M in `small_extensions/small_extensions_contours.csv`; the indices begin at zero, with the scale index preceding the moment index. They are also recorded in `TMOMENT6251_PROPOSAL.json`. The exact checker `certify_Tmoment6251_complete.py` reconstructs (10.8)–(10.14), including every complete contour and both polynomial pieces. Every cell satisfies

$$\mathcal{L}-\mathcal{U}>\frac{14}{10000}>0.$$

Lemma 10.1 excludes the twenty cells.

The scale and moment intervals include all their common endpoints. For $N\ge N_0$, both lower fields increase, while J/N and $1/n$ in the upper bounds decrease. All contour, polynomial, and source-weight bounds are fixed on their closed cells. The contradiction therefore holds on the entire degree ray. □

10.3. **The high-X interval up to $\rho = 31/500$.**

Lemma 10.3. *There is no strict-hole configuration with*

$$\rho_L := \frac{30501}{500000} \le \rho \le \rho_U := \frac{31}{500}, \qquad X_0 := \frac{69}{100} \le X \le 1, \qquad 0 < \alpha \le X.$$

Proof. The distance calculation preceding (8.37) remains valid on this interval. The endpoint inequality

$$\frac{2}{125} - \rho_U \left(\frac{1}{4} + \frac{1}{15625}\right) > 0$$

gives $\eta < \alpha^2/125$ on the whole interval. The nearest-point comparison above gives

$$d^2 > (X - \alpha/2)^2 + (1 - \alpha^2/125)^2 \ge 1 + \frac{117}{500}X^2 =: G(X).$$

Write $G_0 = G(X_0)$. Lemma 9.1 applies because $1/100 < \rho_L \le \rho \le \rho_U < 9/100$; it gives $D > 25$. Proposition 4.10 therefore gives

$$y_0 < 3/d < 3/\sqrt{G_0} < 57/20.$$

The unphased radial identity gives

$$h > \frac{N_0 \rho_L}{2(1 + 57/20)} > 49, \qquad D^2 > 49^2 G_0 > 51^2.$$

In the matrix envelope (4.2), put $S/h = c + is$ with $c \ge 0$. If $|s| \ge 7/10$, then $c < 18/25$ and its right side is at most

$$\frac{21/5}{51^2} + \frac{(11/2)(18/25)}{51} + \frac{1 - (7/10)^2}{4} - \frac{(7/10)^2}{2} < 0.$$

Thus $|\operatorname{Im} S| < (7/10)h$ and $\mu = 1/D < 1/50$.

A joint source and moment estimate. Take $\delta = 3/5$ and set

$$b_0 = \frac{1 - \rho_U}{1 + \delta^2}, \quad f = [1 - \rho_U^2(1 + \delta)^2/2]^{-1}, \quad D_p = 1 + [\rho_U(1 + \delta)/2]^2.$$

The source β and the finite polar inequality give

$$\rho h A_\delta > K, \qquad A_\delta = X - \kappa_\delta(X) y_0, \qquad K = \min_{\rho \in \{\rho_L, \rho_U\}} \frac{n_0 \rho b_0/(n_0\rho + 2D_p b_0) - \rho}{f}. \tag{10.18}$$

The expression minimized is concave in ρ and increases with n, as in (8.29). Exact evaluation gives

$$K = \frac{1793507148275409343}{2882630131640625000} > \frac{31}{50}.$$

Hence $A_\delta > 0$. On the entire X interval,

$$\kappa_\delta(X) > c_0 X^2, \qquad c_0 = 12/25. \tag{10.19}$$

The active endpoint $(1 - \delta^2)/\delta = 16/15$ is beyond one, so this is a single rational-polynomial comparison.

Put $v = Xy_0$. Then $0 < v < 1/c_0$ and $A_\delta < X(1 - c_0 v)$. The phase bound in the radial identity gives $N\rho/2 < hR$, where

$$R := \beta_0 + I_0 y_0, \qquad \beta_0 = 7/10 + (3/10)\rho_U, \qquad I_0 = 1 - \rho_L(1 + X_0^2)/2.$$

Multiplying this estimate and (10.18) for the same configuration gives, for $q = Q/h^2 = y_0/h$,

$$q^2 < \frac{2y_0^2 A_\delta R}{NK} \le \frac{2v^2(1 - c_0 v)[\beta_0/X_0 + I_0 v/X_0^2]}{N_0 K}. \tag{10.20}$$

The second step uses the nonnegative factor $1-c_0v$ and the decrease of $1/X, 1/X^2$ on $X \ge X_0$. The polynomial

$$N_0K - 800v^2(1-c_0v)[\beta_0/X_0 + I_0v/X_0^2] \tag{10.21}$$

is positive throughout $[0, 1/c_0]$. Clearing its positive denominator and using exact Bernstein subdivision proves this sign, together with (10.19). Thus $q < 1/20$. Proposition 5.4 applies with $q < 1/20$ and $\mu < 1/50$. It gives two distinct original zeros, different from β, of the form $u_\pm = \beta + w_\pm/S$, with

$$|w_\pm| < 17/2, \qquad |(w_+ + w_-)/2| < 19/10, \qquad |\operatorname{Im} S| < 7y_0. \tag{10.22}$$

The entire range $h \ge 95$. Suppose $h \ge 95$. Then $D^2 > 95^2 G_0 > 10000$. Proposition 4.11, followed by the wide row of Proposition 5.2, gives

$$Qd/h < 9/4, \quad q < 9/400, \quad \mu < 1/100, \quad |\operatorname{Im} S| < (23/5)y_0.$$

In particular $y_0^2 G_0 < 81/16$ and $y_0 < 427/200$. We apply Lemma 10.1 on the ten moment intervals in the following table, taking the entire scale interval $[\rho_L, \rho_U]$ in each row and $X \in [X_0, 1]$. Here $J = (23/5)u$, where u is the upper moment endpoint.

$[l,u]$	H	δ	$10^6\overline{A}$	$10^{12}U$	contours
$[0,1/4]$	607	1/200	1000001	179392392668	149
$[1/4,1/2]$	335	1/5	924589	129486068112	117
$[1/2,3/4]$	230	3/10	803342	65122807018	95
$[3/4,1]$	175	7/20	679683	43668386653	78
$[1,5/4]$	140	9/20	528001	32941176471	64
$[5/4,3/2]$	117	1/2	428583	26504850362	52
$[3/2,7/4]$	100	11/20	355372	22213966289	44
$[7/4,15/8]$	95	3/5	285411	19094795814	39
$[15/8,2]$	95	3/5	256512	17872445821	37
$[2,427/200]$	95	3/5	227612	16806687307	34

Each H is either 95 or at most $(N_0\rho_L/2 - J)/(\rho_U + u)$. Thus $h \ge H$ follows from the hypothesis $h \ge 95$ and the radial identity. The U satisfy (10.4). At both moment endpoints, the entire source polynomials are checked on $[X_0, 1]$, using the exact active/inactive split in (10.15).

The 709 exact triples are the family X rows of `small_extensions/small_extensions_contours.csv`, with the cells in the printed order; they are also in `T6251_HIGHX_PROPOSAL.json`. The checker `certify_T6251_highX_complete.py` reconstructs all the predicates in Lemma 10.1. In every one of the ten cells,

$$\mathcal{L} - \mathcal{U} > \frac{6}{10000} > 0.$$

These ten cells exclude the entire range $h \ge 95$.

The pair when $49 < h < 95$. It remains to use the pair (10.22). For $\delta = 3/5$,

$$y_0 A_\delta < v - c_0v^2 \le \frac{1}{4c_0} = \frac{25}{48}.$$

The radial identity and $|\operatorname{Im} S| < 7y_0$ give $\rho(N-2h) < 2y_0(h+7)$. Since $A_\delta > 0$ and $N - 2h > 0$,

$$\rho h A_\delta < \frac{2(25/48)95 \cdot 102}{6250 - 190}.$$

Let $B_{95} = 2D_p(95f+1)/6251 < 1$. Here $A_\delta < X \le 1$ by (10.19), so this is a valid bound for the complete polar denominator. Lemma 8.3 gives

$$W < \frac{f\,2(25/48)95 \cdot 102/(6250-190) + \rho_U}{1 - B_{95}}. \tag{10.23}$$

Put $z_\pm = u_\pm - \beta$, $R = 17/98$, and $m = 19/490$. The pair satisfies $|z_\pm| < R$, $|(z_+ + z_-)/2| < m$, and $\operatorname{Re} z_\pm \le 0$. Its insets lie in $[3/4, 6/5]$: for the lower bound use

$$I(\beta + z) \ge (1 - \rho_U)(1 - R) - \rho_U R^2/2 > 3/4,$$

and for the upper bound use $I(\beta + z) \le 1 + R < 6/5$. On this whole interval, $\sqrt{I} \ge I/2 + 49/100$. Indeed both sides are nonnegative, and the difference $(I/2 + 49/100)^2 - I$ is a convex quadratic negative at the two endpoints. The bound on the pair mean gives

$$I_+ + I_- > I_\Sigma := 2(1 - \rho_U)(1 - m) - \rho_U R^2,$$
$$I_\Sigma = \frac{43251409}{24010000} > 0,$$
$$|s - u_+|^2 + |s - u_-|^2 < D_\Sigma := 2(1 + \delta^2) + 4(7/6)m + 2R^2,$$

where $(7/6)^2 > 1 + \delta^2$. Since $I_\Sigma/2 + 49/50 > 0$, Cauchy–Schwarz on these two terms, together with the separate source β, yields

$$E > \frac{1 - \rho_U}{1 + \delta^2} + \frac{(I_\Sigma/2 + 49/50)^2}{D_\Sigma}. \tag{10.24}$$

The exact lower bound in (10.24) exceeds the upper bound in (10.23) by

$$\frac{90148458221177326087571836631833337}{97049415152918630536485588216000000} > \frac{92}{1000}.$$

This contradicts $W \ge E$.

The source bound, the polynomial (10.21), the geometric and phase bounds, and the two terminal bounds are rationally checked by `certify_T6251_highX_bootstrap.py`; its code is also called by the complete ten-cell verifier. For $N \ge 6250$, the initial fields and the source strengthen, while the finite polar and radial corrections decrease. On the terminal range $h < 95$, the denominator satisfies $N - 2h > N - 190 \ge 6060$. All other domains and contours are fixed, and every scale and moment endpoint has been included. This proves the lemma on the entire stated degree ray. □

11. Completion of the proof

Retain $N_0 = 6250$, $\rho_0 = 30501/500000$, and $A_0 = \rho_0/100$ from Section 9.

Proof of Theorem 1.1. Suppose the theorem fails. By Lemma 2.2, there is a strict supporting-edge hole $0 < a < 1$ and an actual simple rightmost zero $b = x + iy$ with $x \ge a$, $y \ge 0$. Put $r = |b|$ and $\rho = \max(x, 1 - y)$.

For $0 < a \le A_0$, Proposition 9.3 excludes $\rho > \rho_0$, and Proposition 8.1 excludes $0 < \rho \le \rho_0$. For $A_0 \le a \le 1/10$, Proposition 9.4 applies when $r \le 47/50$, and Proposition 9.5 applies when $r > 47/50$ outside the origin box. The remaining origin box is already excluded. Propositions 7.12 and 7.13 cover $1/10 \le a \le 39/40$, and Proposition 7.10 covers $39/40 \le a < 1$. Thus all values of a have been excluded.

The parameter intervals are closed and the final contradictions strict. In particular, the high small-witness estimate includes $|b| = 1$; the compact estimate includes the interval $[1/10, 11/100]$; and the two additional small-witness source ranges cover their entire common boundaries with the origin range. Every estimate is uniform for $N \ge N_0$. This proves the theorem. □

The affine invariance of the argument gives the following formulation.

Corollary 11.1. *Let p have degree at least 6251 and suppose its zeros lie in a closed disk of radius $R \geq 0$. Every point of $\operatorname{conv} Z(p)$ lies within distance R of a critical point.*

Proof. For $R > 0$, translate the center to zero and divide the variable by R, then apply Theorem 1.1. For $R = 0$ all zeros and all critical points coincide. □

12. Finite certificates and reproducibility

The electronic appendix contains the fixed rational inputs, verification programs, a file index, and a manifest of SHA-256 hashes. The programs recompute the inequalities from these inputs by the methods below.

12.1. **Elementary enclosures.** All rational calculations use integer numerators and positive integer denominators. For a positive integer m, a square-root enclosure is obtained by taking the integer square root of $\lfloor m^2 a/b \rfloor$ for a rational $a/b \geq 0$; division by m gives the lower bound and addition of $1/m$ gives an upper bound unless the square is exact. Thus the directions of the rounded radical bounds are determined by integer inequalities.

The enclosures $6283/2000 < 333/106 < \pi < 22/7$ follow from Machin's identity

$$\pi = 16 \arctan(1/5) - 4 \arctan(1/239).$$

Indeed $\tan(4 \arctan(1/5)) = 120/119$, and subtraction of $\arctan(1/239)$ gives tangent one with angle in $(0, \pi/2)$. If $A_m(x) = \sum_{k=0}^{m-1} (-1)^k x^{2k+1}/(2k+1)$, the alternating-series bounds give

$$16A_6(1/5) - 4A_3(1/239) < \pi < 16A_7(1/5) - 4A_2(1/239).$$

Both comparisons with the displayed rational endpoints are exact. Finer enclosures are obtained by retaining more terms.

For $x \geq 1$, first divide x by a power of two to put it in $[1, 2)$. With $t = (x-1)/(x+1)$ on this interval,

$$2 \sum_{j=0}^{m-1} \frac{t^{2j+1}}{2j+1} \leq \log x \leq 2 \sum_{j=0}^{m-1} \frac{t^{2j+1}}{2j+1} + \frac{2t^{2m+1}}{(2m+1)(1-t^2)}.$$

The same formula at $x = 2$ encloses the range-reduction contribution. The formulas for $\log(1 \pm r)$ in (4.8) retain their complete alternating or positive geometric remainder. For an integer $m \geq 0$, put $E_m(x) = \sum_{j=0}^{m} x^j/j!$. If $0 \leq x < m + 2$, then

$$E_m(x) \leq e^x \leq E_m(x) + \frac{x^{m+1}}{(m+1)!(1 - x/(m+2))}. \tag{12.1}$$

Negative exponentials are bounded by reciprocals of positive enclosures. The seven rows in Proposition 3.1, for example, use the lower logarithm bound with 48 terms and $e^{-x} \leq 1/E_{160}(x)$. Their checker uses only rational arithmetic.

The interval computations use directed real and complex balls. An asserted strict inequality means that the entire resulting enclosure has the required sign. Input floating-point test points are converted to their exact binary rationals before use.

12.2. **Entire integrals.** The only numerical integrations in the field certificates are of entire functions of the form $\exp(P(t))$, with P a polynomial, over a finite real interval. The preceding analytic lemmas bound the finite-product errors separately from these model integrals.

There is a direct series verification of each such integral. After an affine change of variable, write $P(t) = p_0 + \sum_{j=1}^{d} p_j t^j$ on $[0,1]$ and $\exp(P(t) - p_0) = \sum_{k\ge 0} a_k t^k$. The coefficients are determined by

$$a_0 = 1, \qquad k a_k = \sum_{j=1}^{\min(d,k)} j p_j a_{k-j}.$$

For any rational $R > 1$, Cauchy's estimate gives $|a_k| \le M_R R^{-k}$, where $M_R = \exp(\sum_{j=1}^{d} |p_j| R^j)$. Consequently

$$\left| \int_0^1 e^{P(t)}\,dt - e^{p_0} \sum_{k=0}^{m} \frac{a_k}{k+1} \right| \le |e^{p_0}| \frac{M_R R^{-(m+1)}}{(m+2)(1-1/R)}.$$

Subdividing the interval, or increasing m, gives arbitrarily small error bounds. Formula (12.1) encloses M_R. The field checkers also supply validated analytic quadrature using Arb/Acb; all integrands satisfy its entire-function hypothesis. The separate series implementations in the electronic appendix give the coefficient recurrence and the Cauchy remainder explicitly.

For the Gaussian shells in Proposition 4.5, the rational integration-by-parts and Mills bounds in its proof suffice. An additional verifier evaluates the Gaussian tails by the complementary error function with interval bounds.

12.3. **Polynomial bounds on whole boxes.** The companion centers are defined by finite recurrences, so their coefficient identities are checked by exact polynomial arithmetic. For the Gaussian center, the moments at $s = \pm 2\pi i$ obey $I_0 = 0$ and $I_m = (1 - m I_{m-1})/s$. This derives the center and residual coefficients exactly. The identity $e^s = 1$ is used only at $s = \pm 2\pi i$; the resulting polynomial bounds may then be verified on the larger real interval $6 \le |s| \le 44/7$.

After rescaling each real variable to $[0,1]$, a polynomial of degree at most d has the Bernstein representation

$$\sum_{j=0}^{d} c_j t^j = \sum_{k=0}^{d} \left(\sum_{j=0}^{k} \frac{\binom{k}{j}}{\binom{d}{j}} c_j \right) \binom{d}{k} t^k (1-t)^{d-k}.$$

The basis functions are nonnegative and sum to one. Therefore the extrema of the coefficient list bound the polynomial on the whole interval. Tensor products give the same result on a box. Exact dyadic subdivision and the de Casteljau recurrence preserve these bounds. Bounds on complex moduli are proved by applying this rule to the difference between a proposed squared bound and the squared modulus. The common coefficient-domain verifier checks all twenty Gaussian polynomial bounds by this method.

Where a one-variable polynomial sign is sharper than its Bernstein enclosure, the programs use its exact Sturm chain. Polynomial division is over the rationals; signs at both endpoints are checked separately. The variation difference counts roots in the open interval. Combined with a sign at one point and the endpoint checks, this proves the desired sign on the whole closed interval.

12.4. **Coverage and degree dependence.** The rectangular covers in this paper use either a rational edge decomposition or a dyadic tree. In the former, checking every open slab between consecutive rational edges and every edge itself proves closed coverage; the horizontal intervals are merged in sorted order. In the latter, every deepest pair of leaves must be the two halves of the same coordinate at a common parent. Replacing such a pair by its parent must eventually produce the complete root box.

All inequalities involving the degree are extended from $N_0 = 6250$ by the stated analytic monotonicities. One frequent step is worth making explicit. If $H > (N - c)/F$, with $F > 0$ and $0 \le c < N_0$, then $H/N > (1 - c/N)/F$ increases as a lower bound with N. Together with

$Q \geq H^2/N$ this gives, for every $N \geq N_0$,

$$Q > \frac{(N_0 - c)^2}{N_0 F^2}.$$

This is the energy lower bound used in the product estimates.

The principal finite covers have the following sizes. The count of contours includes complete boundaries, with the inverse-map and amplitude comparisons both checked.

Component	closed cells or rows	additional finite objects
Affine field	$275 + 28$	2750 paired shells
$D \geq 25$ transition	50614	186477 point obstructions
$D \geq 30$ transition	26397	82823 point obstructions
$D \geq 100$ transition	9745	8160 point obstructions
Origin base	9 path ranges	55 inverse contours
Origin $T = 1$ strip	63	3831 inverse contours
Origin $X = 1$ strip	301	14891 inverse contours
Fine small-witness range	1	41 inverse contours
Low and middle small-witness range	23	1809 inverse contours
High small-witness range	20	$411 + 20$ contour constructions
Moment extension	20	954 inverse contours
High-X extension	10	709 inverse contours
Boundary and auxiliary strips	8	whole-strip inequalities
Compact range	13	$2918 + 2$ inverse contours

The file `verification/INDEX.json` maps each mathematical component to its fixed data and executable checker. The entry point `verification/run_all.py` runs the included verifications and writes new results to a separate results directory. The input hashes are checked before execution. The programs use Python's exact rational arithmetic, SymPy for exact polynomial arithmetic, and, where indicated, `python-flint` for rigorous balls. The LaTeX source, all fixed inputs, and the verification code are included.

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